

Analog Electronics

Analog Electronics

*From Ohm's law to an operational amplifier
you have designed, transistor by transistor*

Erik Pihl

Analog Electronics

Copyright © 2026 QRTECH Academy.

This book, and the course material it is typeset from, is released under the MIT License, which is the license of the companion repository. You may use, copy, modify and redistribute it, including commercially, provided the copyright notice and the license text travel with it. The full text is in the repository, as LICENSE.

The repository holds the lecture material, the exercises, the worked solutions that this book does not print, the figure sources, and the test suites that judge the analysis library you write:

<https://github.com/qrttech-academy/analog-electronics>

Every figure in this book is generated, from the Python in `diagrams/` of that repository, and is the same drawing the lectures show. Every number this book quotes as a result is checked against `diagrams/models.py` by the repository's own continuous integration, so a figure that is redrawn and a number that changes cannot silently disagree with the text around them.

The C++ this book specifies is C++17, compiled with GCC. *The Art of Electronics* and *Microelectronic Circuits* are cited as further reading and are the property of their respective publishers. Part numbers such as the 2N3904 appear as examples of devices a design might use.

Set in TeX Gyre Pagella and DejaVu Sans Mono with Lua^AT_EX.

Typeset from the ten lectures of the course, their exercises, and the two written papers. This edition: 12 September 2026.

Contents

Preface	ix
1 Circuits, units, and the nodal solver	1
1.1 Circuits, units, and loading	2
1.2 Nodal analysis, and what to build	6
1.3 Review	9
1.4 Exercises	10
2 Reactance, phasors, and frequency response	15
2.1 Reactance, time constants, and phasors	16
2.2 The complex stamps, and what to build.	20
2.3 Review	22
2.4 Exercises	24
3 Passive filters and the operational amplifier	29
3.1 Passive filters, cascading, and Q	30
3.2 The operational amplifier, and what to build.	34
3.3 Review	37
3.4 Exercises	38
4 Feedback, active filters, and the diode	43
4.1 Feedback, and what one number decides	44
4.2 The diode, and solving for it	47
4.3 Review	52
4.4 Exercises	53
5 The transistor as a device and as a switch	57
5.1 The bipolar transistor, and the switch	58
5.2 The MOSFET, the comparison, and what to build	61
5.3 Review	65
5.4 Exercises	67

6	Biasing, and what an emitter resistor actually buys	71
6.1	The quiescent point, and the divider that loads itself	72
6.2	Thermal drift, the emitter resistor, and what to build	75
6.3	Review	80
6.4	Exercises	81
7	Small-signal analysis: r_e, the emitter factor, and the cascode	85
7.1	The small-signal model, and the three results	86
7.2	The emitter factor, and where it belongs	89
7.3	Review	96
7.4	Exercises	97
8	Followers, and the output stage they turn into	101
8.1	The follower, and the load it exists to survive	102
8.2	The output stage, and the model that finally breaks	106
8.3	What a real power amplifier adds	112
8.4	Review	114
8.5	Exercises	116
9	The differential amplifier	121
9.1	The pair, the tail, and the two factors of two	122
9.2	Rejection, the mirror, and what limits it	125
9.3	Review	130
9.4	Exercises	131
10	Building an operational amplifier, and the capstone	135
10.1	The amplifier, stage by stage	136
10.2	The budget, the missing operating point, and the loop	139
10.3	Review	145
10.4	Exercises	146
A	The Written Papers	149
B	Written Paper A	155
C	Written Paper B	163
D	Model Answers to Paper A	171
	Question 1: A divider, its resistance, and the load that takes a third.	171
	Question 2: One capacitor, two corners, and the resistance nobody drew . .	173
	Question 3: Two sections, one buffer, and the factor of 1.72	174
	Question 4: Loop gain, the error it divides, and a diode that will not converge	175
	Question 5: Sixty millivolts a decade, and a switch that ignores beta	177

Question 6: A bias point that loads itself, and drift that is a loop gain	178
Question 7: The emitter factor, and the node it belongs to	180
Question 8: Eight ohms, two Darlington's, and a bias that is not two drops .	182
Question 9: The tail, the 520 volts, and the resistor that cancels	183
Question 10: A gain budget, and the two stages that amplify nothing	184
E Model Answers to Paper B	187
Question 1: Where a source resistance comes from.	187
Question 2: Why a corner is a corner	189
Question 3: Cascades, virtual shorts, and what an op-amp assumes	191
Question 4: Why everything is a function of the loop gain	192
Question 5: One equation, three regions	194
Question 6: A rule, a load line, and a factor that appears twice	196
Question 7: Three results from one model	197
Question 8: A dead band, a rule, and a thing bolted to a heatsink.	199
Question 9: The tail, the tanh, and the output you choose	200
Question 10: The gain you cannot buy	202

Preface

This book is for embedded and software engineers who read a schematic every week and have never been shown how one is arrived at. You can write C++, you are comfortable rearranging an expression, and you have at some point been handed an amplifier stage with no way to tell whether it was right. It takes you from Ohm's law to a discrete operational amplifier you have designed, sized, predicted and solved, transistor by transistor.

The subject is the transistor amplifier. Circuit theory, passives and the operational amplifier are built in the first four chapters because the last six need them, not for their own sake. That makes the book deliberately narrow: ten hours of lectures is not long enough for everything an analog textbook carries, so this edition keeps what decides a design and says plainly what it dropped, in *What this edition leaves out* in the Preface.

How the book is organised

The book is typeset from a ten-lecture course, and each chapter is one lecture. Its sections are the lecture's appendices, in the same order and with the same subsections, so a chapter here and a lecture directory in the companion repository can be read side by side:

Chapter	Lecture	Part	Topic
1	L01	Foundations	Circuits, units, and the nodal solver
2	L02	Foundations	Reactance, phasors, and frequency response
3	L03	Foundations	Passive filters and the operational amplifier
4	L04	Foundations	Feedback, active filters, and the diode
5	L05	Transistors	The transistor as a device and as a switch
6	L06	Transistors	Biasing, and what an emitter resistor actually buys
7	L07	Transistors	Small-signal analysis: r_e , the emitter factor, the cascode
8	L08	Transistors	Followers, and the output stage they turn into
9	L09	Transistors	The differential amplifier
10	L10	Transistors	Building an operational amplifier, and the capstone

Every chapter opens with what is in it, works through the material, closes with a review of what you should now be able to explain, and ends with its exercises. The appendices hold the two written papers and their model answers.

Three things about the ordering are deliberate, and all three differ from how the subject is usually taught:

- **The solver comes first.** Chapter 1 has you write nodal analysis before you have met a single active device. It costs an hour and it pays for the rest of the book: from then on every claim the material makes about a circuit can be checked against a program you wrote, and no result has to be taken on authority.
- **The operational amplifier comes before the transistor.** You use one as a black box in Chapters 3 and 4, and only in Chapter 10 do you build one. That inversion is the whole arc of the book: the thing you were told to trust in the fourth chapter is the thing you design in the last.
- **The imperfections come before the topologies.** Biasing, thermal drift and the emitter factor are met in Chapters 6 and 7, before any of the clever stages. Meeting them early means that the question “why would anyone use a current mirror as a load” has an answer you can compute rather than remember.

What you should be able to do at the end

Having worked through the book, you should be able to:

- Solve any resistive network by nodal analysis, by hand and in code.
- Read and predict a first-order frequency response, magnitude and phase, from its corner alone, and say what a source resistance does to it.
- Apply the two operational amplifier rules to any standard configuration, and say precisely where each of them fails.
- Compute a loop gain and everything it decides: gain error, distortion, impedances, bandwidth.
- Bias a transistor stage to a specification, including the base current’s effect on the divider that sets it, and state its thermal drift.
- Derive gain, input resistance and output resistance for any stage in the book from one small-signal model, and say which node each result belongs to.
- Choose between a follower, a Darlington and a redesign from the arithmetic rather than from habit.
- Size a class-AB output stage and say why its bias generator is bolted to a heatsink.
- Design a differential pair to a rejection requirement, and say which components do not affect the answer.
- Compute an operational amplifier’s gain budget with loading, close the loop around it, and predict the error.

What you are assumed to know

You are expected to arrive already comfortable with:

- **Ohm's law**, and what a resistor, a capacitor and a voltage source are. Nothing beyond that is assumed about circuits; Chapters 1 and 2 build the rest.
- **Engineering calculations**: rearranging an expression, working in decibels, taking a logarithm, and being willing to carry an approximation and say how wrong it is.
- **Complex numbers** far enough to accept that j rotates by ninety degrees. Chapter 2 uses that and builds the frequency domain out of it.

No semiconductor physics is taught, and none is needed. The transistor is introduced as a device with an exponential, a current gain and a saturation voltage. Depletion regions, carrier transport and the Ebers-Moll derivation are named once and not used.

C++ is used, and this is not a C++ book. You write a small analysis library across the ten chapters. The C++ involved is free functions, plain structs, `std::vector` and `std::complex`. There is no class hierarchy anywhere in it. If you have written embedded C++, you have written harder code than any of it.

There is no instrument here, and that is the design

A book like this one usually comes with a breadboard, a bench supply, an oscilloscope and LTspice. This one does not. There is nothing to install but a C++ compiler, `make` and `git`.

What replaces it is that **you write the instrument**. Across the ten chapters you build `ael`, an analog analysis library in C++: a netlist, a nodal solver, complex stamps for reactance, a Newton-Raphson loop, device models for the diode, the BJT and the MOSFET, and on top of them the closed-form results for every amplifier stage in the book. By Chapter 10 they compose into one program that takes a discrete operational amplifier, predicts every node in it, and then solves it.

None of that code is in this book or in the repository. Every component is specified in prose, with its interface declaration, in the section of the chapter that asks for it, and every one ships with a test suite that judges what you wrote.

That trade cuts both ways:

- **What you lose.** No breadboard, no oscilloscope, no real device spread, no parasitics, no thermal behaviour you can feel with a fingertip, and no experience driving an instrument. Where a number here differs from what a real measurement would give, the text says so and says by how much.
- **What you gain.** Every step is a function you wrote and can read. "The simulator says the gain is -84 " stops being an oracle. You know which term that came from, what it assumed, and what would have to change to move it.

Verification works in three layers, and none of them is a bench:

- **The Cross-check.** Every chapter ends with one exercise where you compute a number by hand, then run your own closed-form code on it, then solve the same circuit numerically with your own nodal solver. Three legs, and they are not equally authoritative: the hand calculation drops terms on purpose, the closed form drops fewer, and the solver drops none but tells you nothing about why. When two of them disagree by 8 per cent, finding out which one is wrong, or that neither is, is the exercise, and each one ends by saying how large a discrepancy to expect.
- **The shipped test suites.** Every component you implement comes with a unit-test suite written against the QAcademy Test framework. They tell you *whether* you are right without telling you *why*, which is the middle ground self-study otherwise lacks.
- **The two models against each other.** From Chapter 7 on, every stage is computed twice: once from the closed form, once by the solver from the same netlist. The closed form is the one you can reason with and the solver is the one that is right. Where they part company, the text names the term responsible.

The toolkit you build

ael lives wherever you like, as long as you tell the test suites:

```
export AEL_DIR=~/ael
make test
```

The layout every suite assumes is `$AEL_DIR/include/ael/...` for the headers and `$AEL_DIR/source/ael/...` for the translation units. **The header path is part of each specification, not a suggestion:** the suites include exactly these paths, and a component written somewhere else stays dormant however correct it is.

Chapter	Header	What it does
1	ael/net/netlist.hpp	Netlist: nodes, resistors, and independent sources
1	ael/mna/solver.hpp	solve: the DC nodal solve
2	ael/ac/sweep.hpp	solveAt and sweep: the same solve over complex admittances
3	ael/filter/response.hpp	Filter responses, resonance, Q, and the loaded cascade
3	ael/opamp/ideal.hpp	The configuration gains and the Schmitt thresholds
4	ael/feedback/loop.hpp	Loop gain, closed-loop gain, and the error
4	ael/device/diode.hpp	The diode equation, its conductance, and the step limiter
4	ael/nr/solve.hpp	The nonlinear DC operating point: the linear solve, in a loop
5	ael/device/bjt.hpp	The transport model, the three regions, and the switch
5	ael/device/mosfet.hpp	The square law, its two regions, and g_m
6	ael/bias/point.hpp	The quiescent point including the droop, and the drift
7	ael/ssm/model.hpp	The r_e model: gain, both resistances, EF, the cascode
8	ael/follower/stage.hpp	Follower gain, both resistances, and the Darlington
8	ael/output/classab.hpp	The 26 mV rule, the bias voltage, and the idle current
9	ael/diffpair/pair.hpp	Both gains, the rejection ratio, the tanh, and the offset
10	ael/report/amplifier.hpp	The capstone: the budget, the open loop, the closed loop

Namespaces mirror the header *directory*, so everything in `ael/ssm/model.hpp` is in `ael::ssm`. Two directories hold more than one component and take the file name as a third level: `ael/device/bjt.hpp` is `ael::device::bjt`, and `ael/opamp/ideal.hpp` is `ael::opamp::ideal`. Units are SI and unqualified throughout: where the text quotes 26 mV, the code carries 0.026.

Two sign conventions, fixed once, because every component in the book reads them. A current source *injects at its second node*: `addCurrentSource(from, to, current)` takes current out of `from` and pushes it into `to`, so one milliamp into a node with one kilohm to ground puts that node at +1 V. And a voltage source *reports the current leaving its positive terminal*, so a 10 V source driving one kilohm reports +10 mA — which is the opposite sign to the raw modified-nodal unknown. Get either backwards and §1.2.5's suite says so on the first run; get either backwards and keep it, and every bias calculation in Part 2 comes out inverted.

What the exercises look like

Six to ten per chapter, each labelled with its kind:

Kind	What it asks of you
RECALL	State a result without looking it up.
HAND CALCULATION	Produce a number with a pencil, making the approximations deliberately and saying what each one costs.
DESIGN	Start from a requirement and end with E12 component values, the error they leave, and whether you would accept it.
CODE	Write a component of the toolkit, specified in prose and judged by the shipped suite.
CROSS-CHECK	Compute one number three ways and reconcile them. Exactly one per chapter, and it is always the last.

The Cross-check is the signature exercise. You compute a number by hand, run your own closed-form code on the same problem, and then solve the same circuit numerically with your own nodal solver. Three legs, none of them authoritative on its own. Every Cross-check ends by saying how large a discrepancy to expect and what each size of discrepancy would mean, because “your two answers differ by 8 per cent” is the lesson rather than a failure.

The solutions are not in this book. They are published in full in the companion repository, under `lectures/LNN/appendix`, including the plausible wrong answer where there is one — this is a self-study course, and a reader with nobody to ask cannot be left with an unanswered exercise. Try each exercise before you read its solution.

Two written papers, and what they are not

Nothing in the course this book comes from is marked. Assessment is the exercises in every chapter, the test suite each component comes with, and the Cross-check that closes every chapter.

Appendices [A](#) to [E](#) hold two four-hour papers with worked solutions, and they check something else: **what you can reconstruct on paper, with the book out of the room.** Ten questions each, one per chapter.

They exist purely so that you can test your own knowledge after working through the book. They gate nothing, they are not a qualification, and no part of the book requires them.

Take one after the last chapter. Every paper has one question per chapter, so sitting one partway through examines material you have not read yet.

What this edition leaves out

About a thousand pages of standard treatment became ten chapters. What went, in full, and where it is:

Cut	Kept here as
Basic electronics as eight chapters: transformer design, rectifier sizing, rms	Chapters 1 and 2, at the level later work needs
Passive filters as six chapters, each with its own full impedance treatment	Chapter 3, one treatment covering all of them
Telescopic cascodes and CMOS amplifier topologies	Not covered; the cascode itself is §7.2.5
Output stages up to 1 kW: triple class-AB, CFP-EF, Zobel networks, protection	§8.3, as reading
Sixteen appendices of full derivations	The results, with the derivations cited
Noise analysis, Miller's theorem as a chapter, and CMOS frequency response	The Miller effect in §7.2.5; noise not at all
Oscillators, voltage regulators, and a CMOS comparator	Not covered

The last two rows are material a full treatment would carry and this edition does not. **Noise in particular is a real omission:** it decides the input stage of any amplifier that matters, and it is the first thing to add if this course is ever extended. PCB layout, datasheet reading and phase margin as a design procedure are in neither.

Where the obvious answer is wrong, the correction is taught rather than applied quietly. The two largest are Chapter 6's thermal argument and Chapter 7's output resistance, and both are worked demolitions: the tempting claim, the arithmetic that breaks it, and the version that survives.

Further reading

There is no set textbook: every derivation, figure and number here is written for this course. Two books are worth having beside it.

- *The Art of Electronics*, Horowitz and Hill, 3rd edition, is the natural companion. It uses the same intrinsic-emitter-resistance approach this book is built on, and each chapter names the parts of it that cover the same material.
- *Microelectronic Circuits*, Sedra and Smith, if you want the hybrid- π treatment instead, and the device physics this book skips.

Setting up

Everything here runs in a Linux terminal: on Linux itself, or on Windows through WSL. You need a C++17 compiler, `make` and `git`, and nothing else — no LTspice, no MATLAB, no breadboard, no instrument. On WSL or Ubuntu:

```
sudo apt -y update
sudo apt -y install git make g++
g++ --version           # 9 or newer; CI builds with the g++ on ubuntu-24.04.
```

Then clone the companion repository, together with the test framework it keeps as a sub-module:

```
git clone --recursive https://github.com/qrtech-academy/analog-electronics.git
```

Three commands at its root cover everything the book asks of it:

```
make help           # List every target.
make test           # Build and run each chapter's suite against your ↵
                    ↵ toolkit.
make lint           # Links, house rules, quoted numbers, and formatting.
```

`make test` needs `AEL_DIR` to know where your toolkit is. **With it unset the suites still build and still run**, and report that only each chapter's reference tests were active: every test for a component you have not written yet sits behind an `#if __has_include`, so it switches itself on the moment the header appears. That is the correct state on day one, and **anything not yet implemented is skipped loudly rather than failing**, which is what makes `make test` meaningful on the first day and on the last.

Circuits, units, and the nodal solver

Circuit analysis as something a program does, built before there is anything interesting to analyse.

IN THIS CHAPTER

- Charge, current, voltage and power, and the sign convention every later result depends on.
- Series and parallel, and the voltage divider as the one circuit worth memorising.
- Kirchhoff's two laws, which are the only physical content in the whole of circuit theory.
- Thevenin and Norton, and loading as the reason either of them matters.
- Nodal analysis: one unknown per node, one equation per node, and the conductance stamp.
- Why a program that solves circuits is easier to write than the equations are to solve by hand.
- Building `ael::net` and `ael::mna`.

This chapter is the circuit theory the rest of the book needs, and nothing more than that. Most of it will be familiar if you have an electronics background, and two parts of it will not be used the way you expect. The first is the **sign convention**: current into a component is positive, and every gain, every bias point and every output resistance in Part 2 inherits that choice, so it is fixed here rather than argued about in Chapter 7. The second is **loading**. A divider gives 1.71 V until something is connected to it, at which point it gives 1.09 V, and that one subtraction is the single arithmetic fact this book returns to six times, in six different costumes, ending with an operational amplifier that loses 11 dB to it.

Then you write a solver. §1.2 has you implement nodal analysis before you have met a single active device, which costs an hour and pays for the rest of the book: from then on every claim the text makes about a circuit can be checked against a program you wrote, and no result has to be taken on authority. The method itself is one idea — one unknown per node, one current-law equation per node, and a *stamp* that turns each element into matrix entries without anyone deriving anything.

Work the chapter in order. §1.1 is the theory, and if you have the background it is a twenty-minute skim, except for §1.1.6. §1.2 builds the solver, and §1.2.5 is its specification.

The exercises in §1.4 are where you write the code and where the Cross-check shows you what three legs agreeing looks like, before a later chapter shows you what it looks like when they do not.

1.1 Circuits, units, and loading

The circuit theory the rest of the book needs. With a background in electronics, read §1.1.1 for the conventions this book fixes, then skip to §1.1.6.

1.1.1 The sign convention, and why it is worth five minutes

Neither current direction nor voltage polarity is discoverable from a schematic. Fixed once:

- **Current into a terminal is positive.**
- **A voltage is a difference.** V_{AB} is the potential at A minus that at B; a single subscript such as V_C is measured against ground.
- **Ground is node zero**, by definition rather than by measurement.

A bias voltage that comes out negative, or an inverting stage whose gain comes out positive, is a convention error before it is a physics error.

1.1.2 The three quantities, and the one that is usually the answer

Quantity	Symbol	Unit	What it is
Current	I	ampere	Charge per second past a point.
Voltage	V	volt	Energy per unit charge between two points.
Resistance	R	ohm	The ratio of the two, when that ratio is constant.
Power	P	watt	VI , and therefore I^2R or V^2/R .

Ohm's law, $V = IR$, defines a resistor rather than stating a law of nature. Every device in Part 2 has a curve instead, and the work is choosing where on it to sit.

Power decides whether a design survives a bench. 10 V across 1 kilohm is 100 mW, which a common surface-mount part will not tolerate, and nothing in the schematic says so.

1.1.3 Series, parallel, and the divider

Series adds resistance; parallel adds conductance.

$$R_{\text{series}} = R_1 + R_2$$

$$\frac{1}{R_{\text{parallel}}} = \frac{1}{R_1} + \frac{1}{R_2}, \quad R_{\text{parallel}} = \frac{R_1 R_2}{R_1 + R_2}$$

The parallel combination is always smaller than either part, which catches most slips.

The **voltage divider** is the one circuit worth memorising: it is inside the bias network of Chapter 6, the feedback network of Chapter 4, and the attenuator in front of every oscilloscope.

$$V_{out} = V_{in} \frac{R_{lower}}{R_{upper} + R_{lower}}$$

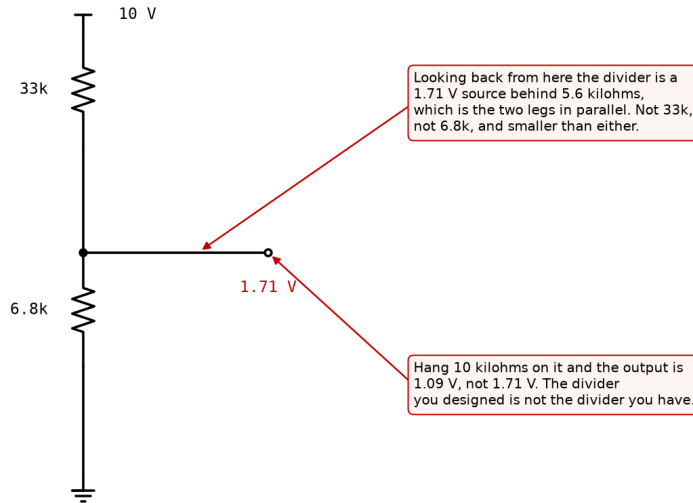


Figure 1.1 A voltage divider from a 10 V supply, 33 kilohm over 6.8 kilohm, with the output node marked at 1.71 V and two annotations giving the Thevenin resistance of 5.6 kilohm and the loaded output of 1.09 V.

33 kilohm over 6.8 kilohm on a 10 V supply:

$$V_{out} = 10 \times \frac{6800}{33000 + 6800} = 1.71 \text{ V}$$

1.71 V, not the 1.65 V it looks like it ought to be. No E12 pair gives 1.65 V from 10 V here, and Chapter 6 designs a transistor stage around the difference.

1.1.4 Kirchhoff's two laws

The only physical content in circuit theory. **Current law:** currents into a node sum to zero, because charge does not accumulate at a junction.

$$\sum_k I_k = 0$$

Voltage law: voltages around a closed loop sum to zero, because potential belongs to a point.

$$\sum_k V_k = 0$$

The current law is the one a program wants: one equation per node, and a netlist has nodes.

1.1.5 Thevenin and Norton

Any network of linear elements, seen from two terminals, behaves exactly like a voltage source in series with a resistance.

- V_{th} is the open-circuit voltage: what appears with nothing attached.
- R_{th} is the resistance looking back in with every independent source killed. A voltage source becomes a short, a current source an open circuit.

Killing the divider's supply grounds the top of the upper leg, so the legs are in parallel:

$$R_{th} = R_{upper} \parallel R_{lower} = 5.64 \text{ k}\Omega$$

R_{th} **does not depend on which leg is the upper one, and is smaller than either.** Two megohms give one megohm; two ten-ohm resistors give five ohms.

Norton is the same source drawn as a current source in parallel with the same resistance, $I_{no} = V_{th}/R_{th}$, and is the form §1.2 wants.

1.1.6 Loading, and why it decides everything later

The divider formula assumes nothing is attached. Attach something, and the lower leg becomes the lower leg in parallel with it. A 10 kilohm load on the 33k/6.8k divider:

$$V_{out} = 10 \times \frac{6800 \parallel 10000}{33000 + (6800 \parallel 10000)} = 1.09 \text{ V}$$

A third of the output gone, to a load most people would call light. R_{th} predicts it: 10 kilohm is not large compared with 5.6 kilohm.

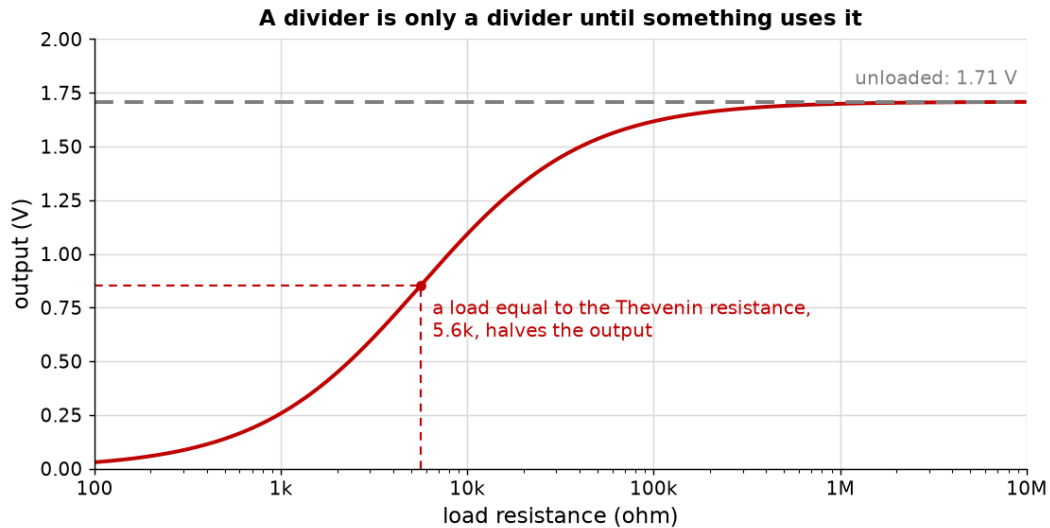


Figure 1.2 Divider output against load resistance, falling from 1.71 V unloaded to below 0.2 V at a 100 ohm load, with a marker showing that a load equal to the 5.6 kilohm Thevenin resistance halves the output.

$$V_{\text{delivered}} = V_{th} \frac{R_{load}}{R_{th} + R_{load}}$$

A load equal to R_{th} halves the output; ten times larger costs about 9 per cent; ten times smaller leaves about 9 per cent.

This is the idea the book is built on. Chapter 3: cascaded filter sections miss the corner you designed, because the second loads the first. Chapter 8: an emitter follower exists only to break a loading chain. Chapter 10: an op-amp loses 11 dB of open-loop gain to it, more than any other term.

1.1.7 What this section is blind to

- **Everything is linear.** From Chapter 4 the solver needs Newton-Raphson.
- **Everything is at DC.** No capacitor, no inductor, no frequency. Chapter 2 adds them by making this same solver complex, not by starting again.
- **Nothing has a tolerance.** Two 5 per cent resistors give a ratio good to about 10 per cent. Ignored until Chapter 9, where the matching of two transistors decides a stage's whole performance.

1.2 Nodal analysis, and what to build

The method that turns a circuit into a matrix, and the specification of the two components this chapter asks you to write. §1.2.5 is the specification; everything before it is what you need in order to write it.

1.2.1 One unknown per node

Solving by hand means picking loops and current directions until the equations match the unknowns. The choices are yours rather than the circuit's, so it does not generalise into code.

Nodal analysis makes them for you: node voltages are the unknowns, the current law at each node supplies the equations, and the count comes out right. Ground is not one of them, so a circuit with n nodes has $n - 1$ unknowns. **Five nodes is four unknowns.**

1.2.2 The conductance stamp

For a resistor between nodes a and b , the current leaving a through it is

$$I_{a \rightarrow b} = \frac{V_a - V_b}{R} = G(V_a - V_b), \quad G = \frac{1}{R}$$

In $\mathbf{G}\mathbf{v} = \mathbf{i}$ one resistor contributes exactly four entries, always these four:

$$G_{aa} += G, \quad G_{bb} += G, \quad G_{ab} -= G, \quad G_{ba} -= G$$

That pattern is a **stamp**, and it is the whole technique: never derive an equation, walk the element list, add each stamp, solve. Two consequences, both tested by the shipped suite:

- **The diagonal accumulates.** Two resistors between the same pair of nodes stamp twice, giving the parallel combination without anyone computing one. A solver that assigns one matrix row per *element* rather than per *node* gets this wrong, and gets it wrong quietly.
- **Ground is skipped.** An entry whose row or column is ground is not written, which is why a resistor to ground contributes one entry rather than four.

A current source does not enter $\mathbf{G}$ at all; it adds to the right-hand side at its two nodes.

1.2.3 The trouble with a voltage source

A voltage source has no conductance: its current is whatever the circuit demands, so it cannot be stamped. Add that current as an extra unknown and its constraint as an extra equation:

$$V_p - V_n = V_{source}$$

One extra row and column per source, and the matrix stops being purely a conductance matrix. That is the whole of the “modified” in **modified nodal analysis**. The extra unknown

says how much a supply delivers, which is how Chapter 6 checks a bias point and Chapter 10 a power budget.

1.2.4 Two conventions this book fixes

Pinned by this chapter’s test suite, because every component in the rest of the book reads them.

A current source injects at its second node. `addCurrentSource(from, to, current)` takes current out of `from` and pushes it into `to`. One milliamp into a node with one kilohm to ground puts that node at +1 V.

A voltage source reports the current leaving its positive terminal. `sourceCurrents[k]` flows out of source k ’s positive terminal into the circuit; a 10 V source driving one kilohm reports +10 mA.

That is the opposite sign to the raw modified-nodal unknown, which comes out as the current flowing from the node into the source. Negating it is your job, and the suite checks that you did.

1.2.5 What to build

Two headers. The paths are part of the specification: the test suite includes exactly these, and a component written anywhere else stays dormant however correct it is.

ael/net/netlist.hpp

A container. It holds elements and hands them to the solver, and it does nothing else.

Member	Contract
<code>using Node = std::size_t</code>	A node index.
<code>constexpr Node Ground{0}</code>	Node zero is ground, always.
<code>addResistor(Node a, Node b, double resistance)</code>	Adds a resistor.
<code>addCurrentSource(Node from, Node to, double current)</code>	Injects at <code>to</code> . See §1.2.4.
<code>addVoltageSource(Node positive, Node negative, double voltage)</code>	Fixes the difference.
<code>nodeCount()</code>	Highest node index mentioned, plus one. An empty netlist returns 1.
<code>resistorCount(), voltageSourceCount(), currentSourceCount()</code>	Element counts, by kind.

Nodes are mentioned rather than declared: there is no “add a node” call, because that would put a bookkeeping step in front of every exercise in the book.

The solver needs to read the elements back out. How you expose them is yours to decide; the tests only require that `solve` can be handed a `const Netlist&`.

ael/mna/solver.hpp

```

namespace ael::mna
{
    struct Solution
    {
        std::vector<double> nodeVoltages{};    ///< Indexed by Node. [Ground] is ↪
        ↪ always 0.
        std::vector<double> sourceCurrents{};  ///< One per voltage source, in ↪
        ↪ insertion order.
        bool solved{false};                    ///< False if the network has no unique↪
        ↪ solution.
    };

    [[nodiscard]] Solution solve(const net::Netlist& netlist) noexcept;
}

```

Assemble from the stamps in §1.2.2 and §1.2.3, then solve by Gaussian elimination with partial pivoting.

solved is not decoration. A floating node, or two voltage sources contradicting each other, gives a singular matrix. A solver that returns zeros there will be believed, and the reader spends an hour looking for the fault in their circuit rather than in their assumptions. Detect the singular pivot and say so. The suite has a test for exactly this.

What good looks like

Around 120 lines for both, of which the elimination is about 20. If yours is much longer, you are probably special-casing element kinds somewhere that a stamp would have handled.

1.2.6 Why write a solver at all

From Chapter 7 this book computes every stage twice: once from a closed form you can reason about, once from a solver that knows nothing about amplifiers. Where they agree, the closed form is trustworthy and simpler. Where they disagree, one is dropping a term, and finding out which is the exercise. That only works if the solver is yours; a number from a tool you did not write is an authority claim.

1.2.7 What this section is blind to

- **Conditioning.** A direct solve handles the three or four orders of magnitude here without complaint. Chapter 4 brings a diode and twelve orders, at which point pivoting stops being a formality.
- **Sparsity.** Real simulators store the matrix sparsely and order it to limit fill-in. Nothing in this book is large enough for that to matter.

- **Anything time-varying.** This is a DC solve; Chapter 2 makes it complex, which covers one frequency in steady state. There is no transient analysis, so an underdamped filter's ringing is something Chapter 3 computes rather than watches.

1.3 Review

What you should be able to do now

After this chapter, you should be able to:

- State the sign convention this book uses, and say what breaks if it is inverted.
- Compute a divider's output, its Thevenin resistance, and its output under any load, by hand.
- Say why a 10 kilohm load on a 33k/6.8k divider is not a light load.
- Write the node equation for any node in a resistive network without deriving it from scratch.
- Stamp a resistor, a current source and a voltage source into a matrix.
- Explain why a voltage source needs an extra unknown, and what that unknown is.
- Implement `ael::net` and `ael::mna` against the shipped suite.

Questions to test yourself

You should be able to answer:

1. A divider made of 1 megohm and 1 megohm, and one made of 10 ohms and 10 ohms, have the same ratio. What is different about them, and which would you put on an oscilloscope probe?
2. Why is a divider's Thevenin resistance the same whichever leg you call the upper one?
3. Nodal analysis gives one equation per node. A network has five nodes. How many unknowns are there, and why is it not five?
4. You stamp a resistor between two nodes and get four matrix entries. Two are positive and two are negative. Which are which, and what would swapping them mean physically?
5. A voltage source between two nodes cannot be written as a conductance. Why not, and what does the extra row in the matrix say?
6. Your solver returns node voltages that are all zero for a network you know is live. Name two causes, and say which one the shipped suite catches.
7. Superposition says two sources add. Your solver never mentions superposition. Why does it hold anyway, and in which chapter does it stop holding?

Further reading

- *The Art of Electronics*, Horowitz and Hill, chapter 1, for a second treatment of the same material at roughly the same level.

Looking ahead

- Chapter 2 makes the solver complex, which is the whole of the frequency domain and costs about thirty lines. Reactance, phasors, and the Bode plot as two straight lines and a corner.

1.4 Exercises

Eight, ending with the Cross-check. Do them in order; the last one needs the code from the two before it.

How to check your work. A `CODE` exercise is judged by the suite that ships with it: run `make test` at the root of the companion repository, with `AEL_DIR` pointing at your toolkit. A suite you have not started is reported as `SKIP` rather than as a failure, so the command is worth running from the first day. The hand calculations and designs are checked against the toolkit functions each one names.

The solutions are not in this book. They are published in full in the companion repository, in `lectures/L01/appendix`, including the plausible wrong answer where there is one. Read them once you have your own answers, including the ones you are unsure about, because being unsure and right is a different thing from being unsure and wrong and the solutions say which is which.

Exercise 1.1 The convention

RECALL

State the sign convention this book fixes for a current source and for a voltage source's reported current. Then answer this: your solver comes back with a divider output of -1.71 V instead of $+1.71$ V, and every voltage in the circuit is negated. Which of the two conventions did you get backwards, and how do you know it was that one rather than the other?

Exercise 1.2 Two dividers

RECALL

A divider made of two 1 megohm resistors and a divider made of two 10 ohm resistors have the same ratio.

1. What is the Thevenin resistance of each?
2. One of them is the front end of an oscilloscope probe and the other is inside a power supply. Which is which, and why?
3. A third divider uses two 10 ohm resistors across a 10 V supply. What is its power dissipation, and is that a problem?

Exercise 1.3 The divider, three ways

HAND CALCULATION

For a 33 kilohm over 6.8 kilohm divider on a 10 V supply, by hand:

1. The unloaded output.
2. The Thevenin resistance.
3. The output with a 10 kilohm load, computed twice: once by substituting the parallel combination into the divider formula, and once from the Thevenin equivalent. Confirm they agree, and say why they must.
4. The load that would halve the output.

Check yourself: divider, divider_output_resistance

Exercise 1.4 Node equations

HAND CALCULATION

A 1 mA current source injects into node 1. A 1 kilohm resistor runs from node 1 to ground, a 2.2 kilohm resistor from node 1 to node 2, and a 4.7 kilohm resistor from node 2 to ground.

1. Write the current-law equation at node 1 and at node 2. Do not solve them yet.
2. Write the two equations as a matrix, and identify each of the four entries as a stamp contribution from a named resistor.
3. Solve for both node voltages.
4. Kill the source and find the resistance looking into node 2. Now divide node 2's voltage from part 3 by the 1 mA source current. Those two numbers are both resistances and both about node 2. Say whether they agree, and if not, say precisely what each one is a resistance *between*.

Exercise 1.5 A reference divider

DESIGN

Design a divider giving 2.5 V from a 10 V supply, subject to:

- Thevenin resistance no greater than 2.2 kilohm, so the next stage does not load it.
- Divider current no greater than 1 mA, because the supply is a battery.
- E12 values only.

Give both resistors, the output voltage you actually achieve, the Thevenin resistance and the current. Then state the percentage error in the output, and say whether you would accept it.

There is a tempting wrong answer to this one. Find the pair whose *ratio* is closest to correct, check it against both constraints, and say what it violates.

Check yourself: divider, divider_output_resistance, nearest_e12

Exercise 1.6 The netlist

CODE

Implement `ael::net::Netlist` to the specification in §1.2.5.

Four of this chapter's seventeen tests are for this. Getting them green needs no solver, so do this first and confirm the suite goes from four passing to eight.

Exercise 1.7 The solver

CODE

Implement `ael::mna::solve` to the same specification.

Assemble the stamps, solve by Gaussian elimination with partial pivoting, and detect a singular matrix rather than returning zeros. All seventeen tests should pass.

Two of them are worth reading before you start rather than after you fail them: `Mna.ParallelResistorsCombine`, which fails for a solver that indexes by element instead of by node, and `Mna.FloatingNodeIsReportedRatherThanGuessed`, which fails for a solver that trusts its own elimination.

Exercise 1.8 The loaded divider

CROSS-CHECK

The signature exercise of this book, and the shape every later one takes. Compute the same number three ways and reconcile them.

Take the 33 kilohm over 6.8 kilohm divider on a 10 V supply, loaded with 10 kilohm.

1. **By hand.** Substitute the parallel combination into the divider formula. Write down the answer to three significant figures before doing anything else.
2. **By your closed form.** Whatever you wrote for Exercise 1.3, as a function, evaluated on the same numbers.
3. **By your solver.** Build the netlist with four elements, one voltage source and three resistors, and read node 1 out of `ael::mna::solve`.

Then reconcile. State the difference between each pair, and say what would have to be true for each difference to be non-zero.

What to expect

Legs 1 and 2 should agree to every digit you wrote down. They are the same formula evaluated by two different pieces of hardware. A disagreement here is an arithmetic slip in leg 1 or a typo in leg 2, and it is not interesting except that finding it is faster than finding it later.

Legs 2 and 3 should agree to about twelve significant figures. Both are solving the same linear problem in double precision, one by algebra you did in advance and one by elimination. The difference is rounding and nothing else, and it should be somewhere around 10^{-13} relative.

A difference of a few per cent means one of them is not solving the circuit you think it is. The usual cause is the load stamped between the wrong pair of nodes, which quietly

gives you a different circuit rather than an error. The second usual cause is the supply node and the output node swapped, which gives you the reciprocal divider.

A difference of exactly a factor of two, or exactly the unloaded answer, means the load is not in the circuit at all. Check `resistorCount()`.

This exercise is deliberately one where all three legs agree. That is worth doing once, so that you know what agreement looks like before Chapter 7, where a closed form and the solver disagree by a factor of ten and the closed form is the one that is wrong.

Reactance, phasors, and frequency response

The frequency domain, arrived at by making the solver complex rather than by deriving a transform.

IN THIS CHAPTER

- The capacitor and the inductor as devices that remember, and the time constant that says how long for.
- Sinusoidal steady state, and why a complex number can stand in for a sinusoid.
- Impedance: resistance that depends on frequency, and the two stamps that follow from it.
- Magnitude and phase, the decibel, and the Bode plot as two straight lines and a corner.
- The phase, which is the half people skip and the half that decides stability.
- The transformer, briefly, and where its model stops being true.
- The complex stamps and `ac`.

A resistor's current depends on the voltage across it now. That is the whole of Chapter 1. A capacitor's current depends on how the voltage has been *changing*, and that one difference is this chapter. Two devices with memory turn a circuit from something with an answer into something with a trajectory, and the method of Chapter 1 cannot solve it.

The way out is one observation rather than a transform. **In a linear circuit driven by a sinusoid, every voltage and current in it is a sinusoid at the same frequency**, so only amplitude and phase differ from node to node, and a single complex number carries exactly those two. Under that substitution differentiation becomes multiplication by $j\omega$, a capacitor becomes a conductance of $j\omega C$, and **nothing about the solver of Chapter 1 changes except its scalar type**. The frequency domain costs about thirty lines, and only because the first solver was written in a way that lets it.

What you get for them is the ability to read a response: the corner, the two asymptotes, and the 45 degrees of phase that are already there when the magnitude has barely moved. The magnitude says whether a loop has gain left; the phase says whether that gain helps or hurts, and Chapter 4 needs both.

Work §2.1 for the physics and the phasor argument, then §2.2 for the stamps and the specification in §2.2.4. The Cross-check that closes §2.4 is the first in this book where two legs are expected to disagree, and by a factor of 6.6.

2.1 Reactance, time constants, and phasors

Two new devices and one trick; the frequency domain falls out of the trick rather than being imposed on top of it.

2.1.1 Two devices that remember

A resistor's current depends on the voltage across it now. That is the whole of Chapter 1. A capacitor and an inductor depend on a rate of change instead:

$$I = C \frac{dV}{dt}, \quad V = L \frac{dI}{dt}$$

Both have memory, so a circuit containing one has no single answer, only a trajectory, and the method of Chapter 1 cannot solve it. The two are duals. This book leans on the capacitor, which is cheap, accurate and small; inductors appear in Chapter 3's filters and then almost nowhere.

2.1.2 The time constant

$$V(t) = V_0 (1 - e^{-t/\tau}), \quad \tau = RC$$

One time constant reaches 63.2 per cent, five reach 99.33 per cent. What is usually forgotten:

Accuracy wanted	Time constants needed
10 per cent	2.3
1 per cent	4.6
0.1 per cent	6.9
0.01 per cent	9.2

$n = \ln(1/\epsilon)$, so each further decade of accuracy costs 2.3 more time constants. **A time constant is not the time it takes, but the time it takes to get e times closer.**

2.1.3 The trick, and what it depends on

In a linear circuit driven by a sinusoid, every voltage and current in it is a sinusoid at the same frequency. Nothing can create a new one, so only amplitude and phase differ from node to node, and a complex number carries exactly those two. Write the excitation as $Ve^{j\omega t}$, the common factor cancels from every equation, and what remains is algebra on **phasors**.

$$\frac{dv}{dt} = j\omega V e^{j\omega t}$$

Differentiation becomes multiplication by $j\omega$, so $I = C dV/dt$ becomes $I = j\omega CV$: Ohm's law with a complex conductance.

It depends on linearity and on the steady state. A diode has no phasor description at all, because it creates harmonics that were not in the excitation. That is why Chapter 4 solves nonlinear circuits a different way, and why Chapter 7 linearises a transistor before putting a phasor near one.

2.1.4 Impedance

$$Z_C = \frac{1}{j\omega C}, \quad Z_L = j\omega L$$

The magnitudes are the **reactances**, and they are what a meter would read:

$$|Z_C| = \frac{1}{2\pi f C}, \quad |Z_L| = 2\pi f L$$

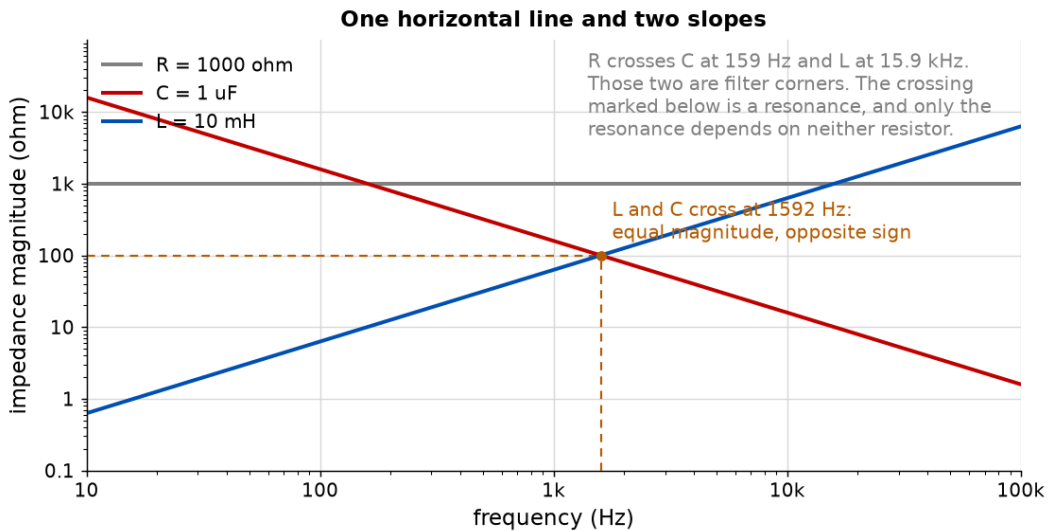


Figure 2.1 Impedance magnitude against frequency for a 1 kilohm resistor, a 1 microfarad capacitor and a 10 millihenry inductor, on logarithmic axes. The resistor is a horizontal line, the capacitor falls and the inductor rises, and the inductor and capacitor cross at 1592 hertz.

A capacitor's impedance falls with frequency and an inductor's rises, both at six decibels per octave. The three crossings are three different things, and keeping them apart is most of Chapter 3:

- **R crosses C** at 159 Hz: an RC corner, set by the resistor.
- **R crosses L** at 15.9 kHz: an RL corner, set by the resistor too.
- **L crosses C** at 1592 Hz: a **resonance**, set by no resistor at all.

The j matters as much as the magnitude. A capacitor's current leads its voltage by 90 degrees and an inductor's lags; that sign decides whether a feedback loop is stable.

2.1.5 The first-order response

The Chapter 1 divider formula still applies, with impedances instead of resistances:

$$H(j\omega) = \frac{Z_C}{R + Z_C} = \frac{1}{1 + j\omega RC}$$

The **corner frequency** is where the reactance equals the resistance:

$$f_c = \frac{1}{2\pi RC}$$

For 1 kilohm and 159 nanofarads that is 1001 Hz.

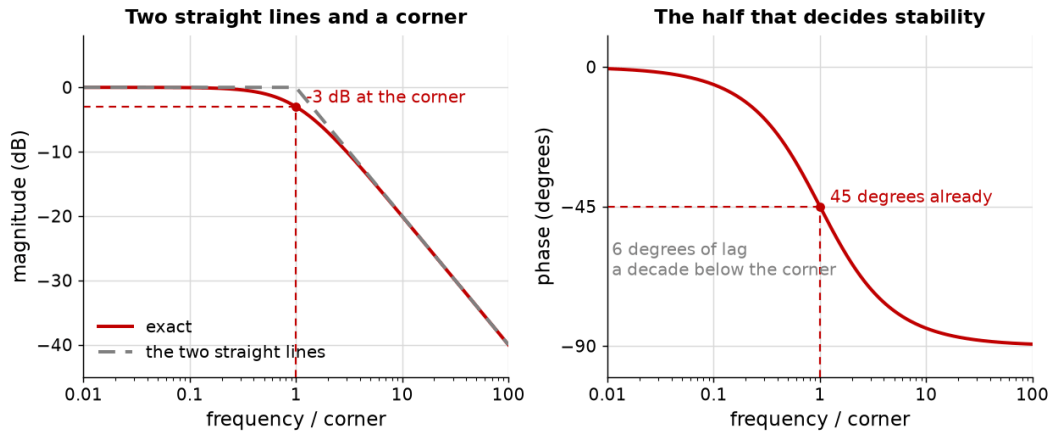


Figure 2.2 Two panels. On the left, the magnitude of a first-order low-pass against frequency normalised to its corner, flat then falling at twenty decibels per decade, with the exact curve and its two straight-line asymptotes and a marker three decibels down at the corner. On the right, the phase falling from zero to minus ninety degrees, passing minus forty five at the corner.

At	Magnitude	Phase
A decade below the corner	0.04 dB down	5.7 degrees
The corner	3.01 dB down	45 degrees
A decade above	20.04 dB down	84.3 degrees
Far above	falling 20 dB per decade	90 degrees

The magnitude is two straight lines and a corner, and drawing it from f_c alone is a skill.

The phase is the half that gets skipped. A decade below the corner the magnitude has lost 0.04 dB, which is nothing, and the phase has moved 5.7 degrees, which is not. The magnitude says whether a loop has gain left; the phase says whether that gain helps or hurts. Chapter 4 needs both.

2.1.6 Decibels, briefly

$$A_{dB} = 20 \log_{10} \left| \frac{V_{out}}{V_{in}} \right|$$

The factor is 20 rather than 10 because the decibel was defined for power and power goes as voltage squared. Every ratio in this book is a voltage ratio, so it is always 20.

Ratio	Decibels
1	0
$\sqrt{2}$	3.01
2	6.02
10	20
100	40
1000	60

The second row is the one that matters: a filter's corner is where it is 3 dB down, which is $1/\sqrt{2}$ in amplitude and half in power.

2.1.7 The transformer, briefly

Two coils sharing a magnetic circuit. The voltage ratio is the turns ratio, the current ratio its inverse, and an impedance on the secondary is reflected to the primary by the ratio squared.

$$\frac{V_2}{V_1} = \frac{N_2}{N_1}, \quad Z_{reflected} = \left(\frac{N_1}{N_2} \right)^2 Z_2$$

The book uses the second form once, as the cleanest example of impedance transformation, which is what Chapter 8's emitter follower does by a completely different mechanism for the same reason.

The model stops being true quickly: real transformers have leakage inductance, winding resistance, a finite magnetising inductance that shorts the primary at low frequency, and a core that saturates. All of that decides whether a design works, and none of it is in the ratio above.

2.1.8 What this section is blind to

- **Transients.** Everything here is steady state. What a filter does in the first few time constants after a step is a calculation this book never makes.
- **Real components.** A capacitor has series resistance and inductance, so its impedance stops falling and starts rising: a real 1 microfarad part is inductive above a few megahertz.
- **Distributed effects.** At high enough frequency a wire is not a node. That boundary is outside this book and inside the working range of a modern transistor.

2.2 The complex stamps, and what to build

The frequency domain costs about thirty lines, and only because the Chapter 1 solver was written in a way that lets it. §2.2.4 is the specification.

2.2.1 Nothing about the method changes

The whole of §1.2 survives intact: one unknown per node, one current-law equation per node, one stamp per element, Gaussian elimination.

Only the number field changes. Kirchhoff's current law does not care whether the currents are real or phasors, because the common $e^{j\omega t}$ divides out of every term. So the right way to write this chapter is not to: template the Chapter 1 solver on its scalar type and instantiate it twice.

```
template <typename T>
[[nodiscard]] std::vector<T> solveLinearSystem(std::vector<std::vector<T>> matrix) {
    ...
    noexcept;
```

If that is hard, the usual culprit is the pivot search, which needs `std::abs` rather than `std::fabs` and must compare magnitudes rather than values.

2.2.2 The two new stamps

A capacitor between nodes a and b at angular frequency ω has admittance $j\omega C$, and stamps exactly like a resistor of that conductance:

$$Y_{aa} += j\omega C, \quad Y_{bb} += j\omega C, \quad Y_{ab} -= j\omega C, \quad Y_{ba} -= j\omega C$$

An inductor has admittance $1/(j\omega L)$ and stamps the same way. That is the entire electrical content of this chapter.

One trap. At zero frequency an inductor's admittance is infinite and a capacitor's is zero, so a sweep that includes DC divides by zero in the inductor stamp and goes singular in any circuit whose only path to ground is through a capacitor. This book's sweeps start above zero; a real simulator does the DC operating point separately, for exactly this reason.


```
}
```

Three details the suite checks, each with a reason:

- **A voltage source is a phasor of its own value at zero phase.** There is no separate AC amplitude, so a divider built for Chapter 1 can be swept without being rebuilt.
- **sweep is logarithmic and inclusive at both ends.** points of 1 returns one point at first. first equal to last returns points identical points rather than dividing by zero.
- **sweep returns empty for a non-positive frequency,** rather than producing infinity in the inductor stamp. The signature is `noexcept`, so reporting by exception is not open to it.

What good looks like

Around forty new lines: ten for the two stamps, twenty for the sweep's frequency generation. If `solveAt` duplicates the elimination from Chapter 1, stop and template that instead. The duplication is the thing this chapter is teaching against.

2.2.5 Reading a complex answer

`nodeVoltages[n]` is a phasor. Two lines turn it into what a plot wants:

```
const double magnitudeDb{20.0 * std::log10(std::abs(v))};
const double phaseDegrees{std::arg(v) * 180.0 / M_PI};
```

The phase sign is the single most common defect in a first AC solver, and it is invisible in a magnitude plot. A low-pass must produce a *negative* phase: the output lags. If yours is positive at every frequency, the likely cause is a capacitor stamped with $1/(j\omega C)$, the impedance, where the admittance is wanted. The suite tests the phase at the corner for exactly this reason.

2.2.6 What this section is blind to

- **DC.** The sweep starts above zero and the stamps break there. Chapter 1's solver still answers a DC question, and from Chapter 5 the two are used together: the DC solve finds an operating point, the AC solve describes small movements around it.
- **Nonlinearity.** Everything here assumes superposition, which a diode or a transistor breaks. Chapter 4 is where that is confronted.
- **Noise.** A phasor describes a deterministic sinusoid. The thermal noise that decides how small a signal an amplifier can usefully handle is not in this book at all.

2.3 Review

What you should be able to do now

After this chapter, you should be able to:

- Compute a time constant, and say how many of them a given settling accuracy needs.
- State what a phasor is, and what has to be true of a circuit before one is allowed.
- Write down the impedance of a capacitor and an inductor at any frequency, from memory.
- Sketch a first-order response from its corner frequency alone, magnitude and phase.
- Say what the phase is doing a decade either side of the corner, to within a degree or two.
- Convert between a ratio and decibels without a calculator for the cases that matter.
- Extend a real solver from real to complex arithmetic without duplicating it.

Questions to test yourself

You should be able to answer:

1. A capacitor's impedance falls with frequency and an inductor's rises. At what frequency are they equal, and what is special about a circuit operating there?
2. Why is $j\omega$ a legitimate substitute for d/dt , and what property of the circuit does that argument depend on?
3. A first-order low-pass is 3 dB down at its corner. How far down is it half a decade above?
4. At what frequency has a first-order low-pass reached 45 degrees of phase lag, and how much magnitude has it lost there?
5. A colleague says a filter "does nothing" a decade below its corner. What has it done to the phase there, and when would that matter?
6. Your AC solver returns the right magnitude and a phase of the wrong sign at every frequency. Name the most likely single cause.
7. An amplifier settles to within 1 per cent in 5 microseconds. How long does it need for 0.01 per cent, assuming one time constant dominates?

Further reading

- *The Art of Electronics*, Horowitz and Hill, chapter 1, sections on reactance and RC circuits, for a second treatment of the same material.

Looking ahead

- Chapter 3 puts the reactance to work. Passive filters, what cascading two of them actually gives you, and then the operational amplifier as a black box with two rules.

2.4 Exercises

Eight, ending with the Cross-check. The Cross-check is the first in this book where two of the three legs are expected to disagree, and by a large amount.

How to check your work. A `CODE` exercise is judged by the suite that ships with it: run `make test` at the root of the companion repository, with `AEL_DIR` pointing at your toolkit. A suite you have not started is reported as `SKIP` rather than as a failure, so the command is worth running from the first day. The hand calculations and designs are checked against the toolkit functions each one names.

The solutions are not in this book. They are published in full in the companion repository, in `lectures/L02/appendix`, including the plausible wrong answer where there is one. Read them once you have your own answers, including the ones you are unsure about, because being unsure and right is a different thing from being unsure and wrong and the solutions say which is which.

Exercise 2.1 What a phasor assumes

RECALL

A phasor replaces a sinusoid with one complex number.

1. State the two properties a circuit must have before that substitution is legitimate.
2. Give a circuit from later in this book that has neither, and say what it does to a sinusoid that a phasor cannot describe.
3. A phasor analysis of a filter says the output is 0.707 of the input at the corner. A colleague switches the circuit on and measures 0.9 of the input. Neither of you is wrong. What is the colleague measuring?

Exercise 2.2 Reading a response

RECALL

Without computing anything, sketch the magnitude and phase of a first-order low-pass with a corner at 1 kHz, from 10 Hz to 100 kHz. Mark on it:

1. The two straight-line asymptotes and where they meet.
2. The magnitude and phase at the corner.
3. The magnitude and phase a decade below the corner.
4. The slope far above the corner, in dB per decade and in dB per octave.

Then say which of those six numbers you had to remember and which follow from the others.

Exercise 2.3 Time constants

HAND CALCULATION

An amplifier's output settles towards its final value with a single dominant time constant, and it reaches 1 per cent of final in 5 microseconds.

1. What is the time constant?
2. How long does it need to reach 0.01 per cent?
3. The specification is tightened from 8 bits to 14 bits of settling accuracy. By what factor does the settling time increase?
4. Somebody proposes doubling the bandwidth to fix it. By what factor does that help, and is it the cheaper of the two options?

Exercise 2.4 Reactance and resonance

HAND CALCULATION

A 10 millihenry inductor and a 1 microfarad capacitor.

1. At what frequency do their reactances have the same magnitude?
2. What is that reactance, in ohms?
3. In series, what is the total impedance at that frequency, and why?
4. A 10 ohm resistor is put in series with both. What is the Q, and what voltage appears across the inductor alone when 1 V is applied at resonance?
5. Which of the answers above would change if the resistor were removed, and which would not?

Check yourself: `lc_resonance`, `inductor_reactance`, `series_rlc_q`

Exercise 2.5 An anti-aliasing filter that has to be honest

DESIGN

A sensor drives an RC low-pass. The requirement is at least 20 dB of attenuation at 10 kHz, and no more than 0.5 dB of loss at 100 Hz.

1. Find the corner frequency that meets both, or show that no single-pole filter can.
2. Choose R and C from E12 values, given that the source driving the filter has an output resistance of 200 ohm and the load after it is 100 kilohm.
3. State the actual attenuation at 10 kHz and the actual loss at 100 Hz for your choice.
4. Say what you would do differently if the 20 dB requirement became 40 dB.

Check yourself: `rc_corner`, `first_order_response`, `nearest_e12`

Exercise 2.6 The complex stamps

CODE

Extend `ael::net::Netlist` with capacitors and inductors, and write `ael::ac::solveAt` to the specification in §2.2.4.

Template your Chapter 1 elimination on its scalar type rather than writing a second one. If that turns out to be difficult, the reason is worth finding: it is almost always `std::fabs` in the pivot search, which does not accept a complex number, where `std::abs` does and returns the magnitude.

Confirm that the Chapter 1 suite still passes. It links the same `Netlist`.

Exercise 2.7 The sweep

CODE

Write `ael::ac::sweep` to the same specification: logarithmic, inclusive at both ends, and well behaved for one point and for `first == last`.

Then use it. Sweep the RC low-pass from §2.1.5 from 10 Hz to 100 kHz and print magnitude in dB and phase in degrees. Confirm by eye that the corner is where you expect and that the phase is negative.

Exercise 2.8 The filter that is not where you put it

CROSS-CHECK

A 33 kilohm over 6.8 kilohm divider on a 10 V supply feeds an RC low-pass: 1 kilohm in series, then 159 nanofarads to ground. The output is taken across the capacitor.

Find the corner frequency, three ways.

1. **By hand, the obvious way.** The filter is 1 kilohm and 159 nanofarads, so $f_c = 1/(2\pi RC)$. Write the number down.
2. **By hand, thinking about it.** What resistance does the capacitor actually see looking back? Recompute.
3. **By your solver.** Build all four elements as one netlist, sweep it, and find the frequency at which the response is 3 dB below its low-frequency value.

Then reconcile all three, and separately state what the low-frequency output voltage is and why it is not 10 V.

What to expect

Legs 1 and 3 disagree by a factor of 6.6, and leg 1 is the wrong one. This is the first Cross-check in the book where the hand calculation is not merely less precise but answering a different question, and the size of the error is the point: this is not a few per cent.

Legs 2 and 3 should agree to about three significant figures, limited only by how finely you sweep. If you sweep 50 points per decade, expect to locate the corner to better than 1 per cent.

What leg 1 got wrong. It used the series resistor and ignored the divider. The capacitor does not care which resistor is which; it sees everything looking back, which is the 1 kilohm

in series with the divider's Thevenin resistance of 5.64 kilohm. The corner is therefore at 151 Hz, not 1001 Hz, and at 1001 Hz the filter is already 16.5 dB down rather than 3 dB down.

If leg 3 gives 1001 Hz, the divider is not in your netlist, or its two resistors are stamped between the wrong nodes so that they are not in the signal path.

If leg 3 gives a low-frequency output of 10 V, the divider is shorted out somewhere. It should give 1.71 V, which is the unloaded divider from Chapter 1, because no direct current flows through the series resistor once the capacitor has charged.

The general lesson, which is Chapter 1's again. A filter's corner is set by the total resistance the capacitor sees, and that includes everything upstream. This is the same arithmetic as §1.1.6 in a different costume, and it will appear a third time in Chapter 3 when two filter sections are cascaded, and a fourth in Chapter 10 where it costs an amplifier 11 dB.

Passive filters and the operational amplifier

The first useful circuits, and the first active device.

IN THIS CHAPTER

- High-pass and low-pass RC, and the impedances they present on both sides.
- Cascading, and why two sections in a row do not give you the corner you designed.
- LC filters, resonance, and Q as two things at once: a sharpness and an overshoot.
- The band-pass, and the trade its two corners cannot escape.
- The operational amplifier as a black box, with two rules and nothing else.
- Inverting, non-inverting, buffer, summing and difference, all from those two rules.
- The comparator, hysteresis, and the Schmitt trigger, which are not amplifiers at all.
- Building `ael::filter`, the VCVS element, and `ael::opamp`.

Swap the two components of an RC section and a low-pass becomes a high-pass; everything else about them, including the corner, is the same. That is the first half of this chapter and it is straightforward. The second half starts when you put two identical sections in a row and measure the corner, and it is not the corner either section has: **the second section is a load on the first**, and this is Chapter 1's arithmetic for the third time, in a circuit where you cannot see it coming.

If loading is the problem, then something with a high input impedance and a low output impedance between the sections is the answer, and that thing is an **operational amplifier**. It arrives here as a black box with two rules — no current into the inputs, no voltage between them — from which every standard configuration is one line of algebra. **You will use it on trust for seven chapters and build it out of transistors in the last one**, which is the whole arc of this book.

The rules are asserted here and derived in Chapter 4, and §3.2.4 is where they do not apply at all: take the feedback away and the circuit that results behaves in a way the rules would call impossible.

§3.1 is the passive filters, §3.2 is the amplifier and the specification in §3.2.5. The Cross-check in §3.4 has four legs rather than three, because the fourth one is the fix rather than another way of measuring the fault.

3.1 Passive filters, cascading, and Q

The first useful circuits in the book, and the third appearance of the one piece of arithmetic that decides everything.

3.1.1 Two components, four filters

A resistor and a capacitor in series make a divider whose ratio depends on frequency. Which component you take the output across decides the kind of filter.

$$H_{LP} = \frac{1}{1 + jf/f_c}, \quad H_{HP} = \frac{jf/f_c}{1 + jf/f_c}$$

Both have the same corner, $f_c = 1/(2\pi RC)$, and both are 3 dB down there. At that corner the low-pass lags 45 degrees and the high-pass leads 45, so **their outputs add to give the input back exactly, at every frequency**. That is the cleanest statement of what complementary means.

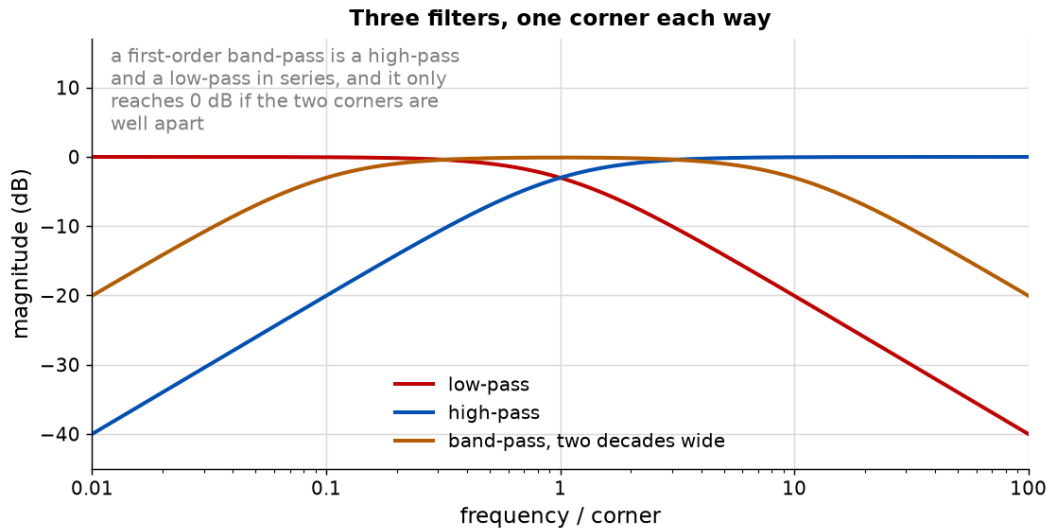


Figure 3.1 Low-pass, high-pass and band-pass magnitude responses against frequency normalised to a common corner, on logarithmic axes. The low-pass is flat then falls, the high-pass rises then is flat, and the band-pass rises and falls, reaching nearly zero decibels between two corners two decades apart.

A **band-pass** is a high-pass and a low-pass in series. It reaches 0 dB in the middle only if the corners are well apart: a decade apart falls 0.83 dB short, equal corners 6 dB.

3.1.2 A filter's own impedances

A filter is a two-port and both of its ports have a frequency-dependent impedance. Both matter for Chapter 1's reason: the source upstream and the load downstream form dividers with them.

$$Z_{in} = R + \frac{1}{j\omega C}, \quad Z_{out} = R \parallel \frac{1}{j\omega C}$$

Z_{in} is capacitor-dominated and large at low frequency, approaching R at high; Z_{out} is R at low frequency and approaches zero at high. **So a low-pass RC presents its own series resistance as an output impedance at low frequency, and the next stage sees it.**

3.1.3 Cascading, and the corner you did not design

Two identical low-pass sections in a row look like they should give the square of one section's response: 6 dB down at the corner, falling at 40 dB per decade. **That is the answer for two sections with a buffer between them, and it is wrong for two connected directly.**

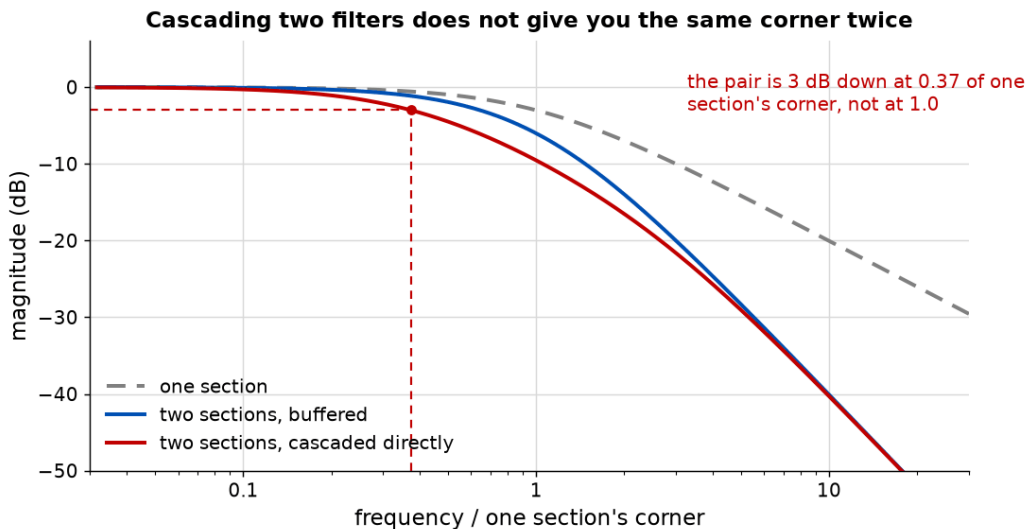


Figure 3.2 Magnitude against frequency for one RC section, for two sections with a buffer between them, and for two sections cascaded directly. The directly cascaded pair falls away earliest, and a marker shows it is three decibels down at 0.37 of one section's corner.

The second section loads the first. Its input impedance parallels the first section's capacitor, so the two poles split apart, their product staying at f_c^2 :

$$f_{low} = f_c \frac{3 - \sqrt{5}}{2} = 0.382f_c, \quad f_{high} = f_c \frac{3 + \sqrt{5}}{2} = 2.618f_c$$

Arrangement	3 dB point
One section	$1.000f_c$
Two sections, buffered	$0.644f_c$
Two sections, cascaded directly	$0.374f_c$

The direct cascade is a factor of **1.72** lower than the buffered pair. **A filter designed by multiplying two responses together and built by soldering two sections together is not the filter that was designed.**

The fix is a buffer: high input impedance, low output impedance, between the sections so the second cannot load the first. An op-amp follower is exactly that, which is why this chapter covers filters and op-amps together.

This is Chapter 1's arithmetic for the third time. A divider loaded by 10 kilohm loses a third of its output; here two filter sections move each other's poles. In Chapter 10 an op-amp loses 11 dB the same way.

3.1.4 LC, resonance, and Q

An inductor and a capacitor have reactances of opposite sign, so at one frequency they cancel:

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

In series they cancel to zero impedance, in parallel to infinite. **Nothing about that frequency depends on any resistance**, which is what makes a resonance different in kind from an RC corner. The resistance decides only the sharpness, for a series RLC taken across the resistor:

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}, \quad \text{bandwidth} = \frac{f_0}{Q}$$

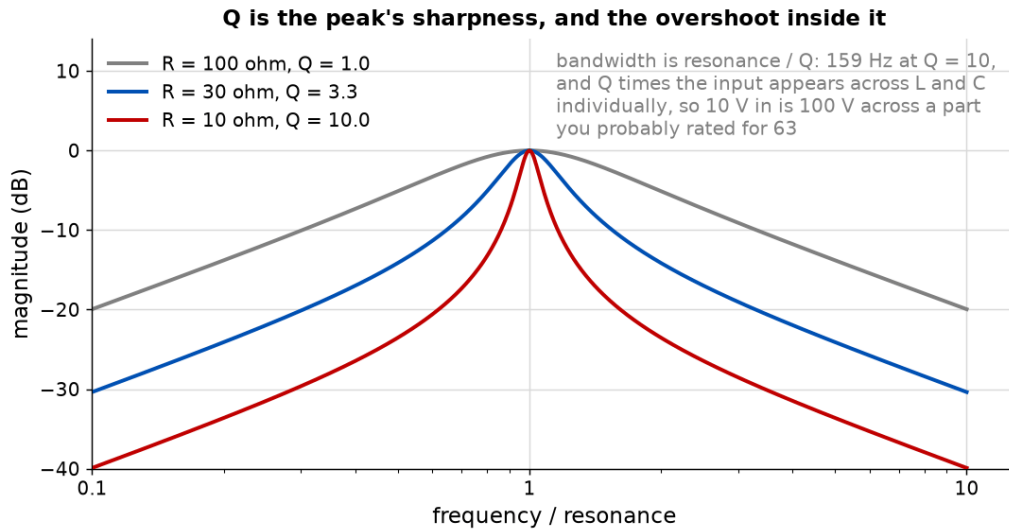


Figure 3.3 Band-pass magnitude for three values of Q on the same resonance, showing a broad peak at Q of one, a narrower one at Q of 3.3 and a sharp one at Q of 10, all reaching zero decibels at resonance.

Q is two things at once, and the second is the one that gets people. It is the sharpness of the peak, and it is the factor by which the inductor's and the capacitor's own voltages exceed the input: at resonance the LC is a short, so the current V/R flows through a reactance Q times R.

$$V_L = Q \times V_{in}$$

A Q of 10 with 10 V applied puts 100 V across a capacitor. That is how a filter correct on paper destroys a component on a bench, and it is nowhere in the transfer function, which describes only the output.

3.1.5 Higher orders, and what this book does not do

Two poles is as far as this book goes, and it gets there by cascading rather than by designing.

Real design starts from a specification, picks the polynomial that meets it with the fewest poles, and realises it as a circuit: Butterworth for a flat passband, Chebyshev for a steeper transition at the cost of ripple, Bessel for flat group delay. Chapter 4 builds one Sallen-Key, the cheapest complex pole pair from one amplifier. Left out: pole placement, the approximation problem, tolerance sensitivity, switched-capacitor realisations.

3.1.6 What this section is blind to

- **Component tolerance.** Every corner here is nominal. A filter of 5 per cent parts has a corner good to about 7 per cent, and a high-Q filter is far worse, because Q depends on a ratio of two square roots.
- **Real inductors.** The Q values assume the resistance is the one you put there. A real inductor's winding resistance is often dominant, and it caps Q where no external resistor can lift it.
- **Transients.** A high-Q filter rings, for roughly Q cycles. This book never computes it.

3.2 The operational amplifier, and what to build

Two rules, five circuits, and one circuit where the rules do not apply. The op-amp is a black box until Chapter 10, where it is built out of the transistors of Chapter 5 to Chapter 9: you use it on trust for seven chapters, then find out what you were trusting.

3.2.1 What is inside, stated once

A differential amplifier with enormous gain, two inputs and one output:

$$V_{out} = A_{OL} (V_+ - V_-)$$

A_{OL} is 10^5 to 10^6 at DC, and that is the entire model here. **A gain that large makes the amplifier useless alone:** 100 microvolts of input difference saturates it. Everything below is a consequence of the feedback, not of the amplifier.

3.2.2 The two rules

With negative feedback around it, and enough gain:

1. No current flows into either input.
2. No voltage appears between the inputs.

Rule 1 is about the amplifier: its input impedance is large. Rule 2 is about the feedback: the output moves to whatever makes the difference zero, leaving the output divided by A_{OL} . Rule 2 is the **virtual short**, and no current flows through it.

Where they fail. Rule 1 fails with bipolar inputs driven from a very high impedance, where the input bias current develops a real voltage; Chapter 9 is where that current comes from. Rule 2 fails with no feedback (§3.2.4), with positive feedback, and where A_{OL} is no longer large, which is Chapter 4.

3.2.3 The configurations, all from the two rules

Each is one line of algebra: rule 2 gives the voltage at the inverting input, rule 1 says whatever current arrives there leaves through the feedback resistor.

Non-inverting. Feedback divider from output to inverting input, signal into the non-inverting input. Gain cannot be less than one; the input impedance is the amplifier's own.

$$A = 1 + \frac{R_f}{R_g}$$

Voltage follower. Non-inverting with $R_f = 0$ and no R_g : gain exactly one, useless as an amplifier and invaluable as the buffer §3.1.3 needed.

Inverting. Signal through R_{in} , feedback through R_f , non-inverting input grounded. Gain can be less than one.

$$A = -\frac{R_f}{R_{in}}$$

Its input impedance is exactly R_{in} , because rule 2 pins that input at ground, which is usually the reason to prefer the non-inverting version.

Summing. Several inputs, each through its own resistor, into the inverting input.

$$V_{out} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \dots \right)$$

They do not interact, because rule 2 holds their common node at ground: the natural resistor-ladder digital-to-analogue converter.

Difference. Two inputs, two matched pairs of resistors.

$$V_{out} = \frac{R_f}{R_{in}} (V_2 - V_1)$$

It rejects what is common to both inputs only as well as the two resistor ratios match. A 1 per cent mismatch limits rejection to about 46 dB. Chapter 9 meets the same limit from transistor matching.

3.2.4 The comparator, where the rules do not apply

Take the feedback away and rule 2 goes with it: any positive difference drives the output to the positive rail, any negative one to the negative rail. That is a **comparator**, a one-bit analogue-to-digital converter. **Real inputs have noise**, so an input crossing the threshold slowly with a millivolt of noise crosses it many times, and one wanted transition becomes a burst.

The fix is **positive** feedback, a fraction of the output back to the non-inverting input:

$$V_{th\pm} = \pm V_{supply} \frac{R_{lower}}{R_{upper} + R_{lower}}$$

The threshold now moves away from the input as soon as the output switches, so the input must travel back across a gap to switch again. That gap is the **hysteresis** and the circuit is a **Schmitt trigger**. With 12 V rails and 100 kilohm over 10 kilohm, thresholds of ± 1.09 V.

What hysteresis costs. The circuit no longer says when the input crossed zero, but when it crossed one of two levels that depend on where it has been. Make the gap just larger than the noise.

3.2.5 What to build

Three pieces. The first is a netlist element, the other two are closed forms.

An addition to `ael/net/netlist.hpp`

```

/// A voltage-controlled voltage source: V(outPositive) - V(outNegative)
///                                     = gain * (V(inPositive) - V(inNegative)).
void addVcvs(Node outPositive, Node outNegative, Node inPositive, Node inNegative,
    double gain) noexcept;
[[nodiscard]] std::size_t vcvsCount() const noexcept;

```

The op-amp as far as a solver is concerned: finite gain, infinite input impedance, zero output impedance. A gain of 10^5 obeys the two rules; a gain of 100 lets them fail, as Chapter 4 does.

The stamp is Chapter 1’s voltage source with two more entries, and the same extra unknown. Only the constraint row differs: instead of $V_p - V_n = V$ it says

$$V_{op} - V_{on} - A(V_{ip} - V_{in}) = 0$$

so it has up to four entries and a right-hand side of zero, the two output-node entries being the same ± 1 as a voltage source’s. About six lines, if Chapter 1’s stamp was a small function.

`ael/filter/response.hpp`

Function	Returns
<code>rcCorner(r, c)</code>	The corner frequency of an RC section.
<code>lowpass(frequency, corner)</code>	The complex first-order low-pass response.
<code>highpass(frequency, corner)</code>	The complex first-order high-pass response.
<code>lcResonance(l, c)</code>	The resonant frequency.
<code>seriesQ(r, l, c)</code>	Q of a series RLC taken across the resistor.
<code>bandpass(frequency, resonance, q)</code>	The second-order band-pass response, unity at resonance.
<code>cascadedCorner(r, c)</code>	The 3 dB point of two identical sections cascaded directly.

`cascadedCorner` is deliberately given no closed form here. Take the two poles from §3.1.3, then locate the half-power point of their product by bisection on a logarithmic axis. Twenty iterations is plenty; ten lines.

`ael/opamp/ideal.hpp`

Function	Returns
<code>nonInvertingGain(feedback, ground)</code>	$1 + R_f/R_g$.
<code>invertingGain(feedback, input)</code>	$-R_f/R_{in}$, negative.
<code>differenceGain(feedback, input)</code>	R_f/R_{in} .
<code>schmittThresholds(supply, upper, lower)</code>	The lower and upper thresholds, as a pair.

Three lines each, and worth having anyway: from here on they are what the solver gets checked against.

3.2.6 What this section is blind to

- **Everything real about an op-amp.** Offset voltage, input bias current, finite bandwidth, slew rate, output current limit, supply rejection and noise. Chapter 4 adds the gain and the bandwidth only.
- **Stability.** Wrapping feedback around an amplifier can make an oscillator. The condition is in Chapter 4 and the treatment is brief.
- **Single-supply operation.** Everything here assumes symmetric rails and signals around ground. Real designs often have neither.

3.3 Review**What you should be able to do now**

After this chapter, you should be able to:

- Turn a filter requirement into a corner frequency, and say when one pole cannot meet it.
- Compute what a filter's own input and output impedance are, and why both depend on frequency.
- Say what two directly cascaded RC sections actually give, and by how much it differs from the answer that assumes they are independent.
- State Q two ways: as the sharpness of a peak, and as the voltage multiplication inside it.
- Apply the two op-amp rules to any of the standard configurations and get the gain in one line.
- Say precisely which assumption each rule is, and name a circuit where each one fails.
- Compute the two thresholds of a Schmitt trigger, and choose them for a given amount of noise.

Questions to test yourself

You should be able to answer:

1. A low-pass and a high-pass built from the same R and C have the same corner. What is different about them at that corner, other than which side of it they pass?
2. You cascade two identical low-pass sections and measure the corner. Is it higher or lower than one section's corner, and by roughly what factor?
3. Adding a buffer between two filter sections changes the response. Which direction does the corner move, and why is that the direction that makes it predictable?
4. A band-pass made of one high-pass and one low-pass never quite reaches 0 dB in its passband. What decides how close it gets?
5. A filter has Q of 10 and you apply 5 V at resonance. What is the largest voltage anywhere in the circuit, and where is it?
6. The two op-amp rules assume infinite gain. Which of the two survives when the gain is 1000, and which one starts to leak?
7. A comparator with no hysteresis and a slowly rising noisy input produces a burst of output transitions. How much hysteresis stops it, and what does that hysteresis cost you?

Further reading

- *The Art of Electronics*, Horowitz and Hill, chapters 1 and 4, for passive filters and for the operational amplifier as a black box.

Looking ahead

- Chapter 4 asks why the two rules work at all. Feedback, the loop gain, what it buys and what it costs, and then the diode, which is the first device in the book that a phasor cannot describe.

3.4 Exercises

Eight, ending with the Cross-check. This one has four legs rather than three, because the fourth is the fix rather than another way of measuring the fault.

How to check your work. A `CODE` exercise is judged by the suite that ships with it: run `make test` at the root of the companion repository, with `AEL_DIR` pointing at your toolkit. A suite you have not started is reported as `SKIP` rather than as a failure, so the command is worth running from the first day. The hand calculations and designs are checked against the toolkit functions each one names.

The solutions are not in this book. They are published in full in the companion repository, in lectures/L03/appendix, including the plausible wrong answer where there is one. Read them once you have your own answers, including the ones you are unsure about, because being unsure and right is a different thing from being unsure and wrong and the solutions say which is which.

Exercise 3.1 The two rules

RECALL

1. State the two op-amp rules, and say which one is a property of the amplifier and which is a property of the feedback around it.
2. Name a circuit in which the second rule does not hold at all, and say what replaces it.
3. An amplifier has an open-loop gain of 10^5 and its output sits at 10 V. How much voltage is actually between its inputs? Is the virtual short a good name?
4. Which of the two rules is the one that fails first as frequency rises, and why?

Exercise 3.2 Complementary filters

RECALL

A low-pass and a high-pass are built from the same resistor and the same capacitor, driven from the same source.

1. What is the phase of each at the corner?
2. Add the two outputs. What do you get, at the corner and at every other frequency?
3. Each is 3 dB down at the corner, and 3 dB down means 0.707. Two things that are each 0.707 of the input sum to the input exactly. Explain, without arithmetic, why that is not a contradiction.

Exercise 3.3 What the second section does to the first

HAND CALCULATION

Two identical RC low-pass sections, each 1 kilohm and 159 nanofarads, connected directly.

1. The corner of one section alone.
2. The two pole frequencies of the pair, from §3.1.3.
3. Their geometric mean. Compare it with part 1 and say what that tells you about what loading does.
4. The 3 dB point of the pair, and the factor by which it differs from the answer you would get by assuming the two sections are independent.

Check yourself: rcCorner, cascadedCorner

Exercise 3.4 What is inside a resonance

HAND CALCULATION

A series RLC: 10 millihenry, 1 microfarad, 10 ohm, driven with 5 V at resonance.

1. The resonant frequency and the reactance of each part at it.
2. The Q, and the bandwidth.
3. The current, and the voltage across each of the three components.
4. You are choosing a capacitor. What voltage rating does it need, and what would a reader who looked only at the transfer function have chosen?
5. The 10 ohm resistor is removed and the only resistance left is 0.5 ohm of inductor winding resistance. What happens to all five answers above?

Check yourself: `lcResonance, seriesQ`

Exercise 3.5 A Schmitt trigger for a noisy signal

DESIGN

A sensor output crosses zero slowly and carries about 100 mV peak-to-peak of noise. It drives a comparator on plus and minus 12 V rails, and the comparator's output must make exactly one transition per crossing.

1. Choose the positive-feedback divider from E12 values. State the two thresholds and the gap.
2. Justify the gap: say why it is larger than the noise and why it is not much larger.
3. The sensor signal is a 1 Hz triangle of 2 V amplitude. By how much does the hysteresis delay the reported crossing, in milliseconds?
4. Somebody proposes fixing the noise with a low-pass filter instead. Give one advantage and one disadvantage against the Schmitt trigger.

Check yourself: `schmittThresholds, nearest_e12`

Exercise 3.6 The filter responses

CODE

Implement `ael::filter` to the specification in §3.2.5.

`cascadedCorner` is the only one that is not a one-liner. Find the two poles, then bisect on a logarithmic frequency axis for the point where the product of the two first-order magnitudes falls to $1/\sqrt{2}$. Do not look for a closed form; the exercise is partly to notice that a numerical answer is fine when the question is “where is this curve equal to that”.

Exercise 3.7 The VCVS and the configurations

CODE

Add `addVcvs` to `ael::net::Netlist` and write `ael::opamp::ideal`.

The VCVS stamp is Chapter 1's voltage-source stamp with a different constraint row. If your Chapter 1 code wrote that stamp inline rather than as a function, this is the moment it becomes worth extracting.

Then check the two against each other: build an inverting amplifier as a netlist with a VCVS of gain 10^5 , solve it, and compare with `invertingGain`. They should agree to about five decimal places, and the discrepancy is the finite gain, which is Chapter 4's subject.

Exercise 3.8 The cascade, and the buffer that fixes it

CROSS-CHECK

Two identical RC low-pass sections, 1 kilohm and 159 nanofarads each, connected directly. Find the frequency at which the pair is 3 dB down.

1. **By hand, assuming the sections are independent.** Square the single-section response and find where the result is 3 dB down. Write the number down.
2. **By your closed form.** `cascadedCorner`, which accounts for the loading.
3. **By your solver.** Build the four elements as one netlist and sweep.
4. **Then fix it.** Insert a VCVS of gain 1 between the two sections, as a follower, and sweep again.

What to expect

Legs 1 and 3 disagree by a factor of 1.72, and leg 1 is wrong. 644 Hz against 375 Hz. This is smaller than Chapter 2's factor of 6.6 and more insidious, because 644 Hz is a plausible answer arrived at by a plausible method: multiplying two transfer functions together is exactly what you would do if nobody had told you the sections interact.

Legs 2 and 3 should agree to three significant figures, limited by your sweep density.

Leg 4 should give 644 Hz, which is leg 1's answer. That is the point of the fourth leg: leg 1 was not wrong about the mathematics, it was wrong about the circuit, and inserting a buffer makes the circuit into the one leg 1 was describing.

If leg 4 still gives 375 Hz, the VCVS is not isolating the sections. The usual cause is that its input is taken from the wrong node, so it is measuring the second section rather than the first.

If leg 4 gives something near 1001 Hz, only one section is in the signal path; check that the follower's output feeds the second section rather than replacing it.

What to take from it. A buffer does not improve the filter. It makes the filter behave like the one you designed, which is a different and more valuable thing. Every op-amp in the rest of this book is there for that reason rather than for its gain, and in Chapter 8 an entire transistor stage exists for nothing else.

Feedback, active filters, and the diode

Why the ideal rules work, what they cost, and the first nonlinear device.

IN THIS CHAPTER

- Loop gain, and the closed-loop gain written so the error term is visible.
- Desensitisation: trading gain you have for accuracy you want.
- What feedback does to input and output impedance, and in which direction.
- Distortion divided by the loop gain, and the nonlinearity that always survives.
- Gain-bandwidth product, and the stability limit that ends the free lunch.
- Active filters, and the loading problem an amplifier solves for them.
- The diode: the exponential, the 0.7 V approximation, and where it stops being one.
- Newton-Raphson, and why plain Newton-Raphson does not work on a diode.
- Building `ael::feedback`, `ael::device::diode` and `ael::nr`.

Chapter 3 asserted two rules and used them. This chapter derives them, and the derivation hands back an error term that says exactly how far from true they are. **One number decides everything:** the loop gain T . The gain error is $1/(1 + T)$, distortion is divided by $1 + T$, impedances are multiplied or divided by $1 + T$. Learn T and the rest is bookkeeping — and learn what it costs, which is the gain you gave up and the bandwidth you did not get, because gain and bandwidth trade one for one and an amplifier is sold by their product.

Feedback also stops working, and it is worth knowing where. Enough phase shift around the loop turns negative feedback positive and an amplifier into an oscillator; one pole can never get there, two only asymptotically, and three get there with gain to spare. That is why Chapter 10's four-stage amplifier has a capacitor in it whose only job is to make the amplifier slow.

The second half of the chapter introduces **the first device in this book that is not linear**. A diode has no phasor description, because it creates frequencies that were not in the excitation, and it cannot be solved in one pass, because its conductance depends on the answer. The machinery for that is Newton-Raphson: guess, linearise, solve with the linear solver you

already have, repeat. **Every nonlinear simulator is that loop**, and the one detail that makes it work in practice — damping a large step by taking it in the logarithm — is the difference between seven iterations and a hundred and sixty-eight.

4.1 Feedback, and what one number decides

Chapter 3 asserted two rules and used them. This section derives them, and the derivation hands back an error term that says how far from true they are.

4.1.1 The loop, and the one number

An amplifier of gain A , a fraction β of its output fed back in opposition:

$$A_{CL} = \frac{A}{1 + A\beta}$$

$A\beta$ is the **loop gain**, written T : what a signal is multiplied by on one trip around the loop, and the only quantity in this section that matters. When T is large,

$$A_{CL} \approx \frac{1}{\beta}$$

which depends on the feedback network alone, and that is the entire reason feedback is used. β is two resistors; A varies by three between devices and drifts with temperature.

The two rules of Chapter 3 are this restated. Gain set by β alone is rule 2, the virtual short. Rule 1 is the amplifier's own property, not a feedback result at all.

4.1.2 The error, which is the useful form

$$A_{CL} = \frac{1}{\beta} \cdot \frac{T}{1 + T} = \frac{1}{\beta} \left(1 - \frac{1}{1 + T} \right)$$

The gain falls short of ideal by one part in $1 + T$, which turns a judgement into arithmetic.

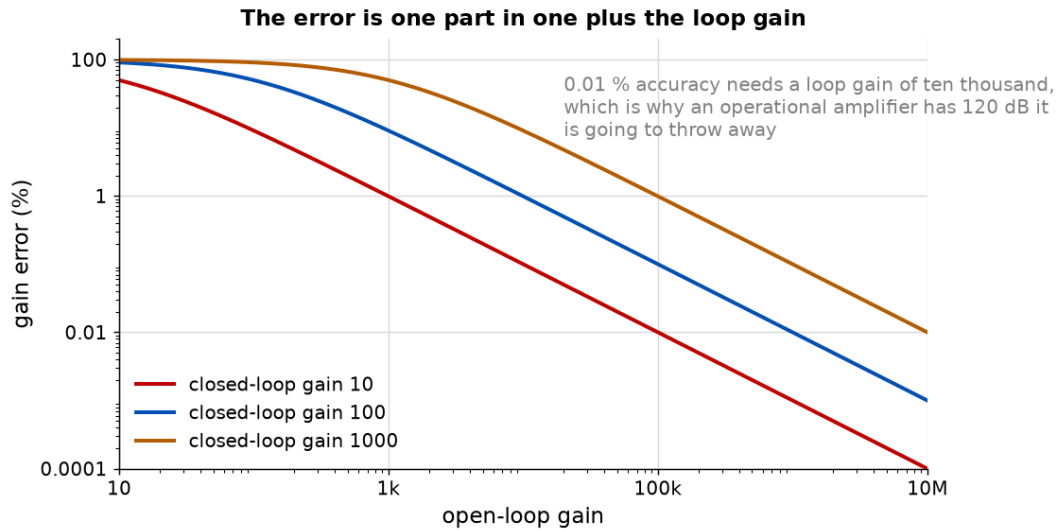


Figure 4.1 Gain error against open-loop gain on logarithmic axes, for closed-loop gains of ten, one hundred and one thousand. Each curve falls as one over the loop gain, and reaching 0.01 per cent error needs an open-loop gain of ten thousand times the closed-loop gain.

Loop gain T	Gain error
10	9 per cent
100	1 per cent
10^3	0.1 per cent
10^4	0.01 per cent

An amplifier with $A = 10^5$ at a closed-loop gain of 10 has $\beta = 0.1$, so $T = 10^4$ and the error is 0.01 per cent.

At a closed-loop gain of 1000 it has $T = 100$ and 1 per cent error, usually unacceptable. **High closed-loop gain is expensive**, which the closed form does not show.

4.1.3 What else $1 + T$ divides

- **Distortion is divided by $1 + T$.** The loop sees the amplifier's own nonlinearity as an error to correct: 1 per cent open loop becomes 0.01 per cent closed.
- **Output impedance is divided by $1 + T$** for voltage feedback: 100 ohm open loop with $T = 10^4$ presents 10 milliohm.
- **Input impedance is multiplied by $1 + T$** for series feedback, which the non-inverting configuration uses. Inverting uses shunt feedback, which divides it: hence its bare R_{in} .
- **Gain sensitivity is divided by $1 + T$.** A 50 per cent change in A moves A_{CL} by 50 per cent divided by $1 + T$.

What feedback does not fix. More feedback cannot drive the remainder to zero, because T itself falls with frequency, and nothing here touches input noise, offset or clipping. **A clipped amplifier has no loop gain at all, so feedback stops working exactly when it would be most useful.**

4.1.4 What it costs: gain-bandwidth

Open-loop gain falls as a single pole from a few hertz upwards, at 20 dB per decade, so the product of gain and frequency is a constant:

$$GBW = A_{CL} \times f_{-3\text{ dB}}$$

A 1 MHz gain-bandwidth product gives 100 kHz at a closed-loop gain of 10, and 10 kHz at 100.

Gain and bandwidth trade one for one, which is why a high-gain stage is usually two stages of moderate gain: two stages of 10 have about six and a half times the bandwidth of one stage of 100, ten per stage less the 0.644 a cascade of coincident poles costs (§4.1.6).

T falls with the open-loop gain, so every benefit in §4.1.3 evaporates as frequency rises: 0.01 per cent gain error at DC is 1 per cent at a hundredth of the bandwidth. **Closed-loop distortion rises with frequency even when the amplifier's own does not.**

4.1.5 Where feedback stops working

Each pole around the loop contributes up to 90 degrees of lag. **At 180 degrees negative feedback adds instead of subtracting, and if T is still above one there, the circuit oscillates.**

One pole can never reach 180 degrees, so a single-pole amplifier with resistive feedback is unconditionally stable; two reach it only asymptotically. **Three reach it with gain to spare**, which is why a three-stage amplifier needs deliberate compensation: Chapter 10's Miller capacitor.

Phase margin, compensation strategy and the Nyquist criterion are a course of their own.

4.1.6 Active filters

An amplifier can do better than isolate two RC sections: it can put feedback around the passive network and produce a response no passive RC network can.

A **Sallen-Key** low-pass is two resistors, two capacitors and one amplifier as a follower, with the first capacitor returned to the output rather than to ground. That capacitor sees the output, feeds energy back into the network, and **the pole pair becomes complex.**

A passive RC cascade makes only real poles, and real poles give a soft corner: two coincident are 6 dB down. A complex pair can be 3 dB down, or peaked, and far sharper.

Realisation	Poles	3 dB point of two sections
Two RC sections cascaded directly	Real, split apart	0.374 of one section's corner
Two RC sections with a buffer between	Real, coincident	0.644
Sallen-Key, $Q = 0.707$	Complex pair	1.0, and a much sharper transition

One part buys a filter the passive components cannot make at any value. That is a better argument for an op-amp than gain is, and the argument Chapter 8 makes for an emitter follower.

4.1.7 What this section is blind to

- **Stability, properly.** The condition is stated in §4.1.5 and never developed. A design that needs a phase margin computed needs a different course.
- **Feedback topologies.** There are four, by whether voltage or current is sensed and fed back. This section assumes voltage sensing and describes two.
- **Noise.** Feedback does not reduce noise generated at the amplifier's input, which in most low-noise designs decides everything.

4.2 The diode, and solving for it

The first device in this book that is not linear, and the machinery that makes a nonlinear circuit solvable at all.

4.2.1 The diode equation

$$I = I_S (e^{V/V_T} - 1)$$

I_S is the saturation current, around 10^{-14} A for a small-signal diode; V_T is the thermal voltage, 26 mV. That is the whole model. Two consequences do all the work.

A decade of current costs 60 mV, from $V_T \ln 10 = 59.9$ mV. A diode at 1 mA and 0.65 V is at 0.71 V for 10 mA and 0.59 V for 100 microamps.

The slope is proportional to the current.

$$g = \frac{dI}{dV} = \frac{I + I_S}{V_T} \approx \frac{I}{V_T}$$

So the incremental resistance is V_T/I , 26 ohm at 1 mA. **That is the same quantity as r_e , the intrinsic emitter resistance the whole of Part 2 is built on,** and for the same reason: a base-emitter junction is a diode.

4.2.2 The 0.7 volt model, and where it fails

The constant-drop model replaces the exponential with a switch that turns on at 0.65 V. Crude, universal, usually good enough, which makes its failure mode worth more than dismissing it.

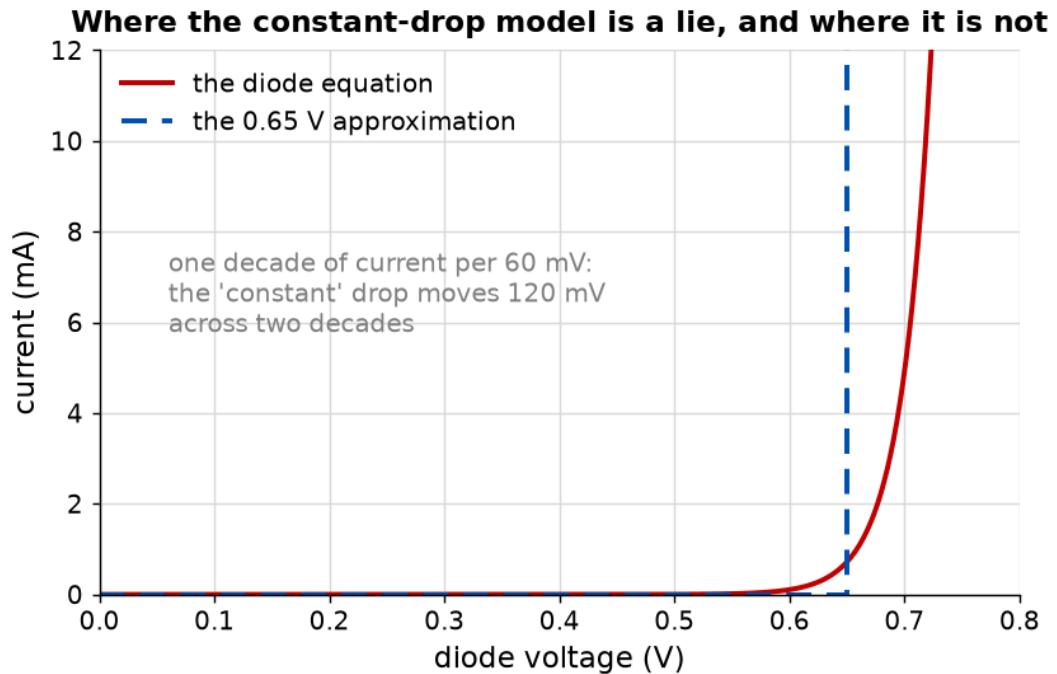


Figure 4.2 The diode equation plotted as current against voltage on linear axes, rising sharply above about 0.6 volts, with the constant drop approximation drawn as a vertical line at 0.65 volts.

It is exact at exactly one current, the one where the real diode sits at 0.65 V. Either side of that it is wrong, and the direction reverses:

Series resistor	Real diode voltage	Real current	Constant-drop current	Error
1 kilohm	0.697 V	4.30 mA	4.35 mA	+1.1 per cent
100 kilohm	0.578 V	44.2 microamps	43.5 microamps	-1.6 per cent
10 megohm	0.458 V	0.454 microamps	0.435 microamps	-4.2 per cent

All from a 5 V supply. **The error in the current stays small and the error in the diode voltage does not:** the drop moves 240 mV, from 0.70 V to 0.46 V, across a model that calls it constant.

That distinction is what matters in Part 2. A bias network assuming a fixed 0.65 V gets the collector current about right, because the drop appears inside a subtraction where a few tens of millivolts do not matter. Where the same millivolts appear inside an exponential, as in Chapter 6's thermal drift, the constant-drop model says nothing at all.

4.2.3 Newton-Raphson

A nonlinear circuit cannot be solved in one pass, because the diode's conductance depends on the answer. Guess, linearise, solve, repeat.

At a guessed V_k , replace the diode with its tangent: a conductance g_k in parallel with a current source I_{eq} that makes the tangent pass through the operating point.

$$g_k = \frac{I_k + I_S}{V_T}, \quad I_{eq} = I_k - g_k V_k$$

Both stamp into the linear solver of Chapter 1, the conductance like a resistor and the equivalent current like a current source. Solve, take the new node voltage as the next guess, iterate until the change is small.

Every nonlinear simulator is that loop. The rest is bookkeeping and better device models.

4.2.4 Why plain Newton-Raphson does not work

Start at zero volts. The diode carries no current, so its conductance is I_S/V_T , about 10^{-12} siemens, an open circuit, and the first solve puts the entire supply across the diode.

At 5 V the exponential's slope is astronomical and the tangent is almost vertical. Each iteration walks back down by about one thermal voltage, so 5 V to 0.7 V takes **168 iterations**, from a method advertised as quadratic.

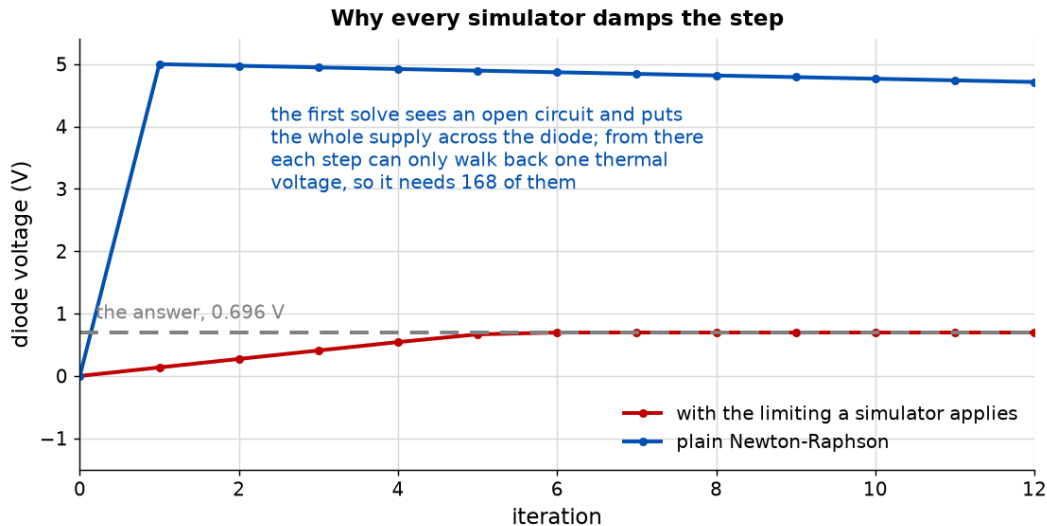


Figure 4.3 Diode voltage against iteration number for two Newton-Raphson runs. The plain method jumps to five volts on the first step and then descends very slowly. The limited method climbs smoothly and reaches the answer of 0.696 volts in seven iterations.

The fix every simulator applies is to damp a large increasing step by taking it in the logarithm:

$$V_{k+1} \leftarrow V_k + V_T \ln \left(1 + \frac{\Delta V}{V_T} \right) \quad \text{when } \Delta V > 2V_T$$

With that, the same circuit converges in **seven** iterations.

This is not a trick to make the demonstration work. It is what SPICE does, and a solver without it will appear to work on every circuit in this section and then fail to converge on the first transistor circuit in Chapter 6.

4.2.5 What to build

An addition to `ael/net/netlist.hpp`

```
/// An ideal diode, anode to cathode, with the given saturation current.
void addDiode(Node anode, Node cathode, double saturationCurrent = 1.0e-14) {
    noexcept;
    [[nodiscard]] std::size_t diodeCount() const noexcept;
```

`ael/feedback/loop.hpp`

Function	Returns
loopGain(openLoop, beta)	$T = A\beta$.
closedLoopGain(openLoop, beta)	$A/(1 + T)$.
gainError(openLoop, beta)	$1/(1 + T)$, the fraction by which the gain falls short.

Three lines each. They are worth writing rather than remembering because §4.1.4 makes them functions of frequency: the open-loop gain handed to them is the gain at the frequency of interest, not the datasheet’s.

`ael/device/diode.hpp`

Function	Returns
current(voltage, saturation)	The diode equation.
conductance(voltage, saturation)	Its derivative, $(I + I_S)/V_T$.
limit(proposed, previous)	The damped step of §4.2.4.

ael/nr/solve.hpp

```

namespace ael::nr
{
    struct Solution
    {
        std::vector<double> nodeVoltages{};
        std::vector<double> sourceCurrents{};
        std::size_t iterations{};    ///< How many it took. Zero means it did not run.
        bool converged{false};
    };

    /// Iterate to a DC operating point. Converged when every node moves less than `
    // tolerance`.
    [[nodiscard]] Solution solve(const net::Netlist& netlist, double tolerance = 1.0e-
    // 9,
                                std::size_t maxIterations = 100U) noexcept;
}

```

Three things the suite checks:

- **iterations is reported**, because it is the diagnostic that says whether the limiting works. Seven is healthy; ninety is converging linearly and will fail on something larger.
- **converged is false rather than silently returning the last iterate** when the loop runs out. This is Chapter 1's solved flag again, for the same reason.
- **A netlist with no diodes gives the same answer as ael::mna::solve**, in one iteration. A nonlinear solver that cannot reproduce the linear one on a linear circuit has a bug that will be much harder to find later.

What good looks like

About fifty lines on top of Chapter 1: fifteen for the loop, five for the limiting. Much more, and the linear solve is probably being rebuilt from scratch each iteration rather than restamped.

4.2.6 What this section is blind to

- **Reverse breakdown.** The equation says a reverse-biased diode carries $-I_S$ forever. A real one avalanches somewhere between 5 V and a few hundred, and a Zener is sold for that behaviour.
- **Capacitance.** A real diode has junction and diffusion capacitance, so a rectifier conducts briefly in reverse when switched off.
- **Temperature.** I_S roughly doubles every 5 degrees and V_T is proportional to absolute temperature. Net, a forward drop falling about 2 mV per degree: the -2 mV/K that dominates Chapter 6.

- **Convergence in general.** The limiting here handles one exponential device. Several interacting nonlinearities can still fail, and real simulators carry more strategies for it.

4.3 Review

What you should be able to do now

After this chapter, you should be able to:

- Compute the loop gain of any of Chapter 3's configurations, and the gain error that follows.
- Say how much open-loop gain a stated accuracy needs, and notice when the answer is unreasonable.
- State what feedback does to input impedance, output impedance and distortion, with the direction right in each case.
- Convert between gain-bandwidth product, closed-loop gain and closed-loop bandwidth.
- Say what a Sallen-Key section buys over two cascaded RC sections with a buffer between them.
- Write down the diode equation, and say what a 60 mV change of its voltage does.
- Explain why plain Newton-Raphson fails on a diode from a cold start, and what a simulator does about it.

Questions to test yourself

You should be able to answer:

1. An amplifier has 10^5 of open-loop gain. You use it at a closed-loop gain of 1000. What is the gain error, and is that amplifier a good choice?
2. Feedback divides distortion by $1 + T$. What distortion remains, and why can it not be removed by more feedback?
3. Two amplifiers have the same gain-bandwidth product. One is used at a gain of 10 and one at 100. Which has more loop gain at 10 kHz, and by how much?
4. Negative feedback becomes positive when the loop phase reaches 180 degrees. How many first-order poles does it take to get there, and what does that say about how many stages a loop can have?
5. A diode carries 1 mA at 0.65 V. What voltage does it need for 10 mA? For 1 microamp?
6. Why does a nonlinear solver need a convergence criterion at all, when a linear one just solves?
7. Your Newton-Raphson loop converges in 6 iterations on one circuit and 400 on another that differs only in a resistor value. What is different about the second one?

Further reading

- *The Art of Electronics*, Horowitz and Hill, chapter 4, for feedback, and chapter 1 for the diode.

Looking ahead

- Chapter 5 begins Part 2. The transistor as a device with an exponential, a current gain and a saturation voltage, and with as little semiconductor physics as can be managed.

4.4 Exercises

Eight, ending with the Cross-check.

How to check your work. A `CODE` exercise is judged by the suite that ships with it: run `make test` at the root of the companion repository, with `AEL_DIR` pointing at your toolkit. A suite you have not started is reported as `SKIP` rather than as a failure, so the command is worth running from the first day. The hand calculations and designs are checked against the toolkit functions each one names.

The solutions are not in this book. They are published in full in the companion repository, in `lectures/L04/appendix`, including the plausible wrong answer where there is one. Read them once you have your own answers, including the ones you are unsure about, because being unsure and right is a different thing from being unsure and wrong and the solutions say which is which.

Exercise 4.1 What the loop gain decides

RECALL

1. Write the closed-loop gain in the form that makes the error visible, and identify the loop gain in it.
2. List four quantities that $1 + T$ divides or multiplies, and give the direction of each.
3. Name two things feedback does not improve, and say why not.
4. An amplifier is driven into clipping. What happens to its loop gain, and what does that say about the distortion at that moment?

Exercise 4.2 The diode

RECALL

1. How much voltage does a decade of current cost, and where does that number come from?
2. A diode carries 1 mA at 0.65 V. What is its incremental resistance, and what else in this book is the same quantity?
3. State the constant-drop model, and say at how many currents it is exact.
4. Between 0.5 microamps and 4 mA a real diode's drop moves by about 240 mV. Give one design in which that matters and one in which it does not.

Exercise 4.3 How much gain do you need

HAND CALCULATION

An amplifier is to have a closed-loop gain of 100, accurate to 0.1 per cent.

1. What loop gain does that need?
2. What open-loop gain?
3. The amplifier available has $A = 10^5$ and a gain-bandwidth product of 1 MHz. Does it meet the accuracy at DC? At 1 kHz?
4. At what frequency does the gain error reach 1 per cent?

Check yourself: loopGain, gainError

Exercise 4.4 The diode at two currents

HAND CALCULATION

A 5 V supply drives a diode through a series resistor. $I_S = 10^{-14}$ A.

1. For a 1 kilohm resistor, find the diode voltage and current, by hand. Two iterations of guess and correct is enough; start from 0.65 V.
2. Repeat for 100 kilohm.
3. Compare both with the constant-drop model's answer, and give the percentage error in the current for each.
4. The error in the current is small in both cases and the error in the diode voltage is not. Explain why, and say which of the two a bias network in Chapter 6 will care about.

Check yourself: current, conductance

Exercise 4.5 A half-wave rectifier with a stated ripple

DESIGN

A 12 V peak, 50 Hz sinusoid feeds a half-wave rectifier into a smoothing capacitor and a 1 kilohm load. The output ripple must not exceed 0.5 V peak-to-peak.

1. Choose the capacitor. State the assumption you are making about the discharge, and say whether it makes your answer optimistic or pessimistic.
2. State the DC output voltage, remembering the diode.
3. What peak current does the diode carry, roughly, and why is it so much larger than the load current?

4. Doubling the capacitor halves the ripple. What does it do to the peak diode current, and what does that suggest about how far this approach scales?

Exercise 4.6 Feedback and the diode model

CODE

Implement `ael::feedback` and `ael::device::diode` to the specification in §4.2.5.

`limit` is four lines and is the only one that is not obvious. Damp only steps that are increasing and larger than two thermal voltages; leave everything else alone. Damping a decreasing step makes convergence worse, not better.

Exercise 4.7 The nonlinear solve

CODE

Add `addDiode` to the netlist and write `ael::nr::solve`.

Do this by calling your existing linear solve inside a loop, not by writing a second solver. Each iteration recomputes the diode stamps from the present guess and re-solves.

Check three things before you believe it:

1. A netlist with no diodes gives the same answer as `ael::mna::solve`, in one iteration.
2. The 5 V through 1 kilohm circuit converges in fewer than ten iterations.
3. Turning the limiting off makes it take about 170. If it does not, the limiting was not doing anything.

Exercise 4.8 The diode, and the model that is nearly right

CROSS-CHECK

A 5 V supply, a series resistor, and a diode with $I_S = 10^{-14}$ A. Find the current, three ways, at two very different operating points.

Point one: a 1 kilohm resistor.

1. **By hand, with the constant-drop model.** $(5 - 0.65)/1000$.
2. **By hand, with the exponential.** Iterate from 0.65 V until it stops moving.
3. **By your solver.** `ael::nr::solve`.

Point two: repeat all three with a 10 megohm resistor.

Then state, for each point, the percentage disagreement between legs 1 and 3, and separately the difference in the diode voltage.

What to expect

At 1 kilohm, leg 1 is 1.1 per cent high. At 10 megohm it is **4.2 per cent low**. The sign reverses, and that is the interesting part: the constant-drop model is exact at exactly one current, the one where the diode really does sit at 0.65 V, and it errs in opposite directions either side.

Legs 2 and 3 should agree to about six significant figures. Both are solving the same transcendental equation, one by hand iteration and one by Newton-Raphson.

The diode voltage disagrees far more than the current does. 0.697 V against 0.65 V at one point, 0.458 V against 0.65 V at the other. The model calls a quantity constant that moves by 240 mV across the range.

Why the current error stays small anyway. The diode voltage is subtracted from 5 V, so a 240 mV error in it is a 5 per cent error in what remains. The current error is bounded by that subtraction, and it would not be if the supply were 1 V instead of 5.

Try that. Repeat point one with a 1 V supply and a 100 ohm resistor. The constant-drop model now errs by a much larger fraction, because the diode drop is most of the supply. That is the condition under which the model should not be used, and it is exactly the condition inside a transistor's base-emitter loop, which is why Chapter 6 computes the drift from the exponential rather than from a constant.

If your solver reports 170 iterations, the limiting is not working. If it reports converged false at 100, the limiting is inverted and it is damping the wrong steps.

The transistor as a device and as a switch

The device, with as little semiconductor physics as can be managed, which is almost none.

IN THIS CHAPTER

- The bipolar transistor as an exponential, a current gain, and a saturation voltage.
- Why this book assumes $h_{FE} = 50$ everywhere, and what that assumption is worth.
- Active, saturation and cutoff, and which one a switch lives in.
- Designing a switch: the base resistor, forced beta, and the saturation voltage you pay for.
- The MOSFET: a threshold, a square law, and two regions.
- BJT against MOSFET, honestly, on gain, input impedance and current.
- Building `ael::device::bjt` and `ael::device::mosfet`.

This chapter opens Part 2, which is the subject of the book. Six chapters on the transistor amplifier: the device arrives here, the operating point in Chapter 6, and the small-signal model in Chapter 7. Nothing is designed until all three exist.

The device is one equation and one ratio. The collector current is an exponential in the base-emitter voltage — **the same exponential as the diode of Chapter 4, with the same V_T and the same I_S , because the base-emitter junction is a diode** — and the base current is that divided by beta. Everything in the next five chapters is a consequence, including the one number the whole of Part 2 is written in terms of: the incremental resistance of that junction, 26 ohm at 1 mA, which arrives here as a property of a diode and is named r_e in Chapter 7.

Beta is not a parameter you may trust. It varies by a factor of three between devices of the same part number and moves with temperature and current, so this book assumes 50, below the specified minimum of most small-signal parts, and every design must work there. The rule that follows runs through the rest of the book: **no design may depend on the value of beta.** Designing a switch is where you first see it enforced, by choosing the base

current from a forced beta rather than from h_{FE} , and the design then stops depending on the parameter you were told not to trust.

The MOSFET gets a section because one fact about it drives every comparison: its transconductance goes as the square root of its current where a bipolar transistor's goes as the current itself, which is a factor of ten at 1 mA. That number decides which device goes at an amplifier's input and which at its output, and both answers appear later.

5.1 The bipolar transistor, and the switch

A practical treatment with as little semiconductor physics as possible. A transistor is an exponential, a current gain, and a saturation voltage, and those three facts carry the whole of Part 2.

5.1.1 One equation, and one ratio

$$I_C = I_S (e^{V_{BE}/V_T} - 1), \quad I_B = \frac{I_C}{\beta}$$

I_S is around 10^{-14} A, V_T is 26 mV, and β , also written h_{FE} , is between 50 and 300.

That is the diode equation of Chapter 4 (Chapter 4), with the same V_T and I_S , because the base-emitter junction *is* a diode. Everything Chapter 4 established carries over:

- **60 mV per decade of current.** A transistor at 1 mA and 0.65 V needs 0.71 V for 10 mA.
- **The incremental resistance is V_T/I , 26 ohm at 1 mA.**

That second one is the single most important number in this book. From Chapter 7 it is r_e , the intrinsic emitter resistance, and every gain, input resistance and output resistance in Part 2 is written in terms of it. It arrives here as a property of a diode.

The emitter carries both currents:

$$I_E = I_C + I_B = I_C \left(1 + \frac{1}{\beta}\right)$$

which at $\beta = 50$ is 2 per cent more than I_C . That 2 per cent is why this book writes $I_E \approx I_C$.

5.1.2 Beta, and why this book assumes 50

h_{FE} is the least trustworthy number on any datasheet. A 2N3904 is specified at 100 to 300 at 10 mA, and the same device at 0.1 mA and at 100 mA can differ by a factor of three.

This book assumes $h_{FE} = 50$ everywhere, below the specified minimum of most small-signal parts, for one reason:

A design that works at $h_{FE} = 50$ works at $h_{FE} = 300$. The reverse is not true.

Hence the rule running through Part 2: **no design may depend on the value of beta.** Where beta appears in a result, the design is arranged so the result is insensitive to it. There

is exactly one exception, the input resistance of an emitter follower in Chapter 8, and the chapter says so.

5.1.3 Three regions

Two junctions, each forward or reverse biased, give four combinations, of which three are used.

Region	Base-emitter	Base-collector	Behaviour
Cutoff	Reverse	Reverse	No current. An open switch.
Forward active	Forward	Reverse	$I_C = \beta I_B$. An amplifier.
Saturation	Forward	Forward	$I_C < \beta I_B$. A closed switch.

The fourth, reverse active, is a transistor used backwards; it works badly and nothing uses it on purpose.

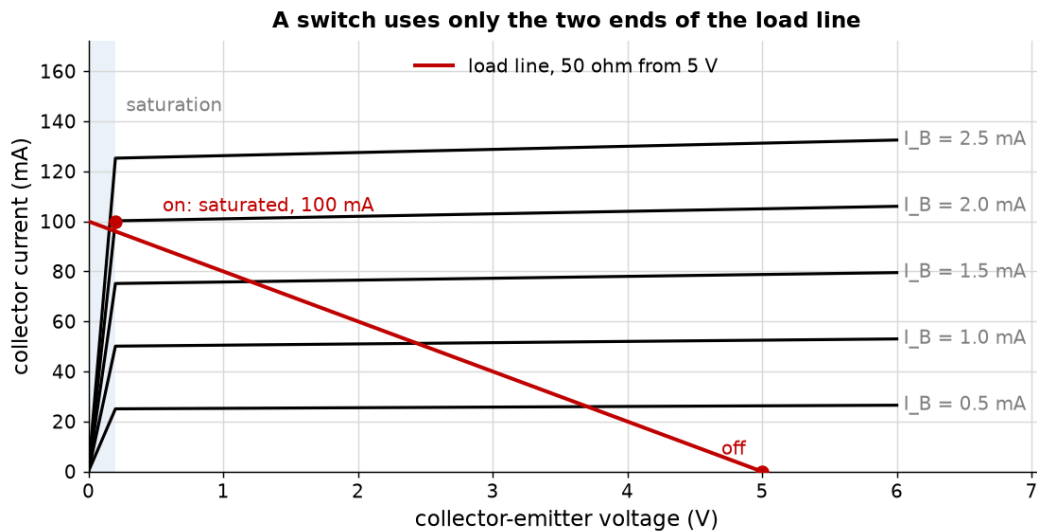


Figure 5.1 Collector current against collector-emitter voltage for five base currents, with the fifty ohm load line drawn across them from five volts on the horizontal axis to a hundred milliamps on the vertical, and the two operating points of a switch marked at the ends of that line.

The curves are what the transistor allows, the straight line what the external circuit allows, and the operating point is where they cross. **A switch uses only the two ends of that line;** everything between them is where an amplifier lives and where a switch dissipates for nothing.

Saturation is where the model stops being simple. In the active region I_C does not depend on V_{CE} , so the transistor is a current source. Below about 0.2 V it does, steeply, so it

is a resistor instead. The simple model draws that knee as a corner and a real device rounds it off.

5.1.4 Designing a switch

A switch must be **on hard**, meaning saturated, meaning a base current above I_C/β . Since β may not be trusted, it is chosen from a **forced beta** instead:

$$\beta_{forced} = \frac{I_C}{I_B}, \quad \text{chosen as 10}$$

Ten times the base current the active region would need guarantees saturation for any device with h_{FE} above 10, which is every silicon transistor ever made.

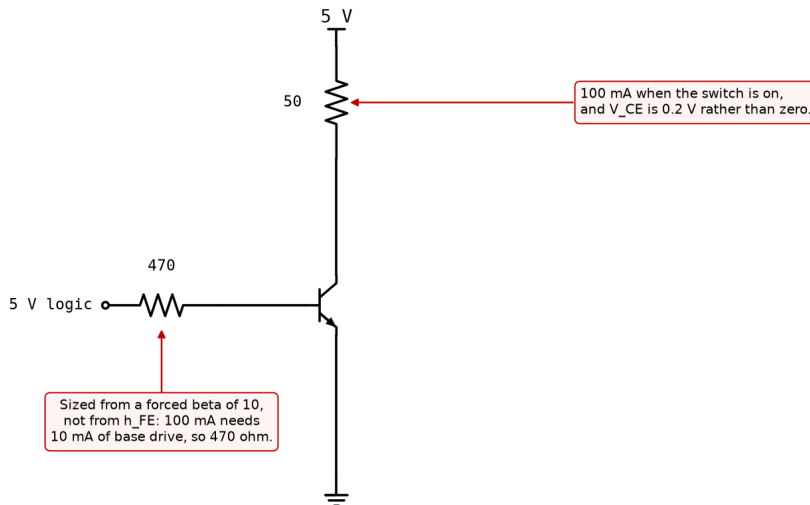


Figure 5.2 A switch: five volt logic through a base resistor into an NPN transistor, whose collector drives a fifty ohm load to a five volt rail and whose emitter is grounded, with annotations giving the base resistor as four hundred and seventy ohms and noting that the saturated collector-emitter voltage is not zero.

For 100 mA of load current from a 5 V logic level:

$$I_B = \frac{100 \text{ mA}}{10} = 10 \text{ mA}, \quad R_B = \frac{5 - 0.7}{10 \text{ mA}} = 430 \Omega$$

so 470 ohm from the E12 series, giving 9.1 mA and a forced beta of 11.

Note the 0.7 V, where every bias point in this book uses 0.65. A forced beta of 10 drives the base ten times harder than the active region would, and 60 mV a decade is what that

costs: the junction sits above the small-signal figure, and `baseResistor` uses 0.7 for exactly this reason. At larger currents it moves again, which is why Exercise 5.5 needs 0.8.

What forced beta costs. A tenth of the load current thrown away in the driver, and stored base charge that must be removed before the device turns off. A hard-saturated transistor is slow to turn off for exactly that reason, and a fast switch is deliberately kept out of deep saturation.

What the saturation voltage costs. 0.1 to 0.3 V across the device, so 10 to 30 mW at 100 mA. A linear stage delivering the same current at half the supply dissipates 250 mW. **That ratio is the entire reason switching exists.**

5.1.5 What the model here is blind to

- **Bulk resistance.** The ideal transport model gives a saturation voltage of about 57 mV for the switch above; a real device gives 100 to 300 mV. **The model is optimistic, and the direction matters:** a design relying on the modelled figure runs hotter than predicted.
- **Speed.** Charge storage, transit time and junction capacitance are all absent, so nothing here says how fast a transistor switches. That is usually what decides the choice of part.
- **Breakdown.** Above some collector-emitter voltage the device conducts whatever the base does, and an inductive load will drive it there unless something is done about that.
- **Second breakdown.** A power transistor can fail at combinations of voltage and current that are individually within specification. Nothing here models it.

5.2 The MOSFET, the comparison, and what to build

A different device with the same three regions, a square law instead of an exponential, and one number that decides every comparison between the two.

5.2.1 The MOSFET in two regions

A gate voltage creates a channel between drain and source. Below the **threshold** V_{TH} there is no channel. Above it, which region applies depends on whether the drain voltage is large enough to pinch the channel off at the drain end.

$$I_D = \begin{cases} 0 & V_{GS} < V_{TH} \\ k \left[(V_{GS} - V_{TH})V_{DS} - \frac{1}{2}V_{DS}^2 \right] & V_{DS} < V_{GS} - V_{TH} \quad (\text{triode}) \\ \frac{1}{2}k(V_{GS} - V_{TH})^2 & V_{DS} \geq V_{GS} - V_{TH} \quad (\text{saturation}) \end{cases}$$

$V_{GS} - V_{TH}$ is the **overdrive**, and almost every MOSFET result is more natural in terms of it than of V_{GS} .

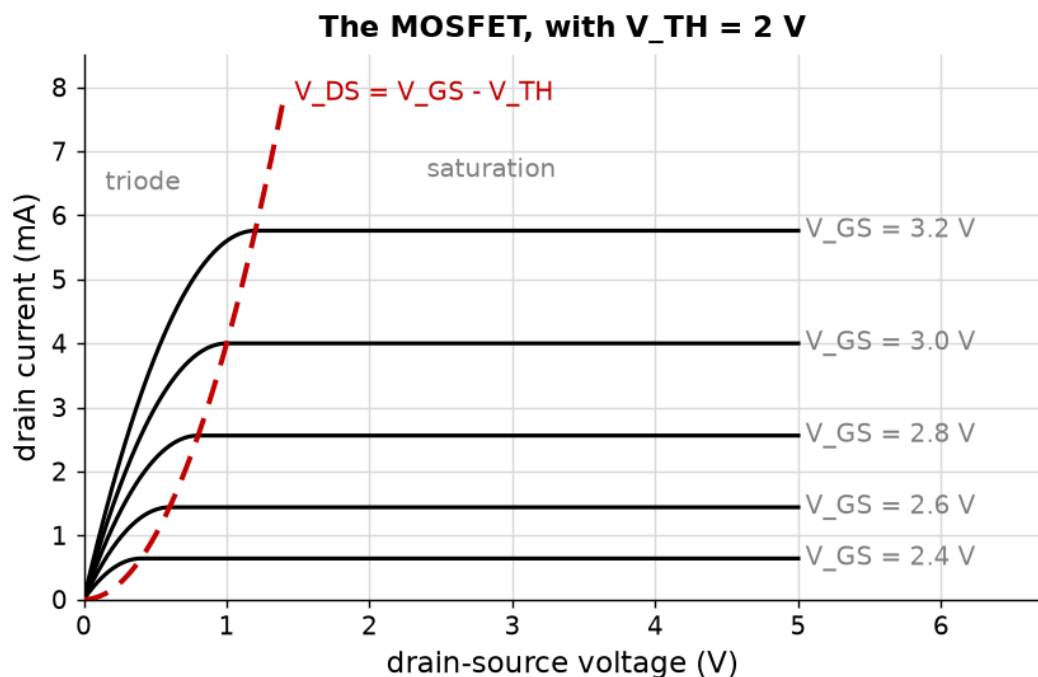


Figure 5.3 Drain current against drain-source voltage for five gate voltages, rising steeply then flattening, with a dashed curve marking the boundary where the drain-source voltage equals the overdrive.

The naming is unfortunate and worth stating once. A MOSFET's *saturation* is where it acts as a current source, which is a BJT's *active*. A MOSFET's *triode* is where it acts as a resistor, which is a BJT's *saturation*. Confusing them is a standard way to misread a datasheet.

The gate draws no current: it is an insulated capacitor plate. That is why a MOSFET's input resistance is effectively infinite at DC, and why Chapter 7's input-resistance results do not carry across between the two devices the way everything else does.

5.2.2 Transconductance, and the factor of ten

$$g_m = k(V_{GS} - V_{TH}) = \sqrt{2kI_D}$$

A MOSFET's transconductance goes as the square root of its current. A BJT's goes as the current itself, $g_m = I_C/V_T$. That difference decides every comparison between them.

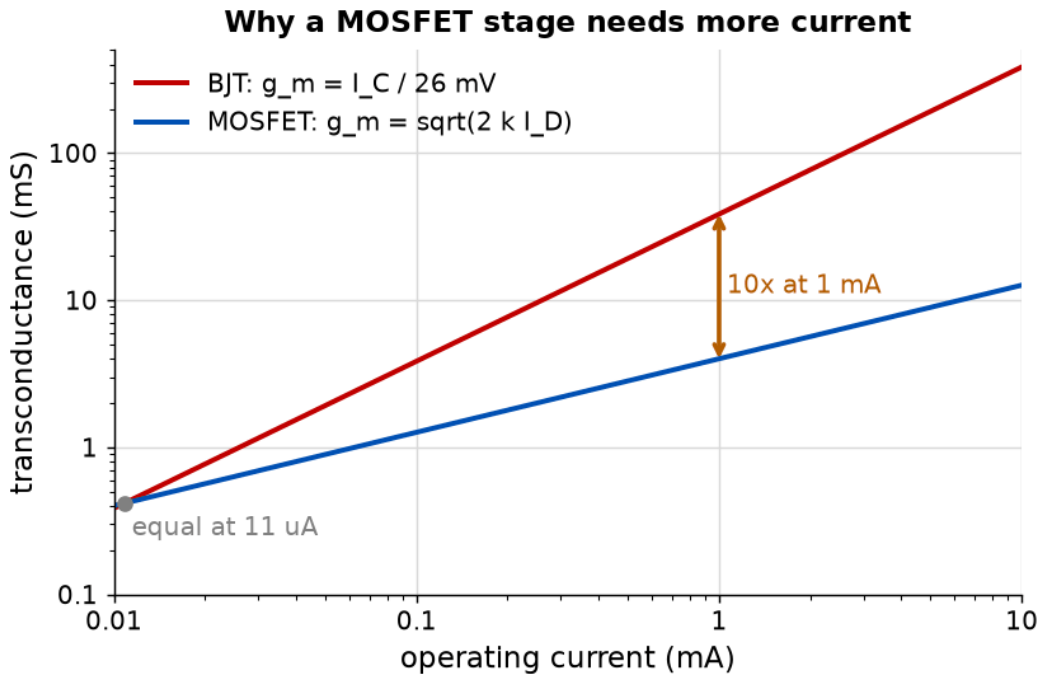


Figure 5.4 Transconductance against operating current for both devices on logarithmic axes, the BJT rising with slope one and the MOSFET with slope one half, with the gap at one milliamp measured as a factor of ten.

At 1 mA a BJT gives 38.5 mS and a representative MOSFET about 4 mS: **a factor of ten.**

- **A MOSFET stage needs about ten times the current** of a BJT stage for the same gain, or a much larger device.
- **The advantage grows with current**, as $\sqrt{I}$. At 10 mA the BJT wins by thirty; at 10 microamps the two are comparable.
- **Chapter 7's source factor is about two where the emitter factor is about ten**, for the same 220 mV across the degeneration resistor. This is the entire reason for it.

One caveat the figure cannot show. The square law says the two are equal near 11 microamps and the MOSFET wins below. A real MOSFET leaves the square law well before then: in weak inversion it conducts exponentially and g_m approaches $I_D / (nV_T)$, the BJT's result divided by one or two. The curves converging is real; the crossing is an artifact of the model.

5.2.3 Choosing between them, honestly

	BJT	MOSFET
Transconductance at 1 mA	38.5 mS	About 4 mS
Input current	I_C/β , real and troublesome	None at DC
Turn-on voltage	0.6 to 0.7 V, and predictable	1 to 4 V, and poorly controlled
As a switch	Saturation voltage, roughly fixed	On-resistance, so the drop falls with current
Matching between devices	Excellent, and exponential in V_{BE}	Poorer, and it limits offset

BJTs for gain and for matched pairs; MOSFETs for switches and for inputs that must draw no current. A discrete amplifier built for gain is usually bipolar, a power switch usually a MOSFET, and an instrument input that must not load its source a MOSFET whatever else it costs.

5.2.4 What to build

ael/device/bjt.hpp

The model is the **transport form**, which handles all three regions with one expression and no branching:

$$I_C = I_S (e^{V_{BE}/V_T} - e^{V_{BC}/V_T}) - \frac{I_S}{\beta_R} (e^{V_{BC}/V_T} - 1)$$

$$I_B = \frac{I_S}{\beta_F} (e^{V_{BE}/V_T} - 1) + \frac{I_S}{\beta_R} (e^{V_{BC}/V_T} - 1)$$

With the collector reverse biased the second exponential vanishes and this collapses to $I_C = \beta_F I_B$. Drive the base hard enough that the collector falls below it and the second exponential takes over, the collector current stops rising, and the device is saturated. **Saturation is not a special case in the code; it is what the equation does.**

Member	Contract
struct Parameters	saturationCurrent 10^{-14} , forwardBeta 50, reverseBeta 2, earlyVoltage 100.
struct Currents	base, collector, emitter.
currents(vbe, vbc, parameters)	The two expressions above, with emitter their sum.
region(vbe, vbc)	Cutoff, Active, Saturation or ReverseActive.
baseResistor(drive, loadCurrent, forcedBeta)	The switch design of §5.1.4.

ael/device/mosfet.hpp

Member	Contract
struct Parameters	threshold 2 V, transconductanceParameter 8 mA/V ² .
drainCurrent(vgs, vds, parameters)	The three-region expression of §5.2.1.
transconductance(drainCurrent, parameters)	$\sqrt{2kI_D}$.
region(vgs, vds, parameters)	Cutoff, Triode or Saturation.

An addition to ael/net/netlist.hpp

```
void addBjt(Node collector, Node base, Node emitter, device::bjt::Parameters &
    parameters = {}) noexcept;
[[nodiscard]] std::size_t bjtCount() const noexcept;
```

The nonlinear solver of Chapter 4 (Chapter 4) has to stamp it. A BJT stamps as **two diodes and one controlled current source**, so it contributes nine matrix entries rather than four, and §4.2.4's limiting must be applied to both junction voltages rather than one.

That last point is not optional. A transistor circuit started from zero has both junctions forward biased at the first solve, and without limiting on both the iteration will not converge at all. This is the circuit Chapter 4 warned would fail.

What good looks like

About eighty lines for the two device models and forty for the stamp. If the stamp branches on which region the device is in, it is fighting the transport model rather than using it.

5.2.5 What this section is blind to

- **Channel-length modulation.** The saturation expression has no V_{DS} dependence, so a MOSFET's output resistance is infinite here. Chapter 7 needs it finite and reintroduces it as r_o .
- **The body effect.** A MOSFET whose source is not at the substrate potential has a higher threshold, and in a follower that is a real loss. Named and not modelled.
- **Short-channel behaviour.** Modern devices left the square law behind decades ago; it is good enough for discrete parts at the currents used here.
- **Temperature.** Both models are at one temperature. Chapter 6 is entirely about what happens when that changes.

5.3 Review

What you should be able to do now

After this chapter, you should be able to:

- Write down the collector current of a BJT from its base-emitter voltage, and the base current from the collector current.
- Say what 60 mV of extra base-emitter voltage does, and why that number is familiar.
- Classify an operating point as cutoff, active or saturated from two voltages.
- Design a switch from a load current and a logic level, without using h_{FE} anywhere.
- Say what forced beta is, why it is used, and what it costs.
- Write down a MOSFET's drain current in both of its regions, and the boundary between them.
- Say why a MOSFET stage needs about ten times the current of a BJT stage for the same gain, and where that stops being true.

Questions to test yourself

You should be able to answer:

1. Two transistors of the same part number have h_{FE} of 80 and 300. Which of your designs should care, and which should not?
2. A transistor has 1 mA of collector current. What is the incremental resistance of its base-emitter junction, and what will that quantity be called from Chapter 7 onwards?
3. Why is a saturated transistor's collector-emitter voltage not zero, and what would make it smaller?
4. A switch is designed with a forced beta of 10. What happens if the transistor turns out to have h_{FE} of 300? What if it has 40?
5. Why does a switch waste less power than a linear stage doing the same job, and where does the power actually go in each case?
6. A MOSFET and a BJT both carry 1 mA. Which has the higher transconductance, by how much, and what would you have to do to the MOSFET to equalise them?
7. At what current do a BJT and a MOSFET have equal transconductance, and why is the answer suspicious?

Further reading

- *The Art of Electronics*, Horowitz and Hill, chapter 2, for the transistor introduced the same way, and chapter 3 for the MOSFET.

Looking ahead

- Chapter 6 puts the device to work as an amplifier, which first means holding it still. Biasing, the quiescent point, and what an emitter resistor actually buys.

5.4 Exercises

Eight, ending with the Cross-check. This one is the first in the book where both computational legs agree with each other and both are wrong about a real device, and saying so is the exercise.

How to check your work. A `CODE` exercise is judged by the suite that ships with it: run `make test` at the root of the companion repository, with `AEL_DIR` pointing at your toolkit. A suite you have not started is reported as `SKIP` rather than as a failure, so the command is worth running from the first day. The hand calculations and designs are checked against the toolkit functions each one names.

The solutions are not in this book. They are published in full in the companion repository, in `lectures/L05/appendix`, including the plausible wrong answer where there is one. Read them once you have your own answers, including the ones you are unsure about, because being unsure and right is a different thing from being unsure and wrong and the solutions say which is which.

Exercise 5.1 The device in three facts

RECALL

1. State the collector current as a function of base-emitter voltage, and name the other component in this book that obeys the same equation.
2. What does 60 mV of extra base-emitter voltage do to the collector current?
3. What is the incremental resistance of the base-emitter junction at 1 mA, and what will that quantity be called from Chapter 7?
4. Give the emitter current in terms of the collector current for $\beta = 50$, and say what error is made by treating them as equal.

Exercise 5.2 The three regions

RECALL

1. Name the three regions used, in terms of which junction is forward biased.
2. Which region does an amplifier live in, and which two does a switch use?
3. A MOSFET's *saturation* region and a BJT's *saturation* region are opposite conditions. Describe each, and say which BJT region corresponds to a MOSFET's saturation.
4. In the active region a transistor is a current source; in saturation it is a resistor. Which fact does a load line rely on, and which does a switch rely on?

Exercise 5.3 Bias points from voltages

HAND CALCULATION

A transistor has $I_S = 10^{-14}$ A, $\beta_F = 50$.

1. What base-emitter voltage gives 1 mA of collector current?
2. What gives 10 mA? What gives 10 microamps?
3. For each of the three, what is the base current, and the emitter current?
4. Sketch the relationship between V_{BE} and $\log I_C$. What is its slope, in volts per decade?

Check yourself: currents

Exercise 5.4 Is it saturated

HAND CALCULATION

For each of these, classify the region and justify it in one sentence.

	V_B	V_C	V_E
a	0.0 V	5.0 V	0.0 V
b	0.7 V	3.0 V	0.0 V
c	0.8 V	0.1 V	0.0 V
d	2.7 V	5.0 V	2.0 V
e	0.7 V	0.0 V	0.0 V

Check yourself: region

Exercise 5.5 A relay driver

DESIGN

A microcontroller pin at 3.3 V must switch a relay coil drawing 150 mA from a 12 V supply.

1. Choose a forced beta and justify it.
2. Compute the base resistor, and choose an E12 value.
3. State the base current your choice actually delivers, and the resulting forced beta.
4. The transistor's h_{FE} is specified as 100 minimum. Does your design depend on that number? Say what would have to be true for it to stop working.
5. The relay coil is inductive. State what that does when the transistor turns off, and what one component fixes it. You are not asked to size it.

Check yourself: baseResistor, nearest_e12

Exercise 5.6 The two device models

CODE

Implement `ael::device::bjt` and `ael::device::mosfet` to the specification in §5.2.4.

The BJT model has no branch on region: write the transport expressions and let saturation happen. If your code tests which region it is in before deciding what to compute, you have written three models rather than one, and the seams will show as discontinuities in Chapter 6.

The MOSFET does need its three cases, because the square law genuinely is piecewise.

Exercise 5.7 The transistor in the netlist

CODE

Add `addBjt` and stamp it in `ael::nr::solve`.

A BJT contributes nine matrix entries: two junction conductances and one controlled source. Take the derivatives of the two transport expressions with respect to both junction voltages, and stamp them as a two-port conductance matrix plus two equivalent current sources.

Apply the step limiting to both junction voltages. Chapter 4's circuit had one junction and would converge without it in 170 iterations; this one has two, and without limiting it does not converge at all. That is the failure Chapter 4 predicted.

Confirm it converges in under twenty iterations on the switch of §5.1.4.

Exercise 5.8 The switch, and the model that agrees with itself

CROSS-CHECK

The switch of §5.1.4: 5 V logic through 470 ohm into the base, 50 ohm from the collector to a 5 V rail, emitter grounded. Find the collector current.

1. **By hand, with the saturation-voltage model.** Assume the transistor saturates at 0.2 V and the collector current is whatever the load resistor allows.
2. **By hand, with the transport model.** Solve the two junction equations for V_{BE} and V_{CE} .
3. **By your solver.** `ael::nr::solve` on the four-element netlist.

Then answer the question the three numbers raise: **which of them would a bench agree with?**

What to expect

Legs 2 and 3 agree to five or six significant figures, at 98.9 mA and a collector-emitter voltage of 57 mV. They are the same model solved two ways.

Leg 1 gives 96.0 mA, about 3 per cent lower, because it assumed 0.2 V across the transistor rather than 57 mV.

And leg 1 is the one closer to a real device. A small-signal transistor at 100 mA saturates somewhere near 0.2 V, not near 0.06 V. The transport model has no bulk resistance in it: no resistance in the collector material, none in the emitter, none in the bond wires. Those are what produce most of a real saturation voltage at 100 mA, and the model omits all of them.

This is the first Cross-check where agreement between the legs is not evidence of being right. Legs 2 and 3 agree because they are the same physics evaluated twice. What they agree on is optimistic by a factor of three in the saturation voltage, and a design that budgeted its heat from 57 mV would run three times hotter than predicted.

What to do about it. Nothing, in this book, except know it. The correction is a series resistance in the model, of a few tenths of an ohm, and it is the first thing a real simulator's transistor model has that this one does not. What matters is the habit: **two legs agreeing tells you your arithmetic is right, and says nothing whatever about whether your model contains the physics that decides the answer.**

Sensitivity, which is the design's real claim. Repeat leg 3 with β_F set to 300 and then to 40. The collector current changes by less than a fifth of a per cent, because the design never used beta. Now set it to 8, below the forced beta of 11, and the transistor leaves saturation: the current falls to 72 mA and the collector sits at 1.4 V. **That is the boundary the forced-beta method buys margin against,** and it is three decades away from any real device.

Biasing, and what an emitter resistor actually buys

The quiescent point, and the first place the obvious argument needs correcting.

IN THIS CHAPTER

- The quiescent point as three numbers, and the load line that constrains them.
- Voltage-divider bias, and the base current that loads the divider you designed.
- Stiffness: why the divider current must be much larger than the base current, and what happens at the usual rule of thumb.
- Thermal drift: the base-emitter voltage falling about 2 mV per degree, and what that does.
- The emitter resistor as local feedback, and the drift divided by the emitter factor.
- An argument that does not survive its own arithmetic, and the correct one.
- The 220 mV rule, and the E12 value it lands on.
- Building a `bias` module, and adding temperature to the transistor model.

An amplifier is a transistor sitting at an operating point with a signal moving it about. This chapter is the sitting still; Chapter 7 is the moving about. The construction is the load line of Chapter 5 used at a different place on it: a switch used the two ends, and an amplifier sits where there is room to move both ways.

Divider bias has a surprise in it immediately. The divider that gives 1.71 V unloaded gives 1.60 V with a base hanging on it, and the collector current comes out 12 per cent below what the design said — **Chapter 1's loading, for the fourth time**, and this time the arithmetic is self-referential, because the base current depends on the collector current which depends on the base voltage which depends on the base current.

Then the problem worth solving. **At a fixed base voltage the collector current rises about 8 per cent per degree**, which over a thirty-degree rise is a factor of ten: a stage biased at 1 mA on a bench at 20 degrees is at 10 mA inside an enclosure at 50, sitting hard against one end of its load line. The fix is an emitter resistor, and **there is a tempting explanation of what it does that does not survive its own numbers**. §6.2.2 sets that argument out in full and then breaks it with one exponential; §6.2.3 gives the version that survives, which is

Chapter 4's: the resistor is a local feedback loop, and it divides the drift by exactly the factor it costs in gain.

6.1 The quiescent point, and the divider that loads itself

An amplifier is a transistor held at an operating point with a signal moving it about. This section is the holding still.

6.1.1 Three numbers and a line

The **quiescent point** is where a stage sits with no signal applied: collector current, collector-emitter voltage, base-emitter voltage. The third is almost always about 0.65 V, so in practice it is two.

The load line of §5.1.3 constrains them:

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

Where on that line to sit is the first design decision. A switch used the two ends; an amplifier wants room to move both ways, so it sits where the headroom above and below matches the wanted swing. Too near the supply clips the positive half of the output, too near saturation the negative half, and both waste the stage.

6.1.2 Divider bias

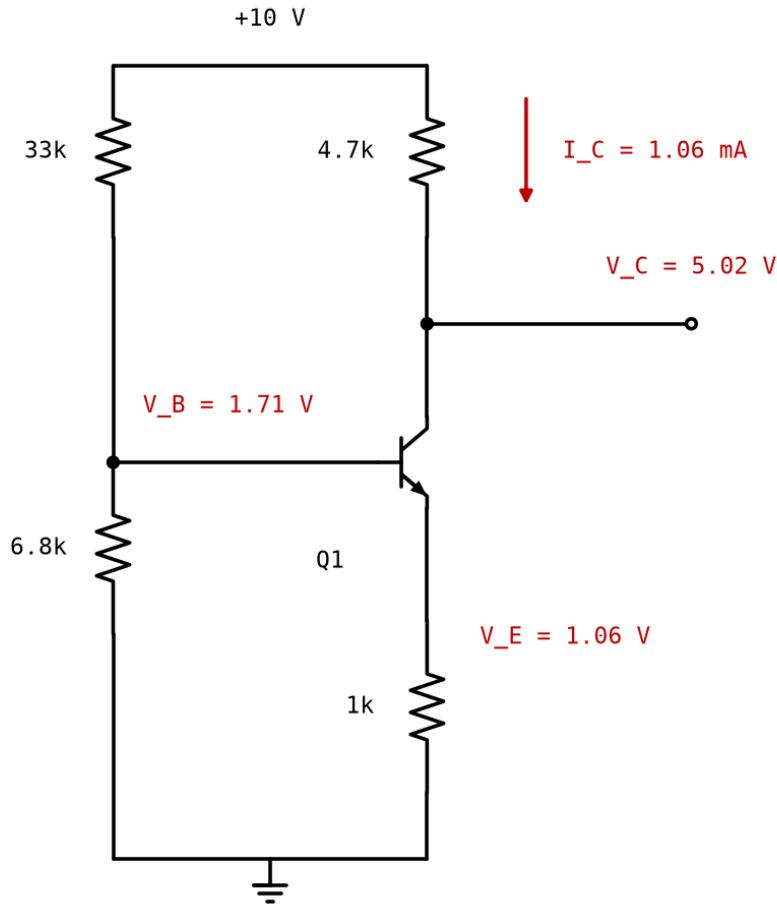


Figure 6.1 A common-emitter stage biased from a 10 V supply through a 33 kilohm and 6.8 kilohm divider, with 4.7 kilohm in the collector and 1 kilohm in the emitter, and the base, emitter and collector voltages marked.

A divider sets the base voltage, the emitter follows one diode drop below, and the emitter resistor turns that voltage into a current:

$$V_B = V_{CC} \frac{R_2}{R_1 + R_2}, \quad V_E = V_B - V_{BE}, \quad I_C \approx \frac{V_E}{R_E}$$

For the values in the figure, 33 kilohm over 6.8 kilohm on 10 V:

$$V_B = 1.71 \text{ V}, \quad V_E = 1.06 \text{ V}, \quad I_C = 1.06 \text{ mA}$$

Those expressions assume nothing is drawing current from the divider. Something is.

6.1.3 The base current loads the divider

$$I_B \approx \frac{I_C}{\beta} = \frac{1 \text{ mA}}{50} = 20 \mu\text{A}$$

as a first estimate, and it flows out of the divider through its Thevenin resistance of 5.64 kilohm (§1.1.5), so the base voltage droops. Solved to convergence the base current settles at 18.7 microamps:

$$\Delta V_B = I_B R_{th} = 18.7 \mu\text{A} \times 5.64 \text{ k}\Omega = 105 \text{ mV}$$

The base sits at **1.60 V**, not 1.71, and the collector current is **0.934 mA**, not 1.06. The emitter resistor carries the *emitter* current, 0.953 mA; the collector gets $\beta/(\beta + 1)$ of it, and at $\beta = 50$ that two per cent is worth keeping in a calculation whose whole subject is the base current.

A 12 per cent error, and Chapter 1's loading arithmetic for the fourth time. A divider loses a third of its output to a 10 kilohm load, a filter corner moves, two filter sections move each other's poles, and here a bias point moves 12 per cent. Chapter 8 (Chapter 8) and Chapter 10 (Chapter 10) are the fifth and the sixth.

It is also self-referential, which is why it needs solving rather than substituting: the base current depends on the collector current, which depends on the base voltage, which depends on the base current. Two iterations by hand converge; a solver does not notice the difficulty.

6.1.4 Stiffness

Make the divider carry much more current than the base takes, so that the base current is a small perturbation on it. That ratio is the divider's **stiffness**:

$$\text{stiffness} = \frac{I_{\text{divider}}}{I_B}$$

Here the divider carries 251 microamps and the base takes 18.7, so the stiffness is **13.4**. The usual rule of thumb is ten.

And the rule of thumb is why the error was 12 per cent. A stiffness of ten accepts roughly one part in ten of the droop, which lands as ten-ish per cent in the collector current. The cure is a stiffer divider and the cost is supply current, one decade for one decade.

Stiffness	Divider current here	Error in I_C
13.4, as designed	251 microamps	12 per cent
50	1.0 mA	4.6 per cent
200	4.1 mA	2.6 per cent

Those currents are not the first row scaled up, and the reason is §6.1.3 a second time: a stiffer divider droops less, so the collector current climbs back towards 1.06 mA and the base current climbs with it. Scaling 251 microamps by 50/13.4 gives 0.93 mA, and 0.93 mA is a stiffness of 46. **What holds exactly** is $I_{divider} = I_C \times \text{stiffness} / \beta$, because the base current is I_C / β by definition.

Stiffness attacks only the R_{th} / β term, so the last row does not reach one per cent. What is left is the $1 / \beta$ by which the emitter current exceeds the collector current, and no divider however stiff can touch it. At 4.1 mA the divider burns four times the current the stage amplifies with, which is where a designer changes bias arrangement instead.

The alternative that removes the problem is to make the base current negligible with a Darlington or a MOSFET, which Chapter 8 and Chapter 9 both do for exactly this reason.

6.1.5 What this section is blind to

- **Beta's spread.** The droop above used $\beta = 50$. A device with $\beta = 200$ droops a quarter as much, so the quiescent current moves with beta. The stiffness requirement makes that dependence small and never removes it.
- **The signal.** Everything here is DC. What the stage does to a signal is Chapter 7, and it depends on the operating point found here.
- **Temperature.** The whole of §6.2.

6.2 Thermal drift, the emitter resistor, and what to build

The problem the emitter resistor solves, an argument for it that does not work, and the one that does.

6.2.1 The problem

A transistor's base-emitter voltage falls about 2 mV per degree at a fixed collector current, because the saturation current rises steeply with temperature and drags V_{BE} down.

Turn that around. **At a fixed base-emitter voltage the collector current rises about 8 per cent per degree**, because $e^{2/26} - 1 = 0.08$.

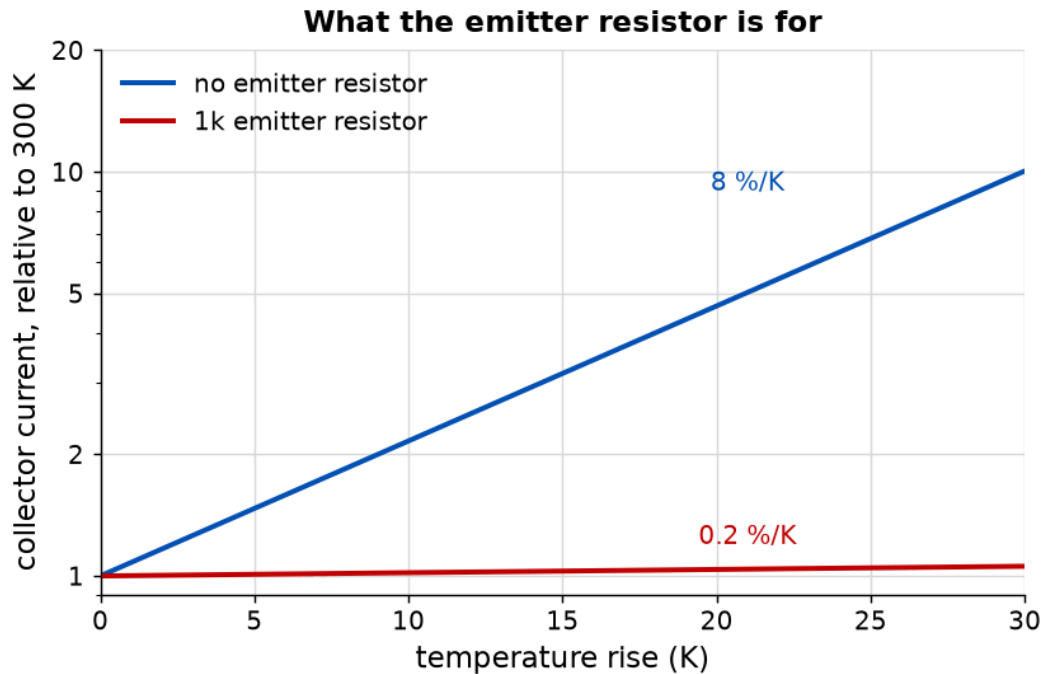


Figure 6.2 Collector current against temperature rise, on a logarithmic vertical axis, for a stage with no emitter resistor and one with a one kilohm emitter resistor. The undegenerated curve climbs by a factor of ten over thirty degrees; the degenerated one is nearly flat.

Over a thirty-degree rise that is a factor of ten. A stage biased at 1 mA on a bench at 20 degrees is at 10 mA inside an enclosure at 50, sitting hard against one end of its load line.

6.2.2 An argument that does not survive its own numbers

A tempting way to explain what the emitter resistor does:

1. A one degree rise lifts the collector current by about 10 per cent, from 1.06 mA to 1.17.
2. The emitter voltage rises with it, by a tenth of the 1.06 V it was already dropping, so by about 100 mV.
3. The base is held by the divider, so the base-emitter voltage falls by that same 100 mV, from 0.65 V to about 0.55.
4. A lower base-emitter voltage means less current, so the operating point is restored.

Every step follows from the one before it, the conclusion is correct, and the argument is wrong.

The 10 per cent is right and the rest does not follow. 100 mV below 0.65 V, at 60 mV a decade:

$$\frac{I_C(0.55)}{I_C(0.65)} = e^{-0.100/0.026} = \frac{1}{47}$$

The collector current would fall by a factor of 47, not by 10 per cent. The circuit never reaches that state: long before the emitter has risen 100 mV the current has been pulled back. The argument describes a system that overshoots wildly and calls it stable. What is wrong is not the physics but the size of the step: **it applies an open-loop change to a closed-loop circuit.**

6.2.3 What actually happens

The emitter resistor is a **local feedback loop**, and the right way to analyse it is Chapter 4's.

The base is held by a stiff divider. The disturbance is V_{BE} drifting down at 2 mV per degree, equivalent to the base being driven 2 mV *up*. The emitter follows within a few millivolts, so the emitter voltage rises 2 mV, and

$$\Delta I_C = \frac{2 \text{ mV}}{R_E} = \frac{2 \text{ mV}}{1 \text{ k}\Omega} = 2 \mu\text{A}$$

On 0.93 mA that is **0.21 per cent per degree**, against 8 per cent without the resistor, and the base-emitter voltage moves by about 2 mV rather than 100. **The suppression factor is the loop gain:**

$$1 + \frac{R_E}{r_e} = 1 + \frac{1000}{28} = 37$$

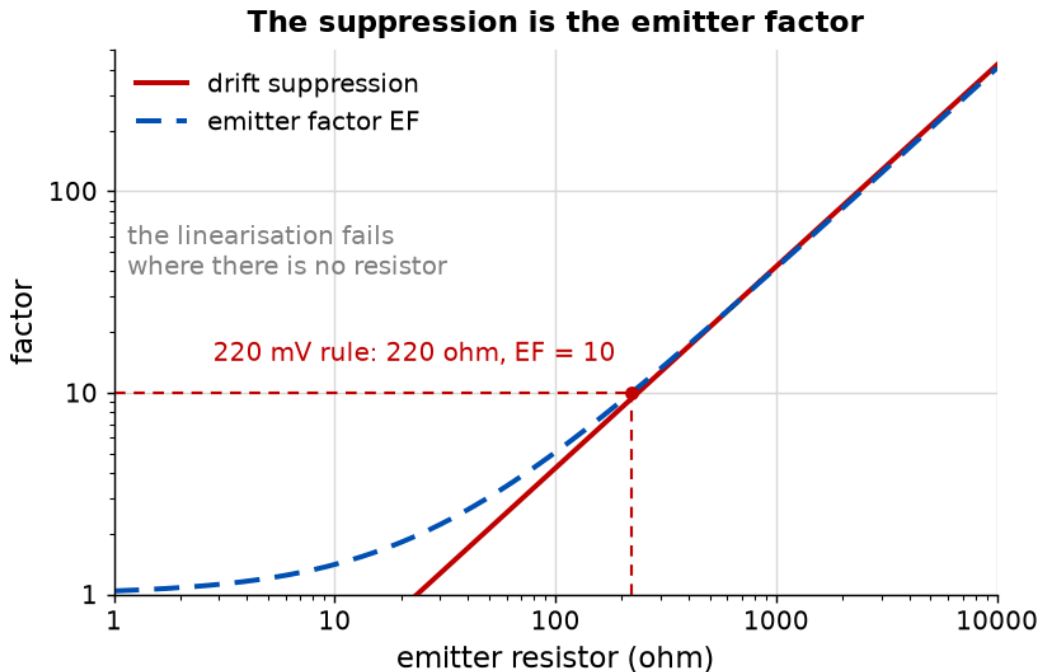


Figure 6.3 Drift suppression and the emitter factor plotted against the emitter resistor on logarithmic axes, the two curves lying on top of each other above about fifty ohms, with the 220 mV design point marked at 220 ohms and a factor of ten.

That number is the emitter factor, which Chapter 7 introduces as the thing that decides gain. The emitter resistor divides the drift by exactly the factor by which it reduces the gain: **stability costs gain, one for one**. B.2's story of voltages in sequence has no place to put a loop gain.

6.2.4 The 220 millivolt rule

Enough emitter resistance for the suppression wanted and no more, since every ohm costs gain. A good rule is to drop about 220 mV across it:

$$R_E = \frac{220 \text{ mV}}{I_C}$$

At 1 mA that is 220 ohm, an E12 value exactly; at 10 mA it is 22 ohm; at 20 mA, 11 ohm. The rule is chosen so that the answer lands on the E12 grid across the decade of currents this book uses.

$$EF = \frac{r_e + R_E}{r_e} = \frac{26 + 220}{26} = 9.5$$

A decade of drift suppression bought with a decade of gain. **22 mV would give 1.85**, barely any suppression; **2.2 V would give 85**, and a stage divided by 85 is not an amplifier worth having.

6.2.5 What to build

ael/bias/point.hpp

```
namespace ael::bias
{
    struct Point
    {
        double baseVoltage{};
        double emitterVoltage{};
        double collectorVoltage{};
        double collectorCurrent{};
    };

    /// The quiescent point of a divider-biased stage, including the base current's ↵
    ↵ droop.
    [[nodiscard]] Point quiescentPoint(double supply, double upper, double lower, ↵
    ↵ double emitter,
    ↵ double collector, double beta = 50.0, double ↵
    ↵ vbe = 0.65) noexcept;

    /// Divider current over base current. Ten is the usual rule of thumb.
    [[nodiscard]] double stiffness(double supply, double upper, double lower,
    ↵ double collectorCurrent, double beta = 50.0) ↵
    ↵ noexcept;

    [[nodiscard]] double driftWithoutDegeneration() noexcept;
    [[nodiscard]] double driftWithDegeneration(double collectorCurrent, double ↵
    ↵ emitterResistor) noexcept;
    [[nodiscard]] double driftSuppression(double collectorCurrent, double ↵
    ↵ emitterResistor) noexcept;
    [[nodiscard]] double degenerationResistor(double collectorCurrent, double drop = ↵
    ↵ 0.220) noexcept;
}
```

quiescentPoint must account for the droop of §6.1.3. It is self-referential, so solve it: two or three fixed-point iterations converge, and a closed form exists if you prefer.

Put the emitter current, not the collector current, through the emitter resistor. Once the base current is in the calculation it is in both places, so $I_E = (\beta + 1)I_B$ and $emitterVoltage =$

$I_E R_E$. At $\beta = 50$ the difference is two per cent, 19 mV here, and the shipped suite checks it to a tenth of a millivolt.

Temperature in ael/device/bjt.hpp

Add a temperature in kelvin to Parameters, defaulting to 300.15, and make both the thermal voltage and the saturation current depend on it:

$$V_T = \frac{kT}{q}, \quad I_S(T) = I_S(T_0) \left(\frac{T}{T_0} \right)^3 \exp \left[\frac{E_g q}{k} \left(\frac{1}{T_0} - \frac{1}{T} \right) \right]$$

with $E_g = 1.11$ eV. **That is where the -2 mV per degree comes from; it is not put in by hand.**

The two together give about -1.77 mV per degree at 1 mA, not exactly -2 . The round figure is what every textbook quotes and what to calculate with; the physics gives a number that depends on the current, and the Cross-check is partly about that difference.

What good looks like

About sixty lines. The temperature addition is six.

6.2.6 What this section is blind to

- **Self-heating.** The transistor's own dissipation raises its temperature, so drift is partly a loop through the package. Negligible here; dominant in Chapter 8's output stage.
- **The divider's own temperature coefficient.** Resistors drift about 100 parts per million per degree, a hundredth of the transistor, and are ignored.
- **Beta's temperature coefficient.** It rises about 0.5 per cent per degree, moving the droop of §6.1.3: second order on first.

6.3 Review

What you should be able to do now

After this chapter, you should be able to:

- Find the quiescent point of a divider-biased stage, including the base current's effect.
- Say what divider stiffness is, choose it, and say what the usual rule of thumb costs.
- State the thermal drift of a base-emitter voltage, and the collector current drift it implies at a fixed base voltage.
- Explain what an emitter resistor does, in terms of feedback rather than in terms of a story about voltages moving.
- Compute the drift suppression, and notice it is the same number as the gain reduction.
- Apply the 220 mV rule and say where it comes from.
- Recognise an argument that is self-contradictory, from its own numbers.

Questions to test yourself

You should be able to answer:

1. A divider is designed for 1.71 V and measures 1.60 V with the transistor connected. Nothing is broken. What is happening, and which Chapter 1 result predicts it?
2. Divider current ten times base current is the usual rule. What error in the collector current does that rule accept?
3. At a fixed base voltage the collector current rises 8 per cent per degree. Where does that number come from, and which of V_T or I_S contributes most of it?
4. An emitter resistor reduces drift and reduces gain. By what factor each, and why is that not a coincidence?
5. The 220 mV rule puts about a fifth of a volt across the emitter resistor. What would 22 mV give, and what would 2.2 V give?
6. Somebody argues that when the current rises 10 per cent, the base-emitter voltage falls from 0.65 V to 0.55 V and pulls it back. Find the contradiction.
7. Your stage is biased at 1 mA and drifts 0.2 per cent per degree. What would it drift at 10 mA with the same emitter resistor?

Further reading

- *The Art of Electronics*, Horowitz and Hill, chapter 2, sections on biasing, for a second treatment of the same material.

Looking ahead

- Chapter 7 is the centre of the book. The small-signal model, the intrinsic emitter resistance the whole of Part 2 is written in terms of, and the emitter factor, including the node it actually belongs to.

6.4 Exercises

Eight, ending with the Cross-check. That one has four legs, and one of them is a tempting argument you are asked to break.

How to check your work. A `CODE` exercise is judged by the suite that ships with it: run `make test` at the root of the companion repository, with `AEL_DIR` pointing at your toolkit. A suite you have not started is reported as `SKIP` rather than as a failure, so the command is worth running from the first day. The hand calculations and designs are checked against the toolkit functions each one names.

The solutions are not in this book. They are published in full in the companion repository, in `lectures/L06/appendix`, including the plausible wrong answer where there is one. Read

them once you have your own answers, including the ones you are unsure about, because being unsure and right is a different thing from being unsure and wrong and the solutions say which is which.

Exercise 6.1 The operating point

RECALL

1. Name the three numbers that describe a quiescent point, and say why it is usually two.
2. Write the load line of a stage with both a collector and an emitter resistor.
3. Where on it does an amplifier sit, and what decides “where”?
4. A stage is biased so that the collector sits at 0.5 V above the emitter. What is wrong with it, and what would the output look like?

Exercise 6.2 Stiffness

RECALL

1. Define divider stiffness, and give the usual rule of thumb.
2. What error in the collector current does that rule of thumb accept?
3. What does buying a factor of ten more stiffness cost?
4. Name two ways of making the base current negligible instead, and say which chapters use them.

Exercise 6.3 The quiescent point, twice

HAND CALCULATION

For the stage in §6.1.2: 10 V, 33 kilohm over 6.8 kilohm, 1 kilohm emitter, 4.7 kilohm collector, $\beta = 50$.

1. The base voltage, emitter voltage and collector current, ignoring base current.
2. The base current that implies, and the droop it causes across the divider’s Thevenin resistance.
3. The corrected base voltage and collector current. One iteration is enough; say why.
4. The percentage error the first answer made, and the stiffness of this divider.

Check yourself: quiescentPoint, stiffness

Exercise 6.4 Breaking the tempting argument

HAND CALCULATION

The argument in §6.2.2 claims a 10 per cent current rise drops V_{BE} from 0.65 V to 0.55 V, which then restores the current.

1. What collector current does $V_{BE} = 0.55$ V actually give, relative to 0.65 V?
2. Is that consistent with “the current rose 10 per cent”?
3. Identify the step in the argument that is wrong. It is not the 10 per cent.
4. Write the correct version in two sentences, using the word “loop”.

Check yourself: currents

Exercise 6.5 Bias a stage to a specification

DESIGN

Design a common-emitter stage: 12 V supply, 2 mA quiescent collector current, the collector sitting at half the supply, and drift below 0.3 per cent per degree.

1. Choose the collector resistor from the quiescent voltage requirement.
2. Choose the emitter resistor from the drift requirement, then check it against the 220 mV rule.
3. Choose the divider for a stiffness of at least 20, in E12 values.
4. State the quiescent point your design actually achieves, including the base-current droop.
5. State the emitter factor, and therefore what this design has given up in gain.

Check yourself: quiescentPoint, stiffness, degenerationResistor, emitter_factor

Exercise 6.6 The bias point

CODE

Implement `ael::bias` to the specification in §6.2.5.

`quiescentPoint` is the only one that needs thought, because it is self-referential: the base current depends on the collector current which depends on the base voltage which depends on the base current. Iterate it. Two or three passes converge to well under a per cent.

Exercise 6.7 Temperature in the device

CODE

Add temperature to `ael::device::bjt::Parameters` and make both the thermal voltage and the saturation current depend on it.

Then check the model against the number everyone quotes: at 1 mA and fixed collector current, the base-emitter voltage should fall by about 1.8 mV per degree. **If yours gives exactly 2.0, you have put the coefficient in by hand rather than letting it emerge**, and the Cross-check below will not work.

Exercise 6.8 The drift, four ways**CROSS-CHECK**

The stage of §6.1.2. How much does its collector current change per degree?

1. **By the tempting argument.** Follow §6.2.2 to its conclusion and state what it predicts. You will not be able to.
2. **By hand, linearised.** The base-emitter voltage drifts -2 mV per degree at a held base, so the emitter voltage rises 2 mV and the current rises by 2 mV over the emitter resistor.
3. **By your solver.** Sweep the temperature from 27 to 37 degrees and measure.
4. **With the emitter resistor removed,** by the solver, for comparison.

What to expect

Leg 1 does not produce a number. Followed honestly it says the current falls by a factor of 47, which contradicts the 10 per cent rise it started from. That is the exercise: an argument can be qualitatively right, easy to believe, and still not close.

Leg 2 gives about 0.21 per cent per degree.

Leg 3 gives about 0.17 per cent per degree, roughly 20 to 30 per cent below leg 2.

That gap is real and has two named causes, and neither is an error:

- The -2 mV per degree is a round figure. The physics gives -1.77 mV per degree at 1 mA, and the coefficient itself depends on the operating current.
- Leg 2 holds the base voltage fixed. The solver lets it droop further as the current rises, because §6.1.3's base current rises with it. That is a second loop the hand calculation does not contain.

Leg 4 gives about 7 per cent per degree, forty times worse, and that is the number the whole chapter exists to prevent.

The ratio between legs 3 and 4 should be about the emitter factor, and checking that is the real content of the Cross-check. If it is not, the suppression is coming from somewhere other than where the theory says.

If leg 3 gives exactly leg 2's answer, the temperature coefficient was put into the model by hand instead of emerging from $I_S(T)$ and $V_T(T)$. That is worth catching: a model that agrees with the hand calculation exactly is a model that contains the hand calculation.

Small-signal analysis: r_e , the emitter factor, and the cascode

The centre of the book, and the model the whole of Part 2 is built on.

IN THIS CHAPTER

- Small-signal as a straight line through an operating point, and everything that discards.
- Building the small-signal schematic: what to short, what to delete, what to replace.
- The intrinsic emitter resistance, and where the 26 mV comes from.
- Gain, input resistance and output resistance, each from one walk around a loop.
- The emitter factor: one number for what degeneration costs and what it buys.
- Which resistance that factor actually multiplies, which is not the obvious one.
- The Early effect, and the current-mirror load that stops the boost being thrown away.
- The source factor and r_s : the same results, transferred to the MOSFET by substitution.
- The Miller effect, and the cascode as the answer to it.
- Building ael : : ssm, checked against the solver on every stage.

Everything before this chapter was preparation; everything after it is this applied. If one chapter is worth rereading, it is this one.

A transistor is exponential. Over a small enough excursion an exponential is a straight line, and the slope of that line at the operating point is all a signal ever sees. That slope has units of conductance, and its reciprocal is r_e , 26 ohm at 1 mA. **Every gain, every input resistance and every output resistance in the rest of the book is written in terms of it**, and each of the three comes from one walk around one loop, done the same way every time.

The organising idea is the **emitter factor**, $EF = (r_e + R_E)/r_e$: one number that answers two questions at once, by what factor the gain falls and by what factor a resistance rises. The first half is exactly right. **The second half is attached to the wrong node in most treatments**, and the difference is a factor of ten. Getting it right is worth doing carefully, because it is what explains the current-mirror loads that would otherwise appear from nowhere in Chapters 9 and 10.

Two more results close the chapter, and neither needs new machinery. The **MOSFET** arrives by naming $r_s = 1/g_m$, after which every result above transfers by substitution, with input resistance the one exception. And the **Miller effect** turns four picofarads into 1.5 nanofarads and rolls a stage off at a hundred kilohertz, for which the answer is a **cascode** — which turns out to be a degenerated stage whose degeneration resistor is another transistor's r_o .

7.1 The small-signal model, and the three results

The centre of the book. One resistance, one construction, three results, and the same method every time.

7.1.1 A straight line through a curve

An amplifier does not use the whole exponential. It sits at an operating point, found in Chapter 6 (Chapter 6), and the signal moves it a little way either side. **Over a small enough excursion, any smooth curve is a straight line**, and that slope is the whole small-signal model.

$$g_m = \left. \frac{dI_C}{dV_{BE}} \right|_Q = \frac{I_C}{V_T}$$

How small is small enough. The exponential's scale is V_T , 26 mV, so a few millivolts is linear to a fraction of a per cent and 26 mV is not linear at all. **A stage handling 10 mV of base signal is already distorting noticeably**, and that is the mechanism Chapter 4's feedback divided down.

What linearising discards: clipping, distortion, the fact that the device turns off, and any signal large enough to move the operating point.

7.1.2 The one resistance

$$r_e = \frac{V_T}{I_C} = \frac{26 \text{ mV}}{I_C}$$

At 1 mA it is 26 ohm.

It is not a resistor. It is the slope of the device's own exponential at the operating point, depending on nothing but current and temperature. It cannot be bought or specified, and if the stage is switched off it does not become large: it ceases to exist with the operating point.

This book uses it in preference to g_m throughout, a resistance in series with the emitter being easier to reason about than a transconductance. The translation is $g_m = 1/r_e$.

7.1.3 Building the small-signal schematic

Four rules, applied in order:

1. **Every DC source becomes a short to ground.** A rail that does not move carries no signal.
2. **Every coupling and bypass capacitor becomes a short.** That is what they were chosen for.
3. **The bias network disappears** wherever it is now in parallel with something much smaller.
4. **The transistor becomes r_e from base to emitter, and a current source from collector to emitter** carrying the current that r_e passes.

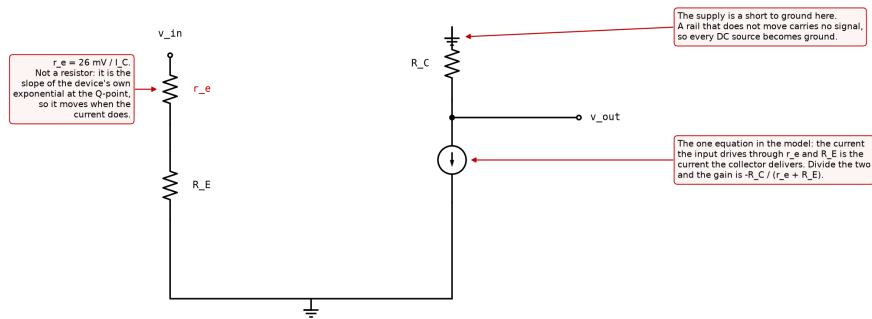


Figure 7.1 The small-signal schematic of a common-emitter stage: the input drives r_e in series with the emitter resistor to ground, and on the output side the supply rail is drawn as a ground with the collector resistor descending to the output node, where a current source draws the collector current to ground.

The result has no transistor in it: one resistance, one current source, the external resistors. **The one equation** links its halves, that the current the input drives through r_e and R_E is the current the collector source delivers. Everything else is bookkeeping.

7.1.4 Three results, one method

Gain. The input drives $v_{in} / (r_e + R_E)$ through the emitter branch; that current comes out of the collector and through R_C , giving $-iR_C$ at the output.

$$A_v = -\frac{R_C}{r_e + R_E}$$

The minus sign is real: more base voltage means more collector current means a lower collector voltage. For 10 kilohm and 1 mA with no emitter resistor that is -385 ; with 234 ohm, -38.5 .

Input resistance. The base draws the emitter current divided by β , at the voltage across the emitter branch:

$$Z_{in(base)} = \beta (r_e + R_E)$$

At 1 mA with 234 ohm and $\beta = 50$, 13 kilohm. **This is the one result in Part 2 that depends on beta**, as §5.1.2 warned, so an input resistance is always a range. The stage's actual input resistance is that in parallel with the bias divider, usually the smaller of the two.

Output resistance is the subject of §7.2, because it is the one result here that the obvious answer gets wrong.

7.1.5 What the emitter resistor does to the gain

Dividing the two gain expressions:

$$\frac{A_v(\text{no } R_E)}{A_v(\text{with } R_E)} = \frac{r_e + R_E}{r_e}$$

That ratio is the **emitter factor**, and the gain falls by exactly it.

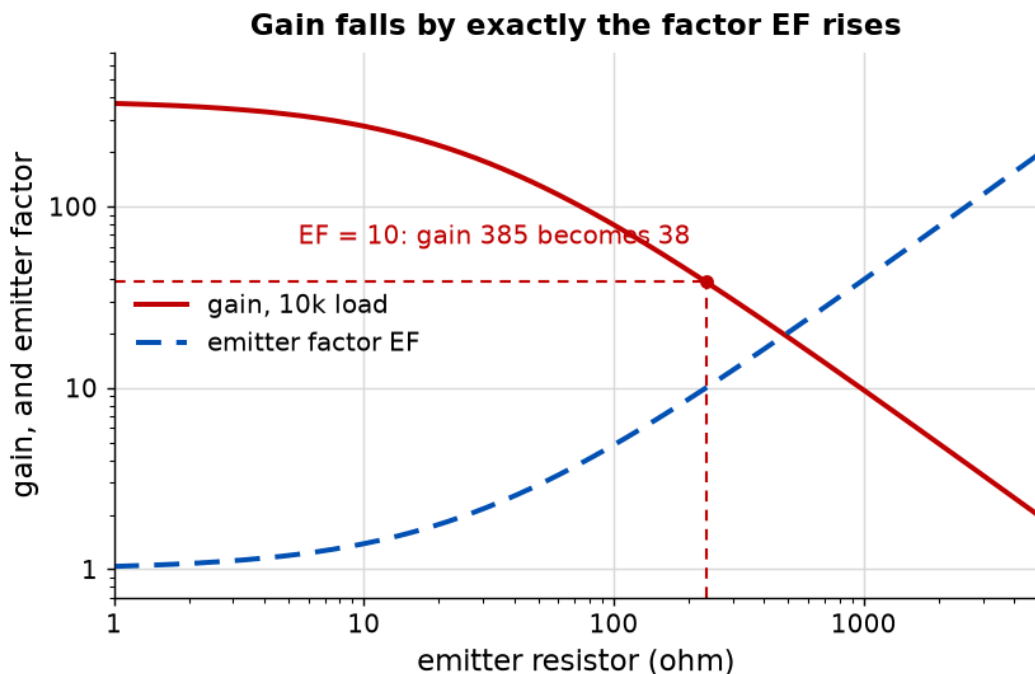


Figure 7.2 Gain and the emitter factor plotted against the emitter resistor on logarithmic axes, the gain falling as the factor rises and the two crossing, with the 234 ohm design point marked where the gain of 385 has become 38.5.

An emitter resistor costs gain one for one. What it buys was Chapter 6's subject, thermal stability, and what else it buys is §7.2's.

Bypassing it with a capacitor recovers the gain at signal frequencies while keeping the DC stability, which is why almost every discrete common-emitter stage has one. It also throws away the distortion reduction, because that was feedback and the capacitor removed it at exactly the signal frequencies.

7.1.6 What this section is blind to

- **Everything nonlinear.** Distortion, clipping and slew rate are outside a model built by assuming a straight line.
- **Capacitance.** Nothing here has a frequency in it. The Miller effect of §7.2.5 is where that starts.
- **The Early effect**, so far. r_o arrives in §7.2.2 and changes the output resistance results and nothing else.
- **Noise.** The model says what a stage does to a signal, not what signal the stage adds itself.

7.2 The emitter factor, and where it belongs

The organising idea of this treatment, the node it belongs to, and the three circuits that follow from getting that right.

7.2.1 The emitter factor

$$EF = \frac{r_e + R_E}{r_e}$$

The factor by which a degeneration resistor raises the *total* emitter resistance, answering two questions at once: by what factor does the **gain fall**, and the **resistance it presents rise**?

The gain half is exactly right (§7.1.5). The second half is the one to be careful about: the obvious reading, that the *stage's output resistance* rises by EF , is wrong for an arithmetic reason (§7.2.3). Chapter 6's 220 mV rule gives $EF \approx 10$.

7.2.2 The Early effect and r_o

Extending the collector-base depletion region shortens the base, raising the collector current slightly with collector voltage. One parameter, the Early voltage:

$$r_o = \frac{V_A}{I_C}$$

With $V_A = 100$ V and 1 mA that is 100 kilohm.

Ignorable until now, at under a per cent of a resistively loaded stage's gain. From here it is the subject of the next three sections.

7.2.3 Where the emitter factor actually belongs

The tempting way to write the stage's output resistance is

$$R_{out} \approx R_C \cdot EF$$

That cannot be right, and the reason is arithmetic. The output resistance is R_C in parallel with what the transistor presents there, and a parallel combination is smaller than either part. **No degeneration can raise a stage's output resistance above R_C** , yet $R_C \cdot EF$ is ten times it.

What degeneration does raise is the resistance **looking into the collector**, with R_C removed:

$$R_{into\ collector} = r_o [1 + g_m(R_E \parallel r_\pi)] + (R_E \parallel r_\pi)$$

very nearly $r_o \cdot EF$: at 1 mA with 234 ohm, 863 kilohm against 1000. So

$$R_{out} = R_C \parallel R_{into\ collector} = 10\text{ k} \parallel 863\text{ k} = 9.89\text{ k}\Omega$$

against **9.09 kilohm** for the same stage with no emitter resistor.

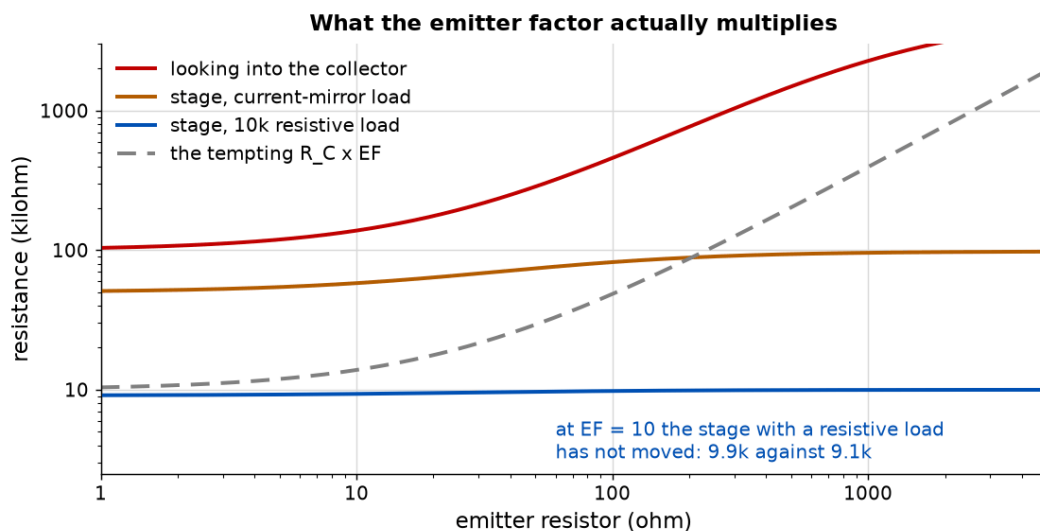


Figure 7.3 Four curves against the emitter resistor on logarithmic axes: the resistance looking into the collector rising steeply, the stage with a current-mirror load rising to a plateau, the stage with a ten kilohm resistive load lying flat, and the tempting R_C times EF rising far above all of them.

With a resistive load, degeneration buys 9 per cent of output resistance, not a factor of ten. The boost is real and it is invisible, because R_C swamps it.

7.2.4 Which is exactly why a current-mirror load exists

If R_C throws the boost away, stop using R_C . A **current mirror** load presents its own r_o , 100 kilohm rather than 10.

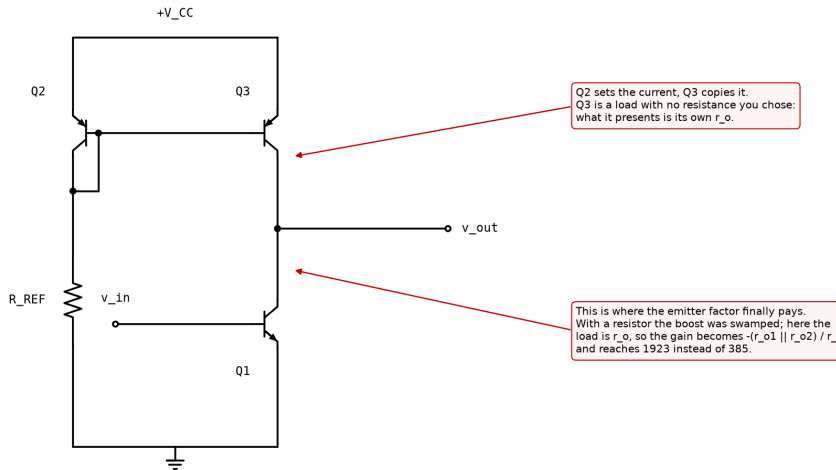


Figure 7.4 A common-emitter stage with a PNP current mirror as its load: a diode-connected reference transistor setting the current, its base tied to the output transistor whose collector feeds the amplifier's collector, with the reference resistor in the left leg.

Load	Output resistance, no R_E	With R_E for $EF = 10$
10 kilohm resistor	9.09 kilohm	9.89 kilohm
Current mirror	50 kilohm	89.6 kilohm

The degeneration now nearly doubles the output resistance, and so the gain. **A mirror load raises gain for two independent reasons**, easy to run together:

- **Its own r_o is large**, loading the output node with 100 kilohm instead of 10. Nothing to do with degeneration.
- **It stops the degeneration boost being swamped**. This is the emitter factor finally paying.

The resistively loaded gain of 385 becomes 1923, five times more, from the first mechanism alone.

7.2.5 Miller and the cascode

The collector-base capacitance bridges input to output, and the output moves the other way by the gain, so the input sees

$$C_{in} = C_{bc}(1 + |A_v|)$$

Four picofarads across a stage with a gain of 385 is **1.5 nanofarads** at the input.

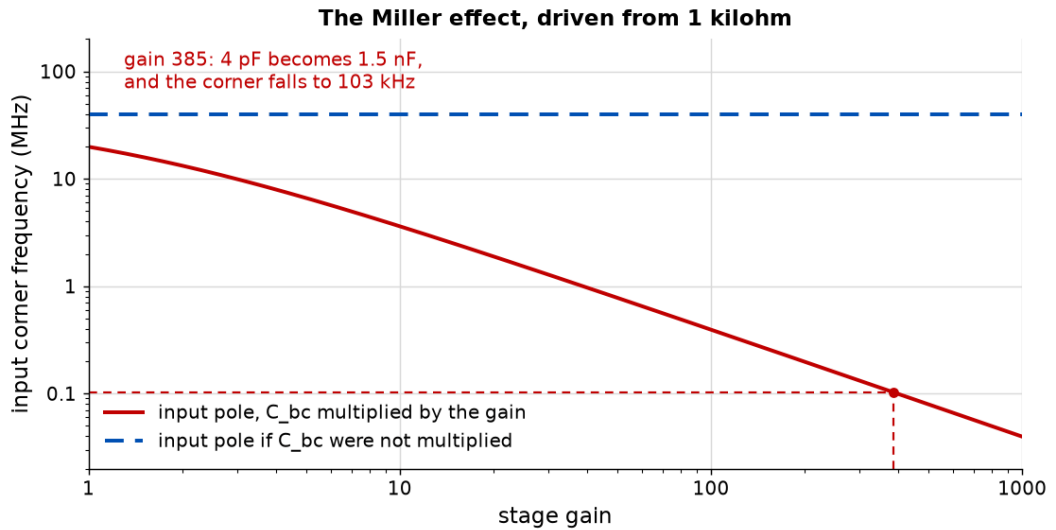


Figure 7.5 Input corner frequency against stage gain on logarithmic axes, falling as the gain rises, against a horizontal line showing where the corner would be if the capacitance were not multiplied.

Driven from 1 kilohm, a corner at **103 kHz** on a device good to hundreds of megahertz. **The Miller effect, not the transistor, limits an ordinary common-emitter stage.**

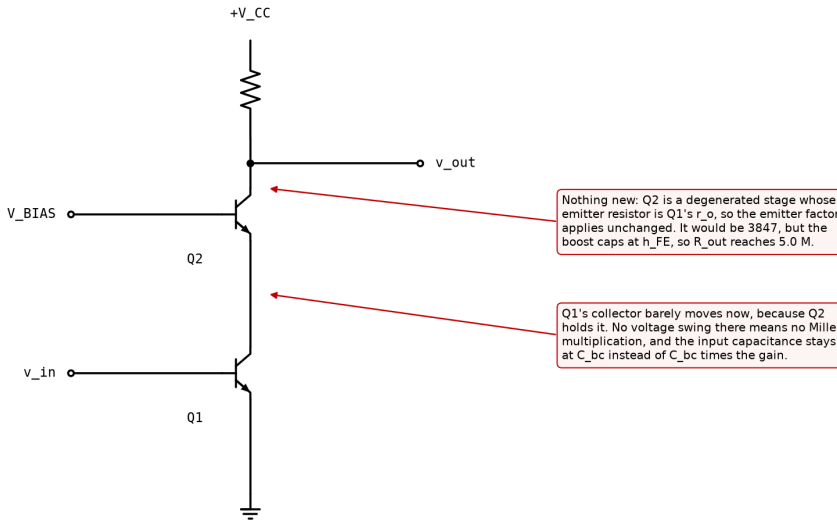


Figure 7.6 A cascode: a common-emitter transistor whose collector feeds the emitter of a second transistor held at a fixed base voltage, with the load resistor on the upper collector and the output taken there.

The cascode is the answer, and a stage you have already analysed. A second transistor above the first, base held fixed, keeps the lower collector at a constant voltage: **no swing there means no Miller multiplication.**

$$R_{out(cascode)} = r_o [1 + g_m(r_o \parallel r_\pi)] + \dots \approx \beta r_o = 5 \text{ M}\Omega$$

That is §7.2.3's expression with $R_E = r_o$: a cascode is a degenerated stage whose degeneration resistor is another transistor's r_o . The emitter factor that would apply is 3847, but the boost caps at β because r_π shunts the degeneration, reaching 5.04 megohm against a βr_o ceiling of 5.00. **No new machinery.**

7.2.6 The MOSFET, in one substitution

No textbook gives the MOSFET a quantity behaving as r_e does, so this book names one:

$$r_s = \frac{1}{g_m}$$

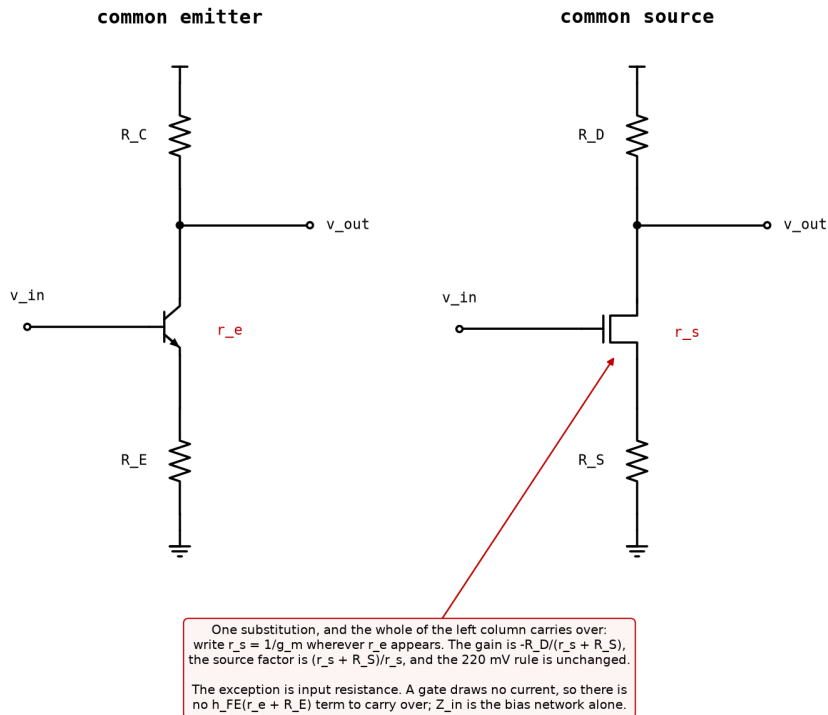


Figure 7.7 Two identical stages side by side, a common-emitter BJT and a common-source MOSFET, with the same supply, load and degeneration resistors, labelled r_e and r_s respectively, and a note that one substitution carries every result across except input resistance.

Every result here transfers by writing r_s for r_e ; the 220 mV rule is unchanged:

$$A_v = -\frac{R_D}{r_s + R_S}, \quad SF = \frac{r_s + R_S}{r_s}$$

The one exception is input resistance. A gate draws no current, so there is no $\beta(r_e + R_E)$ term: a common-source stage's input resistance is its bias network. That is why Chapter 9's input stage and Chapter 10's inter-stage buffer are both MOSFETs.

Why $SF \approx 2$ where $EF \approx 10$, both from 220 mV: r_s at 1 mA is 250 ohm where r_e is 26, transconductance being ten times lower (§5.2.2).

7.2.7 What to build

The Early effect in ael/device/bjt.hpp

Multiply the forward transport current by $(1 + V_{CE}/V_A)$. That is what makes r_o finite and §7.2.3 measurable.

Multiply the collector current only. The base current does not get the factor, so h_{FE} comes out as $\beta_F(1 + V_{CE}/V_A)$ and rises with collector voltage, as a datasheet's curve does.

That placement is not a detail. Putting the factor on the whole transport current, so both currents scale and beta stays flat, is tempting: it leaves Chapter 5's contract untouched. It also makes the base inject extra current into the emitter as the collector rises, which is degeneration through R_E , so the resistance into the collector comes out **18 per cent high** and *above* the emitter factor instead of below it. Your own solver would then contradict §7.2.3. One test catches that version.

One number the chapter does not quote. Differentiating gives $r_o = (V_A + V_{CE})/I_C$: 105 kilohm at a collector 5 V up, not 100. Every closed form here uses the round figure, and that 5 per cent is one of the things the Cross-check's legs disagree about.

ael/ssm/model.hpp

Function	Returns
<code>intrinsicEmitterResistance(ic)</code>	V_T/I_C .
<code>intrinsicSourceResistance(gm)</code>	$1/g_m$.
<code>emitterFactor(ic, re)</code>	$(r_e + R_E)/r_e$.
<code>sourceFactor(gm, rs)</code>	The same, for a MOSFET.
<code>gain(rc, ic, re)</code>	$-R_C/(r_e + R_E)$.
<code>inputResistance(ic, re, beta)</code>	$\beta(r_e + R_E)$.
<code>resistanceIntoCollector(ic, re, beta, va)</code>	The expression of §7.2.3.
<code>outputResistance(rc, ic, re, beta, va)</code>	That, in parallel with the load.
<code>cascodeOutputResistance(ic, beta, va)</code>	<code>resistanceIntoCollector</code> with $R_E = r_o$.
<code>millerCapacitance(gain, cbc)</code>	$C(1 + A)$.

cascodeOutputResistance must be implemented by calling resistanceIntoCollector, not by a separate formula. Separate, and the claim that a cascode is degeneration by r_o is an assertion rather than something the code demonstrates.

What good looks like

About seventy lines, of which none is longer than three.

7.2.8 What this section is blind to

- **The body effect**, which raises a MOSFET follower's threshold and costs real gain in Chapter 8.
- **Base resistance.** Ohmic resistance in the base adds to r_e at high current and limits noise.

- **The second Miller capacitance.** The base-emitter capacitance is not multiplied but is far larger, and at high frequency it, not C_{bc} , sets the limit.
- **High frequency, properly.** One capacitance and one pole is a caricature of a device with three capacitances and a transit time.

7.3 Review

What you should be able to do now

After this chapter, you should be able to:

- Draw the small-signal schematic of any stage in this book, and say what each step discarded.
- Write r_e from a collector current without looking it up.
- Derive gain, input resistance and output resistance for a common-emitter stage.
- State the emitter factor, and both of the things it decides.
- Say which node the emitter factor's boost appears at, and why it is invisible at another.
- Explain why a current-mirror load raises the gain by two independent mechanisms.
- Transfer any result to a MOSFET by substituting r_s , and name the one result that does not transfer.
- Compute a Miller capacitance, and say what a cascode does about it.

Questions to test yourself

You should be able to answer:

1. r_e is 26 ohm at 1 mA. Is it a resistor? What happens to it if the stage is switched off?
2. A stage has $R_C = 10$ kilohm and $R_E = 234$ ohm. What is its gain, and what would it be with the emitter resistor bypassed?
3. The emitter factor is ten. Name the two quantities it relates, and say which of them is a property of the transistor rather than of the stage.
4. Why does adding an emitter resistor barely change the output resistance of a resistively loaded stage, when it multiplies the resistance looking into the collector by ten?
5. A current mirror as a load raises the gain for two separate reasons. Name both.
6. A stage with a gain of 385 and 4 pF of collector-base capacitance is driven from 1 kilohm. Where is its input pole?
7. A cascode is often described as a new circuit. In what sense is it a stage you have already analysed?

Further reading

- *The Art of Electronics*, Horowitz and Hill, chapter 2, for the small-signal model, and *Microelectronic Circuits*, Sedra and Smith, if you want the hybrid- π form instead.

Looking ahead

- Chapter 8 is the stage with no voltage gain at all, and why almost every design contains one. Followers, impedance transformation, and the output stage they turn into.

7.4 Exercises

Eight, ending with the Cross-check. That one settles which resistance the emitter factor multiplies, and the answer differs from the obvious one by a factor of ten.

How to check your work. A `CODE` exercise is judged by the suite that ships with it: run `make test` at the root of the companion repository, with `AEL_DIR` pointing at your toolkit. A suite you have not started is reported as `SKIP` rather than as a failure, so the command is worth running from the first day. The hand calculations and designs are checked against the toolkit functions each one names.

The solutions are not in this book. They are published in full in the companion repository, in `lectures/L07/appendix`, including the plausible wrong answer where there is one. Read them once you have your own answers, including the ones you are unsure about, because being unsure and right is a different thing from being unsure and wrong and the solutions say which is which.

Exercise 7.1 What linearising discards

RECALL

1. State r_e , and say whether it is a resistor.
2. How large may a base signal be before the straight-line model is misleading? Give a number and the reason for it.
3. Name three things a small-signal model cannot describe.
4. A stage is switched off. What happens to r_e ?

Exercise 7.2 The four rules

RECALL

1. List the four steps that turn a stage into its small-signal schematic.
2. Why does a supply rail become a ground?
3. A bias divider of 33 kilohm over 6.8 kilohm sits on the base of a stage whose base looks like 13 kilohm. Which dominates the input resistance, and by how much?
4. Bypassing the emitter resistor recovers the gain. What else does it recover, and what does it throw away?

Exercise 7.3 The three results

HAND CALCULATION

A stage at 1 mA with $R_C = 10$ kilohm, $R_E = 234$ ohm, $\beta = 50$.

1. $r_{e'}$ and the emitter factor.
2. The gain, and what it would be with the emitter resistor bypassed.
3. The input resistance looking into the base.
4. The same three at 10 mA with the same resistors. Which of them changed, and which did not?

Check yourself: `intrinsicEmitterResistance`, `emitterFactor`, `gain`, `inputResistance`

Exercise 7.4 Two nodes, two answers

HAND CALCULATION

For the same stage:

1. The resistance looking into the collector, with R_C removed.
2. The stage's output resistance, with R_C in place.
3. The same two with the emitter resistor removed.
4. State, in one sentence each, what degeneration did to each of the two quantities.

Check yourself: `resistanceIntoCollector`, `outputResistance`

Exercise 7.5 A stage with a stated gain and bandwidth

DESIGN

Design a common-emitter stage: 12 V supply, gain of 40 with the emitter resistor unbypassed, and an input corner above 3 MHz when driven from 600 ohm. Take $C_{bc} = 4$ pF.

1. Choose a collector current and a collector resistor.
2. Choose the emitter resistor for the gain.
3. Compute the Miller capacitance and the input corner. Does it meet the requirement?
4. If it does not, say what you would change, and give the two options with their costs.

Check yourself: `gain`, `millerCapacitance`, `intrinsicEmitterResistance`

Exercise 7.6 The small-signal model

CODE

Implement `ael::ssm` to the specification in §7.2.7.

cascodeOutputResistance must call resistanceIntoCollector with $R_E = r_o$, rather than implementing a separate formula. The chapter claims a cascode is degeneration by r_o ; writing it that way makes the code demonstrate the claim instead of restating it.

Exercise 7.7 The Early effect

CODE

Multiply the forward transport current by $(1 + V_{CE}/V_A)$ in `ael::device::bjt`.

Then check it: solve a stage at two collector voltages a volt apart and confirm the collector current changes by about one part in a hundred, which is $1/V_A$ per volt. If the current does not move at all, r_o is infinite and the Cross-check below cannot work.

Exercise 7.8 What the emitter factor multiplies

CROSS-CHECK

A stage at 1 mA with $R_C = 10$ kilohm and $R_E = 234$ ohm, so $EF = 10$. Find its **output resistance**, four ways.

1. **By the tempting rule.** $R_{out} = R_C \cdot EF$.
2. **By the corrected closed form.** R_C in parallel with the resistance looking into the collector.
3. **By your solver.** Perturb the collector node by a small voltage with the input held, and divide the voltage change by the current change.
4. **Then replace the collector resistor with a current mirror** and repeat legs 2 and 3.

What to expect

Leg 1 gives 100 kilohm. Legs 2 and 3 give about 9.9 kilohm. A factor of ten, and it is the largest disagreement in this book.

Leg 1 cannot be right, and you can see it without computing anything. The output resistance is R_C in parallel with something. A parallel combination is smaller than either part. So the answer cannot exceed 10 kilohm, and 100 kilohm does.

Compare against the same stage with no emitter resistor: 9.09 kilohm. So all that degeneration bought, at the cost of a factor of ten in gain, was **9 per cent** of output resistance. The boost is real, it is a factor of ten, and it is happening at a node where R_C hides it.

Leg 4 is where it stops being hidden. With a current mirror the load is r_o , 100 kilohm rather than 10, and the numbers become 50 kilohm without degeneration and **89.6 kilohm** with it. The emitter factor is finally visible, because there is no longer a resistor swamping it.

And that is the point of the correction. It is not only that leg 1's number is wrong; it is that getting the node right explains why mirror loads exist at all. Without it, a mirror load is a technique that appears from nowhere. With it, it is the only way to collect something you have already paid for.

If leg 3 gives exactly 10 kilohm, the Early effect is missing from your device model, so r_o is infinite and the transistor is a perfect current source. Check Exercise 7.7.

If leg 3 gives 100 kilohm, the collector resistor is not in the netlist.

If legs 2 and 3 differ by a few per cent, that is expected: leg 2's closed form uses $R_E \parallel r_\pi$ where the solver uses the full nonlinear device, and the two agree to about the accuracy of β .

Followers, and the output stage they turn into

The stage with no voltage gain, and why almost every design has one.

IN THIS CHAPTER

- The stage with no voltage gain, and why nearly every amplifier contains several.
- Gain just under one, and the shortfall as a design quantity rather than a rounding error.
- Impedance transformation: h_{FE} going in, r_e coming out.
- The 8 ohm problem, and the arithmetic that says one follower is not enough.
- The Darlington, h_{FE}^2 , and the price of leaning on beta.
- Class A, class B, and the dead band between them.
- Class AB, and the 26 millivolt rule that sizes the emitter resistors.
- Why the bias diodes must be bolted to the same heatsink as the output devices.
- The source follower, the body effect, and the one place r_s does not carry across cleanly.
- Building `ael::follower` and `ael::output`.

This chapter is Chapter 7's model applied at the emitter instead of the collector, and almost nothing in it is derived that was not derived there. A **follower** has a voltage gain of 0.97 and is the most-used circuit in the book, because everything it is for is in its two resistances rather than in its one gain: it shows the stage driving it h_{FE} times what is on its emitter, and shows its load a source resistance h_{FE} times smaller.

The reason that matters is one calculation. Take the stage of Chapter 7, with a gain of 38.5 and an output resistance of 9.89 kilohm, and connect a loudspeaker: it delivers **0.08 per cent** of its gain and the amplifier does not work at all. One follower recovers 4 per cent, which is fifty times better and still useless. A **Darlington** recovers 68. **That is the whole reason output stages exist**, and it is Chapter 1's loading arithmetic for the fifth time in this book.

The second half sizes a real output stage. Two followers sharing the work have a dead band 1.3 V wide whose *position*, not its width, is the problem; closing it costs a small idle current, and the rule that sets it makes the emitter factor exactly 2. Then two things that decide whether the stage survives: a thermal loop that can run away, whose fix is not a circuit

change at all but where a component is bolted, and **the place where the constant-drop model finally breaks** — not by a few per cent, as it has until now, but by a factor of sixty, for a reason that is about the calculation rather than about the model.

§8.3 is **reading, not examinable**: what a real power amplifier adds on top of everything here, and no exercise is set on it.

8.1 The follower, and the load it exists to survive

A stage with a voltage gain of one, and the arithmetic that makes it the most-used circuit in the book.

8.1.1 The same model, at the other terminal

A **follower** is Chapter 7's model (§7.1) with the output taken at the emitter instead of the collector, and no collector resistor.

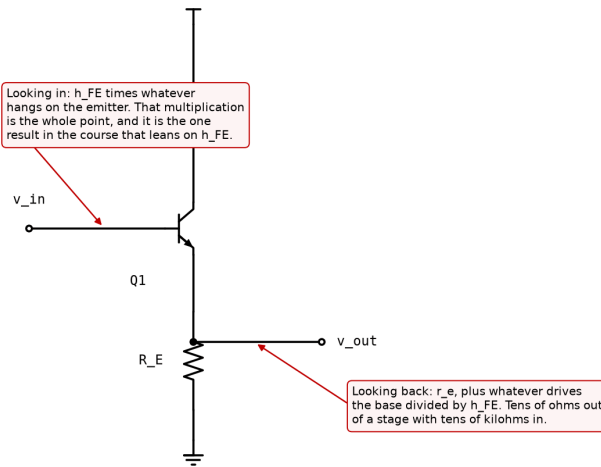


Figure 8.1 An emitter follower: an NPN transistor with its collector tied to the supply rail, its base driven from the input, and an emitter resistor to ground with the output taken at the emitter, annotated at both ends with what the stage does to impedance.

1. **The output is at the emitter**, not the collector.
2. **There is no collector resistor**, because a follower does not turn current into voltage.
3. In high-voltage work the collector is sometimes grounded rather than tied to the rail, which protects the device and changes nothing below.

8.1.2 Gain, and where the missing part went

The input drives r_e and R_E in series and the output is taken across R_E alone: a divider.

$$G = \frac{R_E}{r_e + R_E}$$

Always less than one, and the shortfall is $r_e/(r_e + R_E)$. At 1 mA into 10 kilohm, 0.997. At 120 mA into an 8 ohm loudspeaker, 0.974.

The ratio r_e/R_E is the only thing a follower's gain depends on, and there is one lever on it, $r_e = V_T/I_C$. **The gain is a question about current and nothing else: a follower that is not good enough is a follower that is not biased hard enough.**

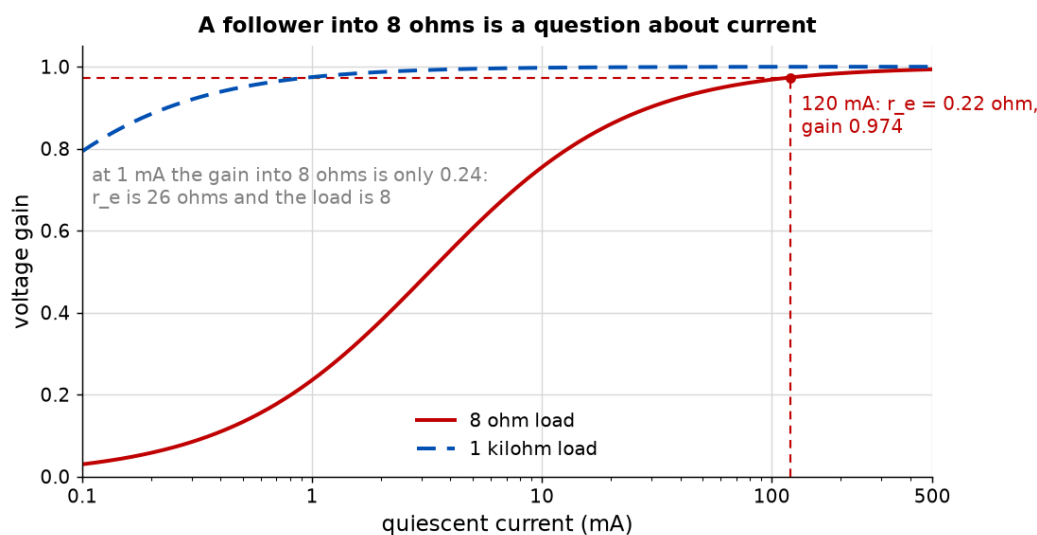


Figure 8.2 Follower gain against quiescent current on a logarithmic axis, for an 8 ohm load and a 1 kilohm load, the 8 ohm curve climbing from about 0.03 at 100 microamps through 0.24 at 1 milliamp towards one, with the 120 milliamp design point marked.

Read the 8 ohm curve at 1 mA: the gain is 0.24. Not 0.99, not 0.9, a quarter: r_e is 26 ohm and the load is 8, so three quarters of the signal drops inside the transistor. That is why §8.2 exists, and why an output stage idles at 120 mA.

8.1.3 The two resistances, which are the point

Looking into the base, the base current is the emitter current divided by h_{FE} :

$$Z_{in} = h_{FE} (r_e + R_E)$$

Looking back into the emitter, the transistor presents r_e and whatever drives the base appears divided by h_{FE} :

$$Z_{out} = r_e + \frac{R_{source}}{h_{FE}}$$

At 1 mA driven from 1 kilohm, $26 + 20 = 46$ ohm.

Together the stage is an impedance transformer: it shows the driving stage h_{FE} times its emitter load, and the load h_{FE} times less than its source, at a signal gain of 0.97.

Both results are proportional to h_{FE} , as §5.1.2 and §7.1.4 warned. §8.1.5 is where it starts to hurt.

8.1.4 The eight ohm problem

Take Chapter 7's stage: 1 mA, 10 kilohm collector, 234 ohm emitter, open-circuit gain 38.5, output resistance 9.89 kilohm. Connect a loudspeaker.

$$G_{loaded} = G \cdot \frac{R_{load}}{R_{out} + R_{load}} = 38.5 \times \frac{8}{9885 + 8}$$

0.031. The stage keeps 0.08 per cent of its gain, and the amplifier does not work at all.

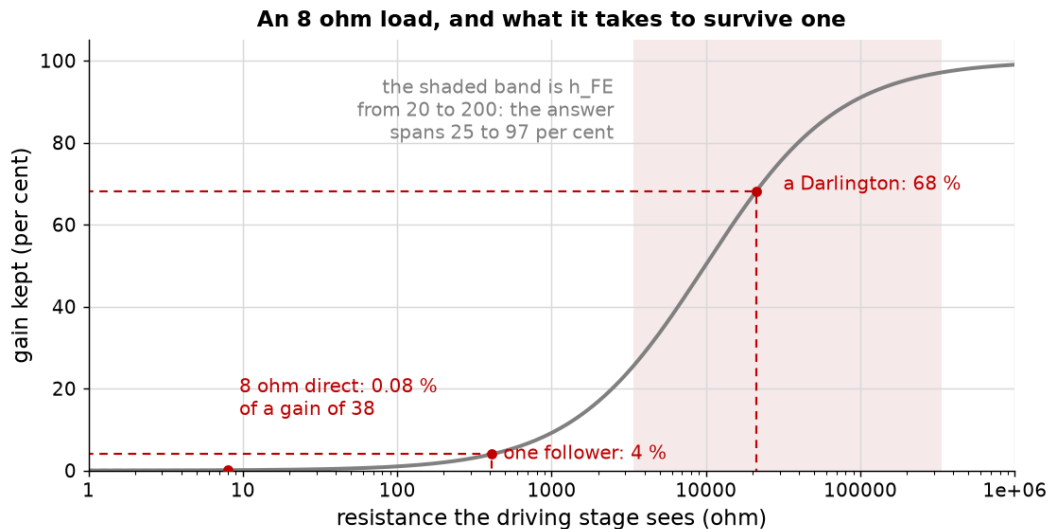


Figure 8.3 Gain kept against the resistance the driving stage sees, on a logarithmic axis, rising from nearly nothing at 8 ohms through 4 per cent at 411 ohms to 68 per cent at 20 kilohms, with a shaded band showing where the Darlington point moves as h_{FE} runs from 20 to 200.

Now put a follower in between, biased at 120 mA: $h_{FE}(r_e + R_{load}) = 411$ ohm, so the driving stage sees 411 instead of 8 and keeps **4 per cent**. **Fifty times better and still useless**. **One follower is not enough**, by two orders of magnitude.

8.1.5 The Darlington, and the price of beta squared

Two transistors, the first driving the base of the second, behave as one device with a current gain of $h_{FE1}h_{FE2}$ and two base-emitter drops. A.3's multiplication then happens twice:

$$Z_{in} = h_{FE}^2 (2r_e + R_{load}) = 2500 \times 8.43 = 21.1 \text{ k}\Omega$$

The resistance the square multiplies is the pair's own $2r_e$, not one device's. The input device runs at the output device's base current, so its r_e is h_{FE} times larger, and is seen through the output device's gain, contributing r_e again. The two h_{FE} s cancel: **a Darlington's effective emitter resistance is exactly twice a single transistor's**. The driving stage now sees 21.1 kilohm against its own 9.89 and keeps **68 per cent**, which is why essentially every discrete output stage is a Darlington.

Now the bill. At $h_{FE} = 20$ the input resistance is 3.4 kilohm and the stage keeps 25 per cent; at 200 it is 337 kilohm and keeps 97. **The same design from the same part number varies by nearly four in gain across an ordinary production spread.**

That is a design **plus** something that removes the dependence, namely Chapter 4's feedback (Chapter 4): at a loop gain of a few hundred the closed-loop gain stops caring. Chapter 10 builds the loop.

The Darlington's other costs, real and not solved by feedback:

- **Two base-emitter drops**, so the swing is 1.3 V short of each rail rather than 0.65.
- **Slow turn-off**, the second base's charge having nowhere to go. A resistor from that base to the emitter is the standard fix, present in every practical circuit.
- **Twice the thermal drift** in the bias, which §8.2.4 deals with.
- **Twice the intrinsic emitter resistance**, so 0.949 into 8 ohms where one follower gives 0.974.

8.1.6 The fifth time

The same arithmetic since Chapter 1. This chapter is the fifth; Chapter 10 (Chapter 10) is the last:

Where	What loaded what	Cost
§1.1.6	10 kilohm on a divider	a third of the output
Chapter 2 (Chapter 2)	source resistance on a filter	the corner moves 6.6 times
Chapter 3 (Chapter 3)	two filter sections on each other	the corner moves 1.72 times
§6.1.3	base current on a bias divider	12 per cent of I_C
This chapter, here	a loudspeaker on a voltage amplifier	99.92 per cent of the gain
Chapter 10, when it is written	each stage on the one before	11 dB of open-loop gain

The subtraction is identical every time. What changes is how much it costs, and here it costs everything.

8.1.7 What this section is blind to

- **Frequency.** A follower with a capacitive load and an inductive source can oscillate, which is what the base resistor in every real output stage is for.
- **Large signals.** 0.974 is a small-signal number at one operating point. A follower driven at its rails runs out of current before voltage; §8.2 is the fix.
- **The second transistor's operating point** in the Darlington: it runs at the first's base current, so its r_e is fifty times larger. In the code, not in the arithmetic above.
- **Power.** 120 mA at 20 V is 2.4 W, a heatsink question rather than a small-signal one.

8.2 The output stage, and the model that finally breaks

Three classes, one dead band, one sizing rule, and the first place in this book where the constant-drop model is not merely approximate but wrong by a factor of sixty.

8.2.1 Class A: the follower you already have

Bias the follower so it conducts for the whole signal. Nothing turns off, so there is no discontinuity in the transfer curve and the distortion is the exponential's curvature.

Its vice is arithmetic. To swing $\pm V$ the device must idle at V/R_{load} , the current it must be able to *stop* delivering at the negative peak, so it burns V^2/R_{load} at idle. Best case **25 per cent efficiency at full output**, falling to zero as the signal falls: **a 100 W class-A amplifier draws 400 W playing silence.**

8.2.2 Class B, and the dead band

Two devices: an NPN follower pushes into the load on positive half-cycles, a PNP pulls on negative ones, and neither conducts at idle. Idle dissipation goes to zero and efficiency rises to $\pi/4$, **78.5 per cent. And it does not work**, for a reason unrelated to efficiency.

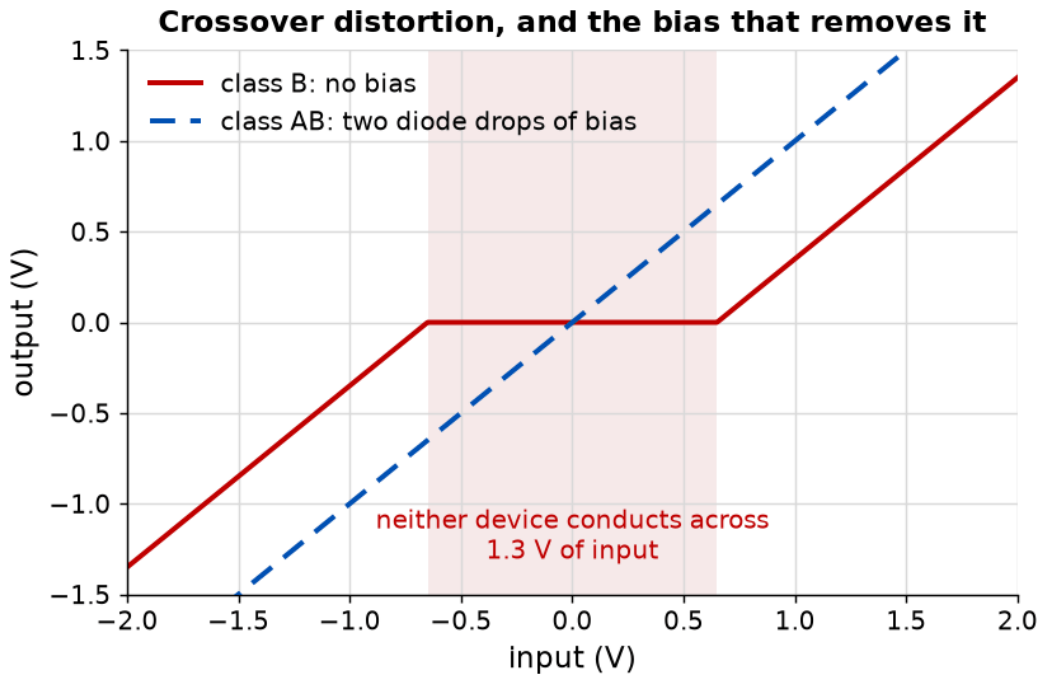


Figure 8.4 The transfer curve of a complementary pair, with the unbiased case flat across a dead band 1.3 volts wide centred on zero, and the biased case a straight line through the origin, the dead band shaded.

Neither conducts until its base-emitter junction is forward biased, so the output stays at zero across an input band **two diode drops wide**, about 1.3 V: **crossover distortion**.

Its size is not the problem; its position is. 1.3 V in a stage swinging 30 V is 4 per cent, tolerable-sounding, but it sits at the origin and music spends most of its time there. **The distortion is worst on quiet passages and vanishes at full output**, the opposite of every other distortion mechanism, and audible far below the level its percentage suggests.

8.2.3 Class AB, and the 26 millivolt rule

Bias both devices so a small current flows at idle: the dead band closes and idle dissipation is small rather than zero. Essentially every audio output stage is **class AB**.

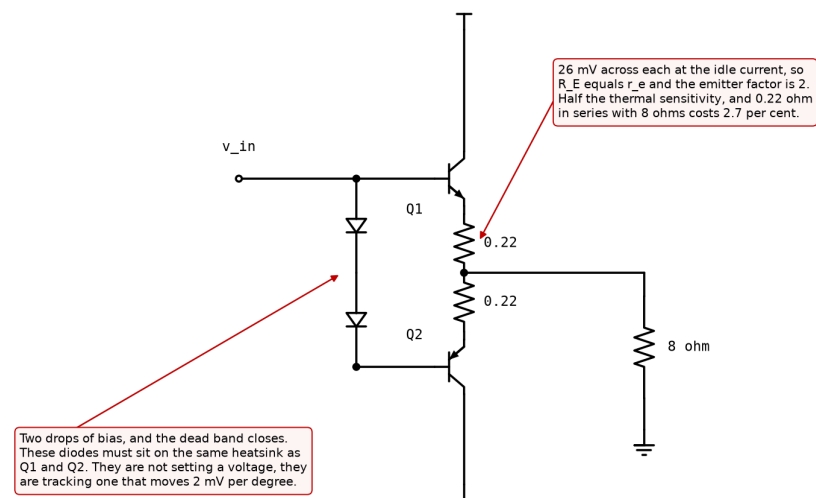


Figure 8.5 A class-AB output stage: an NPN and a PNP follower with their emitters facing each other through two small resistors to the output node, two series diodes between the bases setting the bias, and an eight ohm load to ground.

How much idle current? Enough to close the dead band and no more, stated on the resistors:

$$R_E = \frac{V_T}{I_q} = \frac{26 \text{ mV}}{I_q}$$

At the 120 mA a stage of this size idles at, 0.217 ohm, and 0.22 is an E12 value.

The rule is $R_E = r_e$ the other way round, putting V_T across R_E , so Chapter 7's emitter factor is exactly 2. That is what it is choosing:

	Value
r_e at 120 mA	0.217 ohm
R_E , E12	0.22 ohm
Emitter factor	2.0
Drop across R_E at idle	26 mV
Cost in the load: 0.22 in series with 8	2.7 per cent

Two per cent of the signal for half the thermal sensitivity. At 260 mV the emitter factor is 11 and 2.2 ohm in series with 8 has thrown away a fifth of the output power; at 2.6 mV the resistors do nothing. 26 mV is where those costs cross, the same trade as Chapter 6's 220 mV rule.

8.2.4 Thermal runaway, and a fix that looks like nothing

An output transistor dissipates power, warms, drops its V_{BE} by 2 mV per degree, and draws more idle current at fixed bias. **That loop can run away, and a runaway output stage destroys itself in seconds.** Both junctions drift, so a fixed bias is 4 mV per degree too generous, landing across $2(r_e + R_E)$:

$$\frac{1}{I_q} \frac{dI_q}{dT} = \frac{2|dV_{BE}/dT|}{2(r_e + R_E)I_q}$$

	Per degree	Over a 30 degree rise
No emitter resistors	7.7 per cent	a factor of 9
With the 26 mV rule	3.8 per cent	a factor of 3

A factor of three is still a runaway. The emitter resistors halve the problem and do not solve it: the 26 mV rule buys margin, not stability.

The fix is to make the bias voltage drift too. Bolt the bias diodes to the output devices' heatsink and their forward voltages fall 2 mV per degree as well, cancelling to first order.

So the diodes are not setting a voltage. They are tracking one. Read them as "about 1.3 volts of bias" and mount them on the board next to the driver, and the stage is correct at 25 degrees and destroys itself at 60. **This is the most common way a first output stage fails, and the schematic does not show it: the two circuits are drawn identically.**

Real circuits use a V_{BE} multiplier: one transistor with a divider from collector to base, giving an adjustable multiple of a base-emitter drop, on the heatsink. Adjustable, because the tracking is never exact (§8.2.8).

8.2.5 Where the constant-drop model finally breaks

Since Chapter 5, $V_{BE} \approx 0.65$ V has served for bias points at about 1 per cent error. **Here it fails completely.** What bias voltage does the stage above need to idle at 120 mA?

$$V_{bias} = 2(V_{BE}(I_q) + I_q R_E)$$

V_{BE} at 120 mA is **0.783 V**, not 0.65: it is $V_T \ln(I_C/I_S)$ and 120 mA is **2.2 decades** above the 0.72 mA where 0.65 V is right. The resistors add 26 mV each:

$$V_{bias} = 2(0.783 + 0.026) = 1.619 \text{ V}$$

against **1.353 V** from the constant-drop model: a 16 per cent error in the voltage.

Run it backwards, which is where the size shows. Apply 1.353 V and ask what idle current results. Not 120 mA reduced by a sixth: **1.96 mA**.

A factor of 61. The stage is biased into class B, the dead band is open, and the amplifier sounds broken.

Why here and not before. Every earlier use computed a **current from a voltage across a resistor**, where 133 mV inside a subtraction yielding 1.06 V is 12 per cent, entering linearly. Here it computes a **current from a voltage across a junction**, and 133 mV inside an exponential of scale 26 mV is $e^{5.1}$, a factor of 166. **The emitter resistors turn 166 into 61:** at 120 mA they carry 26 of the 133 mV, at 2 mA nothing.

The lesson is not that 0.65 is wrong, but that a model's error is a property of the calculation it is used in. **And it explains the circuit:** two signal diodes at a few milliamps give 1.40 V and an idle current of 4.8 mA, which is why a real stage uses an adjustable multiplier or diodes at the output current.

8.2.6 The source follower, and the one thing that does not carry across

§7.2.6's substitution of r_s for r_e holds here too:

$$G = \frac{R_S}{r_s + R_S}, \quad Z_{out} = r_s$$

with the input resistance the bias network, because a gate draws no current. **A strict improvement:** the h_{FE}^2 of §8.1.5, factor-of-four spread and all, becomes a resistor you chose. **The cost is the gain:** at 1 mA r_s is 250 ohm where r_e is 26, so into 1 kilohm the source follower gives 0.80 against 0.97.

Ten times the current recovers only three, g_m going as $\sqrt{I_D}$ (§5.2.2). **A MOSFET output stage must be biased much harder for the same gain**, which is why output stages are usually bipolar.

And here the substitution is not exact. A MOSFET's threshold rises when its source sits above its body, as in an integrated CMOS follower it always does. That **body effect** is a source-to-body transconductance g_{mb} , 10 to 30 per cent of the main one, appearing as unasked-for degeneration:

$$G = \frac{R_S}{r_s(1 + \chi) + R_S}$$

With $\chi = 0.2$, 0.80 becomes **0.77**: why a discrete source follower and a CMOS one do not give the same answer from the same equation.

8.2.7 What to build

ael/follower/stage.hpp

Function	Returns
<code>gain(collectorCurrent, load)</code>	$R/(r_e + R)$.
<code>outputResistance(collectorCurrent, sourceResistance, beta)</code>	$r_e + R_{src}/h_{FE}$.
<code>inputResistance(collectorCurrent, load, beta)</code>	$h_{FE}(r_e + R)$.
<code>darlingtonEmitterResistance(outputCurrent)</code>	$2V_T/I$. Twice one device's, and h_{FE} cancels.
<code>darlingtonGain(outputCurrent, load)</code>	$R/(2r_e + R)$: a little worse than a single follower.
<code>darlingtonInputResistance(outputCurrent, load, beta)</code>	$h_{FE}^2(2r_e + R)$.
<code>loadedGain(unloadedGain, outputResistance, load)</code>	The divider of §8.1.4.
<code>intrinsicEmitterResistance(collectorCurrent)</code>	V_T/I_C . The quantity Chapter 7 named, re-exported here so that a follower result can be written without reaching into <code>ael::ssm</code> .

`loadedGain` is three characters of arithmetic and the most-used function in Chapter 10, which calls it eleven times. Write it here rather than inline at each call site.

ael/output/classab.hpp

Function	Returns
<code>degenerationResistor(idleCurrent)</code>	V_T/I_q : the 26 mV rule.
<code>biasVoltage(idleCurrent, emitterResistor)</code>	$2(V_{BE}(I_q) + I_q R_E)$, with V_{BE} from the exponential .
<code>idleCurrent(bias, emitterResistor)</code>	The inverse. There is no closed form.
<code>driftPerDegree(idleCurrent, emitterResistor)</code>	The fraction of §8.2.4.
<code>transfer(input, bias, vbeOn, load)</code>	The dead band of §8.2.2.

idleCurrent must not call a constant-drop approximation. It inverts an equation with an exponential and a linear term that do not separate, and the whole of §8.2.5 is about what happens when you pretend they do. Bisect in the logarithm, or use Newton with the limiter you wrote in Chapter 4.

What good looks like

About fifty lines. `idleCurrent` is the only one with a loop in it.

8.2.8 What this section is blind to

- **The tracking is never exact.** The generator runs at a few milliamps and the outputs at 120, and §6.2.5 established that the temperature coefficient depends on current. It also sits on the heatsink rather than the die, lagging by seconds. Hence the adjustment.
- **Self-heating**, the loop that makes the drift a runaway. §6.2.6 deferred it here; still deferred, because a thermal model is a course of its own.
- **Safe operating area.** 5 A with 40 V across it kills a small transistor whatever the average power, which is what §8.3's protection circuitry is for.
- **A real loudspeaker**, a complex impedance varying by five across the audio band.
- **Distortion, quantitatively.** Crossover distortion is audible; how many parts per million a given bias leaves is not said.

8.3 What a real power amplifier adds

Reading, not examinable. No exercises, nothing in the test suite. Everything in §8.2 still holds; this is the shape of what it leaves out.

8.3.1 The power arithmetic, done properly

50 W into 8 ohm. Confusing rms with peak here propagates into the rail voltage and everything after it.

$$V_{rms} = \sqrt{PR} = \sqrt{50 \times 8} = 20 \text{ V}, \quad I_{rms} = 2.5 \text{ A}$$

$$V_{peak} = \sqrt{2} V_{rms} = 28.3 \text{ V}, \quad I_{peak} = 3.54 \text{ A}$$

Calling 2.5 A the peak current is the easy mistake, and it leads to ± 20 V rails, which deliver at most

$$P = \frac{V_{rail}^2}{2R} = \frac{400}{16} = 25 \text{ W}$$

half the power the example is about. A 50 W amplifier into 8 ohm needs ± 28.3 V before any allowance for saturation, the emitter resistors and supply sag; in practice ± 32 V.

	Idle current	Idle dissipation
Class A	3.54 A, the full peak	226 W
Class AB	120 mA	7.7 W

A factor of 29, for a stage that measures very nearly as well. That is the whole argument for class AB.

Maximum theoretical efficiency, ideal devices at full output: class A **25 per cent** resistively loaded and 50 with a current-source load, class B **78.5 per cent**, which is $\pi/4$. Class AB sits just below. Real amplifiers reach the sixties.

8.3.2 Why one Darlington is not the end of it

§8.1.5 reached 20 kilohm with two transistors. At the 3.5 A a 50 W amplifier needs, a power transistor's h_{FE} is nearer 20:

$$I_{base} = \frac{3.54}{20 \times 20} = 8.8 \text{ mA}$$

from a voltage-amplifier stage running at 1 to 10 mA. **That is the whole of its current, so the driver clips before the output does.** A **third** follower gives h_{FE}^3 : the **triple emitter follower**, with a base current of 440 microamps.

The alternative is the **CFP-EF**, a complementary feedback pair driving an emitter follower. Local feedback round two devices makes the output track the input to within one V_{BE} rather than two, and its thermal behaviour is set by the small driver rather than the hot power device, so it needs much less bias tracking (§8.2.4). Its weakness is a tendency to oscillate. **Both exist for one reason:** current gain that does not depend on a hot power device's beta.

8.3.3 The circuits that are only there to prevent failure

None of these affect the gain, and every one is in every commercial amplifier.

- **Base stoppers**, 10 to 100 ohm in series with each output base. A power transistor and its own lead inductance oscillate at tens of megahertz, where nobody is looking. The resistor damps it.
- **The Zobel network**, typically 10 ohm and 100 nF from output to ground. A loudspeaker is inductive above a few kilohertz; the Zobel keeps something resistive there.
- **The output inductor**, a few microhenries wound over a resistor, isolating a capacitive cable whose phase lag would otherwise sit inside the feedback loop.
- **Snubbers** across the rectifier diodes. Reverse recovery steps current into the transformer's leakage inductance, which rings at radio frequency and reaches the output.
- **Over-current protection**, a transistor across each output base-emitter junction, turned on by the emitter resistor's voltage, stealing base drive above a set current. **So the emitter resistors of §8.2.3 do three jobs:** thermal stability, current sharing between paralleled devices, and current sensing.
- **A DC offset detector** and a relay in series with the loudspeaker. A shorted output device puts the full rail across 8 ohm, 128 W into a voice coil rated for a few.

8.3.4 Heat

Dissipation per device peaks at about 40 per cent of full output, not at full output, where the current is high and the voltage across the device is still large. Here about 17 W: $V_{CC}^2/\pi^2 R_L = 13$ W of class-B dissipation plus 4 W from the idle current.

Thermal resistances add in series like electrical ones: junction to case, case to heatsink through the washer, heatsink to air. 1 plus 0.5 plus 1 is 2.5 °C/W, so 17 W lifts the junction 42 °C: 82 °C at a 40 °C ambient, acceptable for a device rated to 150.

Placement matters as much as size. Every output device and the bias generator on the *same* heatsink, close together, because §8.2.4 needs them at the same temperature. **Two devices on separate heatsinks will not share current, and the hotter one takes progressively more of it:** the same runaway one level up.

8.3.5 Why output stages are usually bipolar

The opposite conclusion to the one §8.2.6 reaches for input stages:

	Bipolar	MOSFET
Output resistance	about ten times lower	higher, by r_s/r_e
Distortion	lower	higher without correction
Thermal stability	needs tracking bias	inherently stable above a crossover point
High-frequency stability	needs stoppers and Zobel	far less prone
Drive	current, so beta matters	voltage, but gate charge matters

The two devices swap places depending on which end of the amplifier they are at, for one reason: a MOSFET's transconductance is ten times lower at the same current. At the input that is irrelevant and the infinite gate resistance decisive; at the output, the reverse. A MOSFET stage with an error amplifier can reach bipolar distortion figures, buying robustness rather than performance.

8.3.6 What to take from this section

1. **A power stage is mostly not the amplifier.** The gain path is a handful of transistors; the rest of the schematic is protection, compensation and thermal management.
2. **The failures are thermal and high-frequency**, neither visible in the small-signal analysis this book teaches. A design correct in every equation of Chapter 7 and this chapter can still destroy itself.
3. **The corrections in §8.3.1 are the kind that matter.** An rms mistaken for a peak is a rail 40 per cent too low, and nothing further down recovers it.

8.4 Review

What you should be able to do now

After this chapter, you should be able to:

- Derive a follower's gain, input resistance and output resistance from the same model as Chapter 7.
- Say what a follower costs and what it buys, in ohms rather than in adjectives.
- Compute what a given load does to a given stage, and choose between a follower, a Darlington and a redesign.
- State the three amplifier classes by what conducts when, and give the efficiency of each.
- Explain crossover distortion, and why it is worse than its size suggests.
- Size the emitter resistors of a class-AB stage from the idle current.
- Say why the bias generator must be thermally coupled to the output devices, and what happens if it is not.
- Carry every follower result to a MOSFET, and name the one thing that stops it being exact.

Questions to test yourself

You should be able to answer:

1. A follower's gain is 0.974. Where did the other 2.6 per cent go, and what would you change to get it back?
2. Looking into a follower's base with 8 ohms on its emitter, what do you see? Now say it again with the beta spread included.
3. Why does a stage with 10 kilohms of output resistance lose 99.9 per cent of its gain into a loudspeaker, when the same load costs a follower only 3 per cent?
4. Class B is 78 per cent efficient and class A is 25. Why is class A used at all?
5. A class-AB stage idles at 120 mA. What emitter resistors does the 26 mV rule give, and what is the emitter factor it produces?
6. The bias generator is two silicon diodes. Name the failure that follows from mounting them on the circuit board instead of the heatsink.
7. A source follower's gain is lower than an emitter follower's at the same current. Give the number and the reason.

Further reading

- *The Art of Electronics*, Horowitz and Hill, chapter 2, for followers, and chapter 2x for output stages at more length than this book has room for.

Looking ahead

- Chapter 9 is the stage that subtracts: the differential pair, what its tail decides, and why the current mirror of Chapter 7 turns up again as its load.

8.5 Exercises

Eight, ending with the Cross-check. That one retires a model this book has used since Chapter 5, at the exact calculation where it stops working.

How to check your work. A `CODE` exercise is judged by the suite that ships with it: run `make test` at the root of the companion repository, with `AEL_DIR` pointing at your toolkit. A suite you have not started is reported as `SKIP` rather than as a failure, so the command is worth running from the first day. The hand calculations and designs are checked against the toolkit functions each one names.

The solutions are not in this book. They are published in full in the companion repository, in `lectures/L08/appendix`, including the plausible wrong answer where there is one. Read them once you have your own answers, including the ones you are unsure about, because being unsure and right is a different thing from being unsure and wrong and the solutions say which is which.

Nothing here is set on §8.3, which is reading.

Exercise 8.1 The stage that does nothing

RECALL

1. Write the follower's gain, input resistance and output resistance.
2. The gain is always less than one. Where does the missing part go, and what is the only lever on it?
3. Which two of those three results depend on h_{FE} ?
4. A follower has no voltage gain. Name two distinct things it is nevertheless used for.

Exercise 8.2 The three classes

RECALL

1. Define class A, class B and class AB by what conducts when.
2. Give the maximum theoretical efficiency of class A and of class B.
3. What is crossover distortion, how wide is the band it occupies, and why is it more objectionable than its percentage suggests?
4. Class B has better efficiency and worse distortion than class AB. What does class AB give up to get the distortion back?

Exercise 8.3 A follower at three currents

HAND CALCULATION

An emitter follower drives an 8 ohm loudspeaker. $\beta = 50$.

1. Its gain at 1 mA, at 10 mA and at 120 mA.
2. Its output resistance at each, driven from a 1 kilohm source.
3. The resistance looking into its base at each.
4. One of the three columns is nearly constant. Which, and why?

Check yourself: gain, outputResistance, inputResistance

Exercise 8.4 What the loudspeaker costs

HAND CALCULATION

The Chapter 7 stage: gain 38.5, output resistance 9.89 kilohm.

1. Its gain driving 8 ohm directly.
2. Its gain driving a follower biased at 120 mA, which drives the 8 ohm.
3. The same with a Darlington in place of the follower.
4. Repeat part 3 with $\beta = 20$ and with $\beta = 200$, and say in one sentence what that spread means for the design.
5. A Darlington's effective emitter resistance is twice a single transistor's, and h_{FE} does not appear in that statement. Derive it.

Check yourself: loadedGain, inputResistance, darlingtonInputResistance

Exercise 8.5 A class-AB output stage

DESIGN

A stage is to idle at 100 mA and drive 8 ohm from ± 25 V rails.

1. Size the emitter resistors by the 26 mV rule, and choose E12 values.
2. State the emitter factor you have bought and what fraction of the load your resistors now are.
3. Compute the bias voltage the stage needs, from the exponential rather than from 0.65 V.
4. State the maximum sine power into 8 ohm, ignoring saturation. Then say what you would actually specify the amplifier at, and why the two differ.

Check yourself: degenerationResistor, biasVoltage, nearest_e12

Exercise 8.6 Thermal

DESIGN

The stage from Exercise 8.5, with the bias generator held at a fixed voltage.

1. The fractional change in idle current per degree.
2. What the idle current becomes after a 30 degree rise, if nothing tracks.
3. Recompute part 1 with the emitter resistors removed, and state what they bought.
4. State where the bias generator must be mounted and what it is doing there. One sentence.

Check yourself: `driftPerDegree`

Exercise 8.7 The follower and the output stage

CODE

Implement `ael::follower` and `ael::output` to the specification in §8.2.7.

idleCurrent may not use a constant base-emitter drop. It is the inverse of an equation with an exponential and a linear term in it and they do not separate. Bisect in the logarithm, or use the Newton iteration with the step limiter you wrote in Chapter 4. A version that assumes 0.65 V will pass one test in the shipped suite and fail four.

Exercise 8.8 The bias voltage a class-AB stage actually needs

CROSS-CHECK

The stage of §8.2.3: two complementary followers, 0.22 ohm emitter resistors, idling at 120 mA. Find the bias voltage between the two bases, three ways.

1. **By hand, with the constant-drop model.** Two base-emitter drops at 0.65 V, plus the two resistor drops.
2. **By hand, with the exponential.** $V_{BE} = V_T \ln(I_C/I_S)$ at the idle current, plus the two resistor drops.
3. **By your solver.** You do not need a PNP model for this. **At idle the stage is symmetric:** no current flows in the load, the output node sits at zero, and each device carries the same current from half the bias voltage. So simulate one half, an NPN with its base held at $V_{bias}/2$ and its emitter resistor returned to ground, and bisect on that base voltage until the collector current is 120 mA.

Then run it backwards, which is where the size of the disagreement shows: **apply leg 1's answer to the real stage and ask your solver what idle current results.**

What to expect

Legs 2 and 3 agree to about 1 per cent, at 1.619 and 1.604 V. Leg 1 gives 1.353 V, a quarter of a volt below both.

And 1.353 V applied to the stage gives an idle current of two or three milliamps, not 120. A factor of about fifty. The stage is in class B, the dead band is open, and every quiet passage is distorted.

Legs 2 and 3 differ by 14 mV and their idle currents differ by 31 per cent. Say why, and note that it is the same effect the exercise is about, one order of magnitude down.

Explain, in your own words, why a model that was worth 1 per cent in Chapter 5 and 12 per cent in Chapter 6 is worth a factor of fifty here. The answer is in what is being computed, not in the model.

If your leg 3 needs no adjustment to reach 120 mA, your solver's transistor is using a constant drop somewhere, and you have measured leg 1 twice.

The differential amplifier

The input stage of every operational amplifier, and the first circuit that rejects rather than amplifies.

IN THIS CHAPTER

- The first circuit in the book built to **reject** rather than to amplify.
- The pair, the tail, and the two halves that share it.
- Differential gain: the common-emitter result with a two in it, and where the two comes from.
- Common-mode gain, and the tail acting as a degeneration resistor of twice its value.
- CMRR, and the fact that the collector resistor cancels out of it entirely.
- Why every real pair has a current source in its tail and not a resistor.
- The current-mirror load, and the two independent factors it wins.
- The large-signal transfer: nine millivolts of linearity, and hard limiting beyond.
- Offset, matching, and why almost every modern input stage is MOSFET.
- Building `ael::diffpair`.

A stage with two inputs and one job: amplify the difference, ignore what the two have in common. Everything else follows from that sentence, and from one component neither transistor is connected to at the top — the **tail**. The tail fixes the total current; the inputs decide only how it is divided. When the two inputs move oppositely the tail node does not move at all, and when they move together it has to move with them. **That asymmetry is the circuit**, and both gains fall straight out of Chapter 7's model with nothing new derived.

Rejection then has a surprise in it. Write the ratio of the two gains and **the collector resistor cancels, exactly and completely**: no choice of load improves rejection, and neither does the gain. The only lever is the tail — and because a resistance in the tail implies a voltage across it, 80 dB of rejection from a resistor needs a 520 V supply. That is why the tail is a current source, and it generalises into the most useful sentence in Part 2: **a current source has a large resistance without a large voltage across it**.

The same device then appears as a load, where it wins two independent factors that textbooks routinely run together, and the chapter separates them. Two things the small-signal model cannot see close it: the pair is linear over about nine millivolts and hard-limits

beyond, which is where an amplifier's slew rate comes from; and the offset that mismatch produces, whose largest term is not a mismatch between the transistors at all.

9.1 The pair, the tail, and the two factors of two

The first stage in this book whose purpose is to ignore something.

9.1.1 Two halves and one current

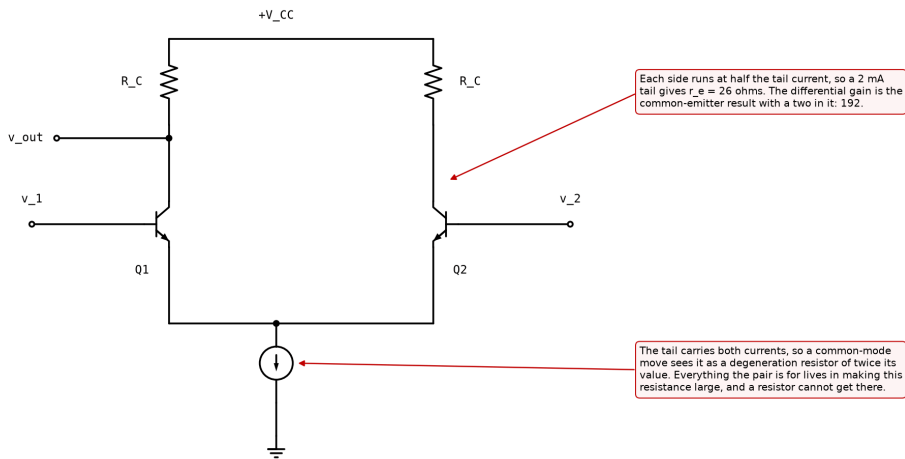


Figure 9.1 A differential pair: two matched NPN transistors with their emitters joined to a tail current source, collector resistors to the positive rail, both bases driven, and the output taken at one collector.

Two matched transistors with their emitters tied together and a **tail** current source below. The tail fixes the total, $I_{C1} + I_{C2} = I_{tail}$; the inputs decide only how it is **divided**. Both properties of the circuit follow from that:

- **A differential input redistributes the current** between the halves. The tail node barely moves, because the sum is unchanged.
- **A common-mode input moves both halves together**, which the tail cannot supply, so the tail node has to move instead.

Each side runs at half the tail current. For a 2 mA tail, 1 mA per side:

$$r_e = \frac{V_T}{I_{tail}/2} = 26 \, \Omega$$

26 ohm, not 13. Reading r_e from the tail current rather than the side current is the commonest slip in this chapter, and it is a factor of two on top of the two below.

9.1.2 Differential gain, and both factors of two

Apply $+v_d/2$ to one base and $-v_d/2$ to the other. The tail node does not move, so each half is a common-emitter stage with its emitter at signal ground: **exactly Chapter 7's circuit**, no new derivation.

$$A_1 = -\frac{R_C}{r_e} \cdot \frac{v_d/2}{v_d} = -\frac{R_C}{2r_e}$$

For 10 kilohm and a 2 mA tail, -192 .

- **The first two is the input split.** A differential input of v_d puts only $v_d/2$ on each base. Unavoidable: it is what differential means.
- **The second two is the output.** Take the difference between the two collectors and the gain is $-R_C/r_e = -385$, the collectors moving oppositely. Take **one** collector and the other half is thrown away.

The second two is recoverable and the first is not. Recovering it is what the current mirror of §9.2.4 does, and one of the two reasons that mirror exists. This book takes the **single-ended** output throughout, because that is what feeds the next stage in an operational amplifier.

9.1.3 Input and output resistance

Looking into either base, Chapter 7's result with the same factor of two:

$$Z_{in(diff)} = 2h_{FE}r_e$$

2.6 kilohm at a 2 mA tail with $\beta = 50$: the two bases in series as far as a differential source is concerned. Low, and the reason §9.2.6 prefers MOSFETs here.

Looking into a collector, Chapter 7's `resistanceIntoCollector` with the tail as the degeneration resistor. With a current-source tail that is enormous, so the output resistance is R_C , or the mirror's r_o when there is a mirror.

And the base current is not zero, which matters more here than anywhere else. Each base draws $I_{tail}/2\beta$, 20 microamps at a 2 mA tail. Those currents flow through whatever drives the inputs, and **if the two source resistances differ, the difference becomes an input voltage the amplifier cannot distinguish from signal**.

9.1.4 What the small-signal model cannot see

Two exponentials dividing one current is a hyperbolic tangent:

$$I_{C1} - I_{C2} = I_{tail} \tanh\left(\frac{v_d}{2V_T}\right)$$

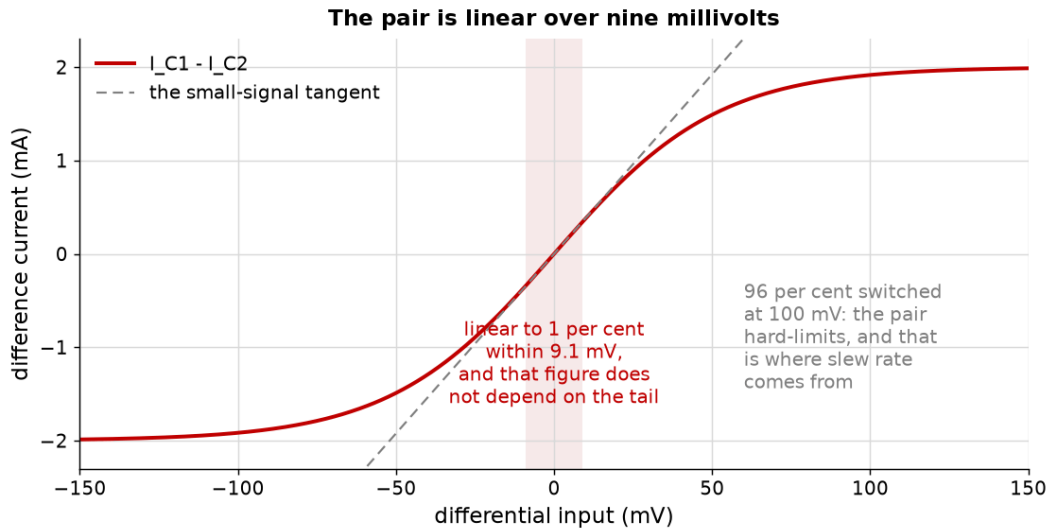


Figure 9.2 The difference between the two collector currents against differential input, an S-shaped curve saturating at plus and minus the tail current, with the small-signal tangent drawn through the origin and the region where the two agree to one per cent shaded.

The pair is linear over nine millivolts. Beyond ± 9.1 mV the tanh has fallen 1 per cent below its tangent, and that is the pair's honest input range.

And that figure does not depend on the tail current at all. The tail scales the whole curve and cancels out of the ratio. **Biasing the pair harder buys gain and no linearity whatever,** which is not what anyone expects and is worth checking in the code.

The pair hard-limits. At ± 100 mV, 96 per cent of the tail has moved to one side and the other is off. Past that the output stops responding: a ceiling, not soft compression.

That ceiling is where slew rate comes from. The pair drives a compensation capacitor, and once fully switched the largest current it can deliver is I_{tail} , so the output ramps at I_{tail}/C and no feedback loop can make it faster. Slew rate is not a small-signal parameter, and this is why.

9.1.5 What this section is blind to

- **Mismatch.** Everything here assumes the two halves are identical. They are not, and §9.2.6 is about what that costs.
- **The tail's own behaviour.** Ideal here; the whole of §9.2 is about it not being.
- **Frequency.** The pair has any stage's Miller problem, mitigated by each half seeing only half the signal. Not treated.
- **Noise,** where a differential input stage earns much of its reputation, and which this book does not cover anywhere.

9.2 Rejection, the mirror, and what limits it

Where the rejection comes from, which is not where most people guess, and the load that wins two independent factors at once.

9.2.1 Common mode, and the tail doubled

Drive both bases with the same v_{cm} : both halves want more current, the tail cannot supply it, so the tail node rises and the current barely changes. **The tail carries the sum of both currents**, so a resistance R_{tail} in the tail behaves as $2R_{tail}$ in one half's own emitter, and it is Chapter 7's degenerated stage unchanged:

$$A_{cm} = -\frac{R_C}{2R_{tail} + r_e}$$

With 10 kilohm in the tail and a 2 mA pair, -0.50 : a gain of *one half* for signals both inputs share, against 192 for signals they do not.

Nothing new has been introduced: the emitter factor of §7.1.5 with $R_E = 2R_{tail}$. The pair is two old circuits wired so one input sees degeneration and the other does not.

9.2.2 CMRR, and the term that cancels

$$CMRR = \frac{A_{dm}}{A_{cm}} = \frac{R_C/2r_e}{R_C/(2R_{tail} + r_e)} = \frac{2R_{tail} + r_e}{2r_e}$$

Look at what happened to R_C . It cancelled, exactly and completely.

$$CMRR \approx \frac{R_{tail}}{r_e}$$

No choice of collector resistor improves rejection, and neither does the gain. Doubling R_C doubles both gains and rejects exactly as badly. The instinct is universally the other way, rejection feeling like it should be about matching or about gain: **it is about the tail and only the tail.**

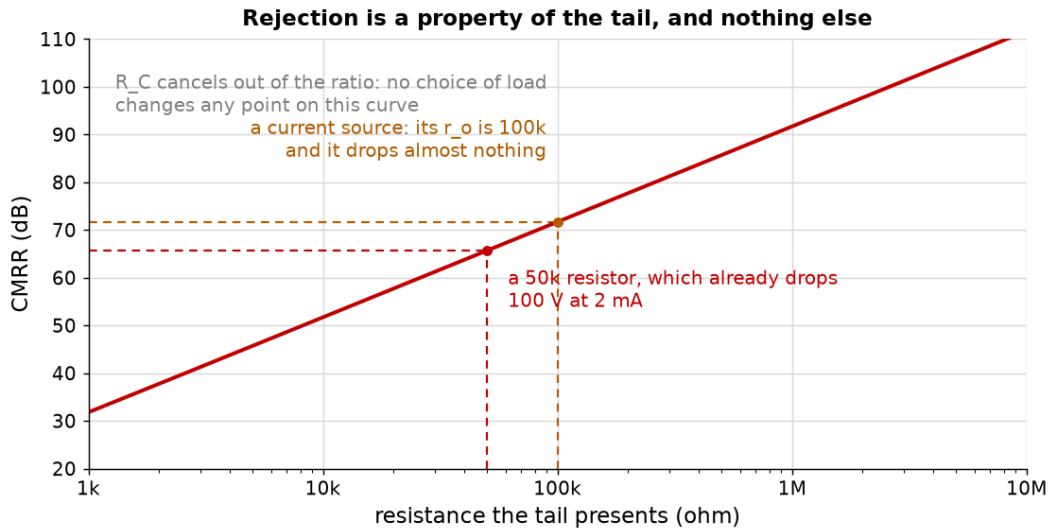


Figure 9.3 Common-mode rejection ratio in decibels against the resistance the tail presents, on a logarithmic axis, rising 20 dB per decade, with the resistor-tail and current-source-tail regions marked and the supply voltage each would need annotated.

9.2.3 Which makes it a supply-voltage question

A resistor in the tail carries the tail current, so a tail resistance implies a voltage, $V_{tail} = I_{tail}R_{tail}$.

Tail	CMRR	In decibels	Volts across it at 2 mA
1 kilohm	39	32 dB	2 V
10 kilohm	385	52 dB	20 V
100 kilohm	3847	72 dB	200 V
260 kilohm	10000	80 dB	520 V
1 megohm	38462	92 dB	2000 V

80 dB of rejection from a resistor needs 520 V of supply, and 80 dB is unremarkable. A current source presents hundreds of kilohms of *incremental* resistance while dropping a volt or two of *actual* voltage, because its resistance is r_o . **That is the whole reason the tail is a current source**, and it generalises: **a current source has a large resistance without a large voltage across it.**

A **cascoded mirror** raises r_o by another β (§7.2.5): another 34 dB for one transistor and 0.7 V of headroom, the usual integrated arrangement.

9.2.4 The mirror load, and its two mechanisms

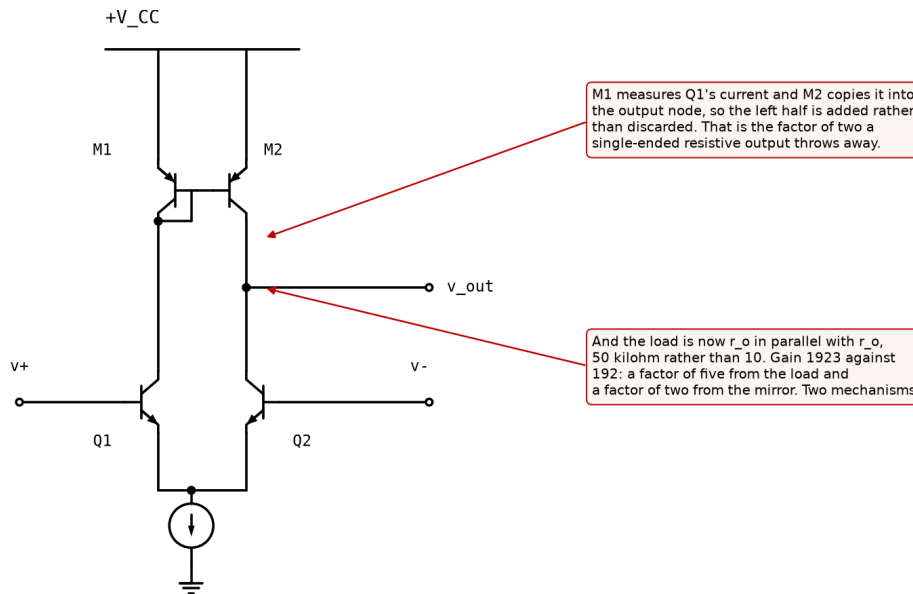


Figure 9.4 A differential pair with a PNP current mirror as its load: the two mirror transistors' emitters to the positive rail, their bases tied, the left one diode-connected to the left pair collector, and the output taken at the right collector.

Replace the collector resistors with a **current mirror**: two PNP devices, emitters to the rail, bases tied, the left diode-connected. M1 carries whatever Q1 carries and M2 copies it into the output node, so a differential input has the mirror pushing the output up while Q2 pulls it down. **Both halves now drive the output.**

$$A_{dm} = -\frac{r_{o(n)} \parallel r_{o(p)}}{r_e}$$

1923 against 192, and it is two separate factors:

- **A factor of two from the mirror as a mirror.** The two in $R_C/2r_e$ came from throwing one collector away, and the mirror stops throwing it away. Nothing to do with resistance.
- **A factor of five from the load.** $r_o \parallel r_o$ is 50 kilohm where the resistor was 10. Nothing to do with mirroring.

They multiply, and textbooks run them together. Only the second is available to a resistively loaded stage, and only the first survives when the next stage's input resistance dominates the output node. **Third distinct job for one device:** stopping Chapter 7's degeneration boost

being swamped, giving the tail resistance without voltage, and recovering the discarded half.

9.2.5 The other output, and why the answer changes completely

Everything above takes the output at **one collector**. Take the **difference between the two collectors** instead and the rejection is set by something else entirely, 46 decibels away.

Why. A common-mode input moves both collectors down equally. Single-ended, that motion *is* the output, so the tail alone decides its size. Differentially it appears on both collectors and **subtracts out exactly**, provided the halves match, and what survives is the mismatch:

$$CMRR_{diff} = \frac{2R_{tail}}{\delta r_e}$$

Output taken	Limited by	With a 10 kilohm tail
One collector	the tail alone	52 dB
Both, 5 per cent mismatch	tail and matching	84 dB
Both, 1 per cent mismatch	tail and matching	98 dB
Both, 0.1 per cent mismatch	tail and matching	118 dB

Forty-six decibels, from 1 per cent resistors and where you put the voltmeter. Same circuit.

So “the CMRR of a differential pair” is not a statement about a circuit unless it says which output is taken: different limits, different fixes, two orders of magnitude apart at ordinary tolerances.

And it explains where integrated amplifiers get their 100 dB: not exotic tails, but two devices adjacent on one die matching to a fraction of a per cent, output differentially so that matching becomes rejection. The same schematic in discrete parts, single-ended, gives 52 dB.

Why this book still takes the single-ended output: an operational amplifier’s second stage has one input, and taking the difference with a current mirror is §9.2.4.

9.2.6 Offset, matching, and why modern input stages are MOSFET

With both inputs at zero the output should sit at its quiescent value. It does not, and the difference referred back to the input is the **input offset voltage**.

Source	Size	Input-referred offset
1 mV of V_{BE} mismatch	typical of two discrete devices	1 mV , directly
1 per cent of R_C mismatch	ordinary resistors	0.52 mV
20 microamps of base current through a 10 kilohm source imbalance		200 mV

The third row is not a misprint, and it dominates the others by two orders of magnitude. 20 microamps through 10 kilohm is 200 mV, and it cancels only if the two sources match.

The classical fix is to make the source resistances equal, so the base currents produce the same voltage and it becomes common-mode: hence the resistor textbook op-amp circuits put in the non-inverting input. It leaves the *difference* between the base currents, and a 1 per cent beta mismatch is 0.2 microamps, 2 mV through 10 kilohm.

The modern fix is to have no base current. A MOSFET gate draws nothing, so the row disappears rather than being cancelled, and with it the requirement that the sources match.

The trade, plainly: a MOSFET input pair has a tenth of the gain at the same current, r_s being ten times r_e (§7.2.6), and none of the input-current problem. **Almost every modern operational amplifier takes that trade** and recovers the gain in the second stage. The third option, best of both for two more devices, is **bipolar inputs with source followers in front of them**.

9.2.7 What to build

ael/diffpair/pair.hpp

Function	Returns
<code>intrinsicEmitterResistance(tailCurrent)</code>	$V_T/(I_{tail}/2)$. Half the tail.
<code>differentialGain(load, tailCurrent, degeneration)</code>	$-R_C/(2(r_e + R_E))$, single-ended.
<code>commonModeGain(load, tailCurrent, tailResistance, degeneration)</code>	$-R_C/(2R_{tail} + r_e + R_E)$.
<code>commonModeRejection(load, tailCurrent, tailResistance, degeneration)</code>	The ratio, as a plain number.
<code>commonModeRejectionDifferential(tailCurrent, tailResistance, mismatch)</code>	§9.2.5's figure, for both collectors.
<code>mirrorGain(tailCurrent, load)</code>	$-R_{load}/r_e$: no factor of two.
<code>transfer(differentialInput, tailCurrent)</code>	$I_{tail} \tanh(v_d/2V_T)$.
<code>linearRange(tolerance)</code>	The input at which transfer falls tolerance below its tangent.
<code>inputOffset(loadMismatch, vbeMismatch, load, tailCurrent)</code>	Input-referred, from both causes.
<code>decibels(ratio)</code>	$20 \log_{10}$, because every figure above is quoted both ways.

commonModeRejection must be implemented as the ratio of the other two, not as a separate formula. The chapter's central claim is that R_C cancels out of that ratio, and a shipped test asks for the rejection at two very different loads and requires agreement to machine precision.

linearRange must not have the tail current as a parameter. If your derivation produced one, it has a factor that should have cancelled.

What good looks like

About sixty lines, of which `linearRange` is the only one that iterates.

9.2.8 What this section is blind to

- **Temperature.** One temperature, both halves at it. Offset drifts about 3 microvolts per degree per millivolt of offset, the specification that matters in a precision amplifier.
- **The mirror's own mismatch,** which appears directly as offset and is why the mirror devices are matched as carefully as the input pair.
- **Noise,** the main reason to choose one pair over another in practice.
- **Common-mode input range.** The pair stops working as the inputs approach either rail, and the tail source needs a volt or two of its own.
- **Frequency, again.** The mirror adds a pole the resistive load does not.

9.3 Review**What you should be able to do now**

After this chapter, you should be able to:

- Draw a differential pair and say what its tail current fixes.
- Derive the differential gain and account for both factors of two in it.
- Derive the common-mode gain and say why the tail appears doubled.
- State CMRR, and say which components in the circuit do **not** affect it.
- Choose a tail arrangement from a CMRR requirement, and justify the supply voltage it needs.
- Give the two independent reasons a current-mirror load raises the gain.
- State the pair's linear input range in millivolts, and say what it does outside it.
- Compute an input-referred offset from a stated mismatch, and say which mismatch dominates.

Questions to test yourself

You should be able to answer:

1. A pair with a 2 mA tail: what is r_e on each side, and why is it 26 ohm rather than 13?
2. The differential gain to one collector is half what it is to both. Where did the other half go, and what would recover it?
3. Two designers argue about CMRR. One wants a larger collector resistor. Who is right?
4. What resistance would the tail need for 80 dB of rejection, and what supply voltage does that imply? Now say what is used instead.

5. A current-mirror load raises the gain by a factor of ten here. Split that ten into its two causes.
6. An input signal of 50 mV differential. Is the small-signal gain the right answer? Show why not.
7. A 1 per cent mismatch between the two collector resistors: what offset does it produce at the input, and how does that compare with 1 mV of V_{BE} mismatch?

Further reading

- *The Art of Electronics*, Horowitz and Hill, chapter 2, sections on the differential amplifier, for a second treatment of the same material.

Looking ahead

- Chapter 10 assembles Chapter 6 to this chapter into one operational amplifier, opens the black box of Chapter 3, closes the loop with Chapter 4's arithmetic, and is the capstone.

9.4 Exercises

Eight, ending with the Cross-check. That one asks a design lever to keep helping, and finds that past a certain point it stops.

How to check your work. A `CODE` exercise is judged by the suite that ships with it: run `make test` at the root of the companion repository, with `AEL_DIR` pointing at your toolkit. A suite you have not started is reported as `SKIP` rather than as a failure, so the command is worth running from the first day. The hand calculations and designs are checked against the toolkit functions each one names.

The solutions are not in this book. They are published in full in the companion repository, in `lectures/L09/appendix`, including the plausible wrong answer where there is one. Read them once you have your own answers, including the ones you are unsure about, because being unsure and right is a different thing from being unsure and wrong and the solutions say which is which.

Exercise 9.1 The pair and its tail

RECALL

1. What does the tail current fix, and what do the inputs decide?
2. A pair has a 2 mA tail. What is r_e on each side? Show the step that is easy to get wrong.
3. Describe what the tail *node* does for a differential input and for a common-mode input.
4. Why does a resistance R_{tail} in the tail behave as $2R_{tail}$ to one half of the pair?

Exercise 9.2 The two factors of two

RECALL

1. The differential gain to one collector is $-R_C/2r_e$. Account for the two.
2. There is a second factor of two available. Where, and what recovers it?
3. Which of the two is a property of what “differential” means, and which is a choice?
4. A current-mirror load raises the gain by a factor of ten in this chapter. Split that into its two independent causes and say what each is worth.

Exercise 9.3 The pair, resistively loaded

HAND CALCULATION

A pair with a 2 mA tail, $R_C = 10$ kilohm, $\beta = 50$, and a 10 kilohm tail resistor.

1. r_e , the differential gain to one collector, and the gain to both.
2. The common-mode gain to one collector.
3. CMRR, as a ratio and in decibels.
4. Recompute all of parts 1 to 3 with $R_C = 100$ kilohm. Which numbers changed and which did not?

Check yourself: differentialGain, commonModeGain, commonModeRejection

Exercise 9.4 The large signal

HAND CALCULATION

Same pair.

1. The difference current at $v_d = 5$ mV, 26 mV and 100 mV.
2. What fraction of the tail has moved at each.
3. The input at which the small-signal answer is 1 per cent optimistic.
4. Repeat part 3 for a 20 mA tail. Explain the result.

Check yourself: transfer, linearRange

Exercise 9.5 A pair to a rejection requirement

DESIGN

An instrumentation front end needs 80 dB of common-mode rejection at a 2 mA tail, with a single-ended output.

1. What tail resistance does that require?
2. What voltage would a resistor of that value drop, and what supply does that imply?
3. State the arrangement you would actually use, and what its incremental resistance has to be.
4. Now suppose you may take the output differentially instead. What matching would give you 80 dB with a 10 kilohm resistor tail? Comment on which of the two designs you would build.

Check yourself: commonModeRejection, commonModeRejectionDifferential, decibels

Exercise 9.6 Offset

DESIGN

The same pair, driven from a source of 10 kilohm on one input and 1 kilohm on the other.

1. The base current in each input, and the offset voltage the *imbalance* produces.
2. The offset from a 1 per cent collector-resistor mismatch, referred to the input.
3. The offset from 1 mV of V_{BE} mismatch, referred to the input.
4. Rank the three, then say what single change removes the largest one entirely and what it costs.

Check yourself: `inputOffset`

Exercise 9.7 The pair

CODE

Implement `ael::diffpair` to the specification in §9.2.7.

commonModeRejection must be the ratio of your own two gain functions, not a separate formula. The chapter's central claim is that R_C cancels out of that ratio, and writing it as a ratio is what makes your code demonstrate the claim rather than restate it.

LinearRange must not take the tail current as an argument. If your derivation produced one, something cancelled that you did not cancel.

Exercise 9.8 Common-mode rejection, and the lever that reverses

CROSS-CHECK

The pair of Exercise 9.3: 2 mA tail, 10 kilohm collector resistors, 10 kilohm tail resistor to a negative rail. Find its CMRR, four ways.

1. **By hand.** $(2R_{tail} + r_e)/2r_e$.
2. **By your closed form, twice**, once with $R_C = 10$ kilohm and once with $R_C = 100$ kilohm.
3. **By your solver.** Apply a small differential input, measure the change at one collector; apply a small common-mode input, measure the change at the same collector; divide.
4. **Then replace the tail resistor with an ideal current source and repeat leg 3.**

What to expect

Legs 1, 2 and 3 agree to about a decibel, at 52, 52 and 51. **Leg 2 gives the same number twice**, because the collector resistor cancels; if yours does not, `commonModeRejection` is not a ratio of your two gains.

Leg 4 is the exercise, and it does not return infinity. An ideal current source has infinite incremental resistance, so the expected answer is that the common-mode gain is zero. What you will get is a small, **stable, positive** common-mode gain and a CMRR of about **101 dB**.

That number is not noise: check it by changing your perturbation size by three decades and watching it stay put. **Find out where it comes from.** Two experiments settle it:

- Set the Early voltage to 10^9 and repeat. The common-mode gain should collapse to your solver's arithmetic floor.

- Set β to 5000 and repeat. It should fall by about a hundred.

Then sweep the tail. Put the ideal source in parallel with a resistor and step that resistor from 1 megohm to 10. The closed form says CMRR rises without limit. Report what the solver says, and where the maximum is.

Finally, answer this. Chapter 5's Cross-check had two legs that agreed with each other and were both wrong about a real device. This one has a leg that is exactly right about the model and answers a question nobody asked. **What is the single statement about models that covers both?**

If leg 3 gives a CMRR near 1, you have applied the common-mode input to only one base.

If leg 3's differential gain is 385 rather than 192, you have measured the difference between the two collectors rather than one of them, which is §9.2.5's question rather than this one.

Building an operational amplifier, and the capstone

The black box of Chapter 3, opened and rebuilt from the six chapters since.

IN THIS CHAPTER

- The black box of Chapter 3, opened: eleven transistors, and every one of them already analysed.
- What each stage is responsible for, and why that is the only way to size any of them.
- The gain budget, and the eleven decibels that loading takes.
- The two stages with no voltage gain, which are worth 31 dB each.
- Why the amplifier has no open-loop operating point at all, and what that means for the solver.
- Miller compensation: making one pole dominant on purpose.
- Closing the loop, and Chapter 4's arithmetic arriving where it was always going.
- The capstone: one program, every stage predicted and solved, printed side by side.

Everything in this chapter is already built. The pair is Chapter 9, the mirror is Chapter 7, the followers are Chapter 8, and the gain stage is Chapter 7 again. This chapter adds no new circuit; it adds the **accounting**, and the accounting is where the surprises are.

There are three of them. The first is the **budget**: two stages of 65.7 dB should give 131, and the amplifier delivers 120, with 10.6 of the missing 11.5 in one place — the gain stage driving the output stage, and that is *after* the output stage was made a Darlington. This is Chapter 1's loading for the sixth and last time. The second is that **the two stages with no voltage gain are the most valuable in the amplifier**: they are six of the eleven transistors, they amplify nothing, and each is worth about 31 dB. Gain is easy; connecting two stages together is what is hard.

The third is that **the amplifier has no open-loop operating point**. A gain of a million and rails of ± 15 V means that 15 microvolts of input offset puts the output against a rail, and no pair of transistors matches to 15 microvolts. A solver that reports the output at -15 V has not failed, and a stage that cannot be biased on its own is not a stage — it is part of a loop.

Then close the loop, and Chapter 4’s arithmetic arrives where it was always going: 120 dB open, a gain of 20 closed, an error of 0.002 per cent, and a closed-loop gain that moves by nine parts in a hundred thousand while the open-loop gain moves by a factor of fourteen. The capstone in §10.4 is one program that predicts the whole amplifier, solves the whole amplifier, and reconciles the two row by row. **The differences are the book**, and a reader who can account for every row has understood the ten chapters.

10.1 The amplifier, stage by stage

Eleven transistors, and not one of them is new.

10.1.1 The black box, opened

Chapter 3 (Chapter 3) used an operational amplifier on the promise that this chapter would build one.

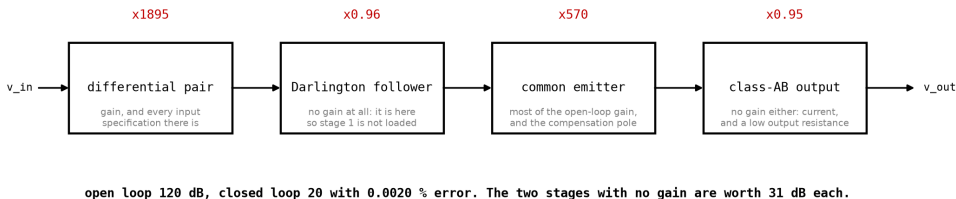


Figure 10.1 A block diagram of four stages: a differential pair, a Darlington follower, a common-emitter stage and a class-AB output, with the gain of each above it and the open-loop and closed-loop figures below.

Stage	What it is	What it is responsible for	Gain
1	Mirror-loaded differential pair	Gain, and every input specification there is	1895
2	Darlington emitter follower	Nothing, except not loading stage 1	0.96
3	Common-emitter, current-source load	Most of the open-loop gain, and the dominant pole	570
4	Class-AB Darlington output	Current, and a low output resistance	0.95

Two of the four have no voltage gain. They are six of the eleven transistors, and §10.2.2 is about why they are the most valuable part of the amplifier.

10.1.2 The whole thing

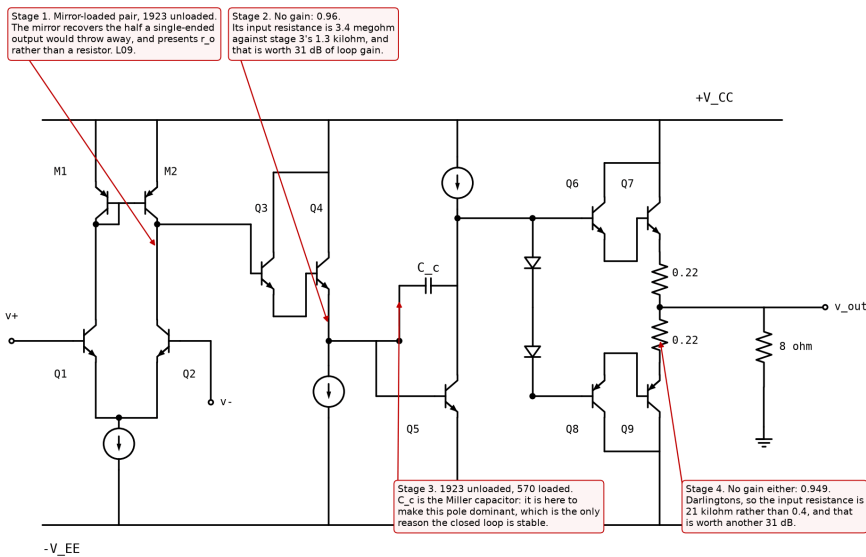


Figure 10.2 The complete amplifier: a differential pair with a PNP current-mirror load and a current-source tail, feeding a Darlington emitter follower, feeding a common-emitter stage with a current-source load and a Miller capacitor, feeding a class-AB output stage of two complementary Darlington's with a two-diode bias and emitter resistors, driving an eight ohm loudspeaker between supply rails.

Every block is a circuit from an earlier chapter:

- **Q1, Q2, M1, M2 and the tail** are Chapter 9 (Chapter 9)'s mirror-loaded pair, entire.
- **Q3 and Q4** are Chapter 8 (Chapter 8)'s Darlington follower.
- **Q5** is Chapter 7 (Chapter 7)'s common-emitter stage with the current-source load of §7.2.4.
- **Q6 to Q9, the diodes and the 0.22 ohm resistors** are §8.2.3's class-AB output stage, each half a Darlington.
- C_c is the only component here this book has not built, and is §10.2.4's subject.

The three current sources are drawn as current sources rather than as mirrors because that is what they are, small-signal: a resistance of r_o with a current through it that does not depend on the node voltage. Each is two more transistors in a real build, and §9.2.3 established why they are not resistors.

10.1.3 Stage 1: the pair, and why it is first

The input stage decides every input specification the amplifier has: offset, input current, input resistance, common-mode range, common-mode rejection, most of the noise. **Nothing downstream improves any of them**, because feedback divides down what happens *after* the input, not *at* it. It also has to have gain, because the loop gain starts here.

Mirror-loaded, for the two reasons of §9.2.4. The mirror recovers the half a single-ended output would discard and presents r_o rather than a resistor. 1923 unloaded.

10.1.4 Stage 2: the buffer, which does nothing

Stage 3 is a common-emitter stage at 1 mA, so its input resistance is βr_e : **1.3 kilohm**. Stage 1's output resistance is **50 kilohm**. Connect them directly and stage 1 keeps

$$\frac{1300}{50000 + 1300} = 2.5 \text{ per cent}$$

of its gain. **1923 becomes 48.7, and the amplifier has lost 32 dB before doing anything.**

So a follower goes between them, a Darlington one: a single follower would present $\beta(r_e + 1300) = 66 \text{ kilohm}$, only a little more than the 50 it is protecting. The Darlington presents $\beta^2(2r_e + 1300) = 3.4 \text{ megohm}$ and stage 1 keeps 98.5 per cent.

It costs 0.34 dB of signal and it is worth 31.5 dB of loop gain.

10.1.5 Stage 3: where the gain is

A common-emitter stage at 1 mA with a current-source load, r_o against r_o , so 50 kilohm and an unloaded gain of $50000/26 = 1923$: the same as stage 1 and for the same reason.

Undegenerated, because every ohm in its emitter divides the gain (§7.1.5) and this stage exists to produce gain. The cost is §10.2.3: it has no stable operating point of its own, and only the loop gives it one.

It is also where the compensation goes, the highest-impedance node being the cheapest place to put a dominant pole (§10.2.4).

10.1.6 Stage 4: the output, which also does nothing

Class AB from Chapter 8 (§8.2), idling at 120 mA with 0.22 ohm emitter resistors and two diodes of bias that must sit on the heatsink.

Each half is a Darlington, for stage 2's reason: an 8 ohm loudspeaker behind a single follower presents 411 ohm to a stage with 50 kilohm of output resistance, which would keep 0.8 per cent of its gain. Behind a Darlington it presents **21 kilohm** and stage 3 keeps 30 per cent.

Thirty per cent is still the largest single loss in the amplifier, 10.6 dB: the price of driving 8 ohms from a 50 kilohm node, and no arrangement of two transistors gets it back. A third would, which is §8.3.2's triple emitter follower.

10.1.7 What this section is blind to

- **Every frequency except one.** The compensation pole is discussed; the other poles, and so the actual stability margin, are computed nowhere.
- **Slew rate**, the tail current over the compensation capacitor, and the one large-signal specification a reader could compute from what is here (§9.1.4).
- **Start-up.** The three current sources have to start, and a mirror whose reference derives from its own output has a stable state at zero current. Real designs include a start-up circuit.
- **Supply rejection.** The rails are drawn as ideal. They are not.
- **Everything about layout**, which decides how much of the above survives being built.

10.2 The budget, the missing operating point, and the loop

Where the gain goes, why the amplifier cannot be biased, and what closing the loop buys.

10.2.1 Two stages of 66 dB give 120, not 131

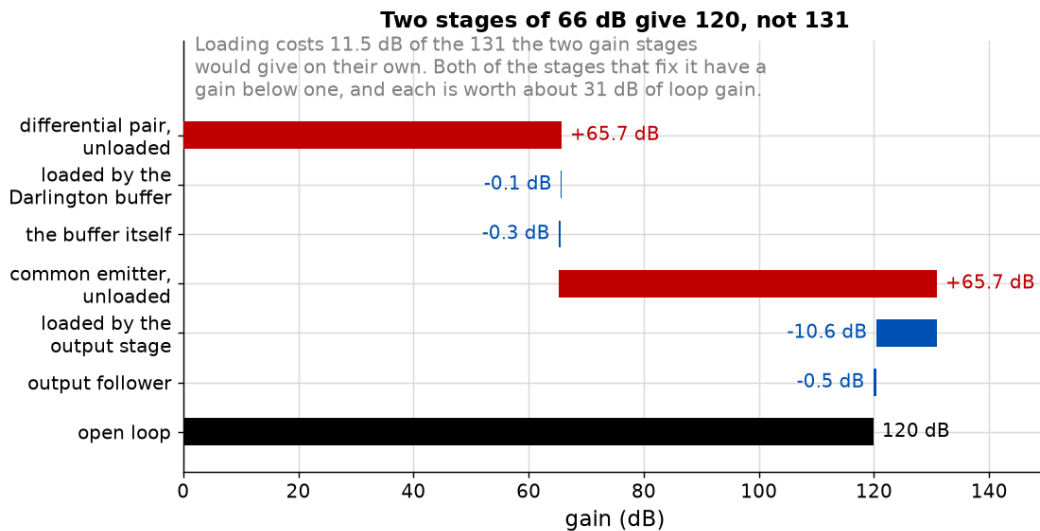


Figure 10.3 A waterfall chart of the gain budget in decibels: the differential pair at plus 65.7, small losses to the buffer, the common-emitter stage at plus 65.7, a loss of 10.6 to the output stage, and a total of 120 decibels.

	dB
Stage 1, unloaded	+65.7
Loaded by the buffer	-0.1
The buffer itself	-0.3
Stage 3, unloaded	+65.7
Loaded by the output stage	-10.6
The output follower itself	-0.5
Open loop	119.9

11.5 dB of the 131.4 the two gain stages would give is lost to loading, and not evenly: 10.6 is one number, stage 3 driving the output stage.

This is Chapter 1's loading arithmetic for the last time, the sixth appearance of one subtraction. **An amplifier's gain budget is not the product of its stage gains, and a budget computed without loading is wrong by a factor of 3.7 in the optimistic direction.**

10.2.2 The two stages with no gain

Stage 2 has a voltage gain of 0.96 and stage 4 of 0.95. Together they are six of the eleven transistors and they amplify nothing. **Take each away and measure:**

	Open loop
As designed	119.9 dB
With stage 1 driving stage 3 directly	88.4 dB
With a single follower instead of the output Darlington	88.9 dB

Each of the two stages with no gain is worth about 31 dB, more than any two-transistor gain stage in the book could deliver. A stage that loses 0.3 dB of signal buys 31.5 dB of loop.

That is the lesson the whole book has been building to. Gain is easy: one transistor and a large load gives 66 dB. What is hard is **connecting two of them together**. Three transistors amplify, Q1, Q2 and Q5; six connect.

10.2.3 The operating point that does not exist

Try to solve the amplifier for its DC operating point with the loop open. It will not work, and the reason is not the solver. The open-loop gain is 986,000 and the rails are ± 15 V, so an input offset of

$$\frac{15}{986000} = 15 \mu\text{V}$$

puts the output hard against a rail. **A pair of transistors matched to 15 microvolts does not exist**, and neither does an amplifier whose open-loop output sits anywhere but a rail.

So the DC solve either fails to converge or converges on a saturated output, and both are correct answers. A solver reporting the output at -15 V has not failed.

Which is why the stage gains have to be measured stage by stage. Bias each stage alone, perturb its input, measure its output, multiply. The whole amplifier can only be solved with the loop closed, and then what you measure is the closed-loop gain of 20.

And it is why stage 3 is undegenerated. A designer looking at it in isolation would call it unbiasable and add an emitter resistor. Inside the loop it does not need one: the loop sets its operating point. **The stage is not a stage; it is part of a loop, and it cannot be judged alone.**

10.2.4 Compensation, and the pole that is made dominant

Four stages give at least four poles, each up to 90 degrees of lag, and Chapter 4 (Chapter 4) established that 180 degrees with a loop gain above one is an oscillator. **The fix is not to make the amplifier fast, but to make it slow, deliberately, in one place.**

C_c bridges stage 3's collector and base. It is a Miller capacitor (§7.2.5), so the input sees $C_c(1 + 1923)$, which against that node's resistance puts a pole at a few tens of hertz. That resistance is not stage 3's 1.3 kilohm base resistance alone: the buffer's 72 ohm is in parallel, so the node sees 68 ohm. **The buffer is what makes C_c large.** Every other pole is at hundreds of kilohertz or above.

So by the time the other poles contribute phase, the loop gain is already below one. The cost is bandwidth: the open-loop gain falls 20 dB per decade from that pole, and the closed-loop bandwidth is the gain-bandwidth product over the closed-loop gain. **This is Chapter 7's Miller effect used on purpose:** what ruined a common-emitter stage's bandwidth makes a four-stage amplifier usable.

What this book does not do is compute the other poles, so it cannot say what C_c should be. That needs the capacitances of §7.2.8.

10.2.5 Closing the loop

Wrap a divider from the output back to the inverting input with a feedback fraction of $\beta_f = 1/20$, and Chapter 4 (Chapter 4)'s arithmetic applies unchanged:

	Value
Open-loop gain	986,000
Feedback fraction	1/20
Loop gain	49,300
Closed-loop gain	19.99959
Error from 20	0.002 per cent

And the thing that matters more than the error. The output stage's input resistance goes as h_{FE}^2 , which varies by a factor of a hundred across an ordinary beta spread:

h_{FE}	Open loop	Closed-loop gain
20	106.4 dB	19.998096
50	119.9 dB	19.999594
200	129.2 dB	19.999862

The open-loop gain varies by a factor of 14. The closed-loop gain varies by nine parts in a hundred thousand. That is the whole argument for feedback, on this book's own numbers. §8.1.5 said an output stage works inside a loop rather than on its own; this is the measurement.

Output resistance goes the same way. Open loop, three terms: the 0.22 ohm emitter resistor, the Darlington's $2r_e$ of 0.43 ohm, and stage 3's 50 kilohm over h_{FE}^2 , which is 20 ohm and dominates. **20.6 ohm over one plus the loop gain: 0.42 milliohm.** Not a real number, since the wire to the loudspeaker limits the answer first, and saying so is the point of quoting it.

10.2.6 What to build

PNP support in ael/net/netlist.hpp

The amplifier has four PNP devices and your netlist has none. Add a polarity:

```
enum class Polarity { Npn, Pnp };

void addBjt(Node collector, Node base, Node emitter, device::bjt::Parameters &
    parameters = {},
    Polarity polarity = Polarity::Npn) noexcept;
```

A PNP is an NPN with every voltage and current negated: the stamp you have, evaluated at $(-v_{BE}, -v_{BC})$ with the currents negated. Longer than fifteen lines and you are writing a second device model.

ael/report/amplifier.hpp

One structure, one function predicting every stage from Chapter 7 to Chapter 9's closed forms, one formatter.

```
namespace ael::report
{
    struct Design
    {
        double tailCurrent{};    ///< Stage 1's tail, split between the two sides.
        double bufferCurrent{};  ///< Stage 2, the Darlington follower.
        double gainStageCurrent{}; ///< Stage 3, the common-emitter stage.
        double idleCurrent{};    ///< Stage 4's quiescent current, per half.
```

```

double load{};           ///< The loudspeaker.
double supply{};         ///< One rail, so the amplifier runs on plus and ↵
↳ minus this.
double beta{50.0};
double earlyVoltage{100.0};
};

struct Stage
{
    std::string name{};
    double unloaded{};    ///< The stage on its own.
    double loadedBy{};    ///< The input resistance of the stage after it.
    double loaded{};      ///< What it delivers into that.
};

struct Budget
{
    std::vector<Stage> stages{};
};

/// The four stages, each loaded by the next stage's input resistance.
[[nodiscard]] Budget budget(const Design& design) noexcept;

/// The product of the loaded gains, and nothing else.
[[nodiscard]] double openLoopGain(const Design& design) noexcept;

/// L04's expression, applied to it.
[[nodiscard]] double closedLoopGain(const Design& design, double fraction) ↵
↳ noexcept;

/// A table, one row per stage, predicted against measured.
[[nodiscard]] std::string format(const Budget& budget) noexcept;
}

```

The member names are part of the specification, because the suite reads `budget.stages[0].loaded` and sets `design.tailCurrent` by name. `beta` and `earlyVoltage` are in `Design` rather than fixed, because §10.2.5's whole argument is what happens when they move.

budget must call the functions from Chapter 7, Chapter 8 and Chapter 9 rather than restating their formulas. It is a composition, and if it contains an expression with r_e in it then it has stopped being one. One shipped test checks that the stage-1 figure agrees with `ael::diffpair::mirrorGain` to machine precision.

What good looks like

About eighty lines, of which format is half. There is no physics in this component at all, which is what makes it the last one.

10.2.7 What the capstone is for

The exercises end with one program that prints this:

stage	predicted	measured	difference
1 input pair	1895.0	1837.2	-3.1 %
2 buffer	0.96	0.96	-0.1 %
3 voltage gain	570.4	548.6	-3.8 %
4 output	0.95	0.94	-0.9 %
open loop	986000	910000	-7.7 %

The differences are the book, and each has a name: the closed forms take V_T as 26 mV where the device computes kT/q ; they take r_o as V_A/I_C where the device gives $(V_A + V_{CE})/I_C$; beta is 50 in one and the transport model's incremental value in the other. **A reader who can account for every row has understood the ten chapters. A reader whose rows agree exactly has run the same code twice.**

10.2.8 What this book is blind to

- **Frequency response, beyond one pole.** No device capacitances, no phase margin, no Bode plot of a real amplifier. The largest omission, and where a second course would start.
- **Noise.** Not mentioned anywhere. It decides the input stage of every precision amplifier and most of the choices in §9.2.6.
- **Distortion, quantitatively.** Where it comes from, never a number.
- **Temperature, beyond V_{BE} drift.** No self-heating, no thermal models, no drift figures.
- **Integrated-circuit design.** Matching, layout, process variation and area are the subject next door; this book is discrete.
- **Anything above a few megahertz**, where transmission lines, package parasitics and electromagnetic compatibility take over from circuit theory.

And one methodological blind spot. Every number here comes from a model with eight parameters. Real devices have a hundred and real amplifiers are measured rather than computed. What the book teaches is how to *predict*, which is worth having because it says what a measurement should be, and therefore when a measurement is telling you something.

10.3 Review

What you should be able to do now

After this chapter, you should be able to:

- Name every stage of a discrete operational amplifier and say what it is responsible for.
- Compute a gain budget with loading, and say why one without loading is not conservative but wrong.
- Explain why an amplifier with no voltage gain in half its stages needs those stages.
- Say why an open-loop DC operating point does not exist, and what your solver does about it.
- Explain what a Miller capacitor is compensating and what would happen without it.
- Close the loop and predict the closed-loop gain, the error, and the output resistance.
- Run one program that predicts and solves the whole amplifier and reconciles the two.

Questions to test yourself

You should be able to answer:

1. The pair gives 1923 and the common-emitter stage gives 1923. Why is the open-loop gain not 3.7 million?
2. The Darlington buffer has a voltage gain of 0.96. What is it worth, in decibels, and why?
3. What is the input resistance of the output stage, and what would it be if it were a single follower instead? What does that difference cost?
4. You solve the amplifier open-loop and the output sits at the negative rail. Is that a bug?
5. What would happen to the closed loop if the Miller capacitor were removed?
6. The closed-loop gain is 20 and the open-loop gain is a million. How far is the closed-loop gain from exactly 20, and what would make it closer?
7. The output stage's input resistance depends on h_{FE}^2 , which varies by a factor of a hundred between devices. Why does the closed-loop gain not?

Further reading

- *The Art of Electronics*, Horowitz and Hill, chapter 4x, for a walk through several real operational amplifier schematics of the kind this chapter builds.

Looking ahead

- The exam papers (Appendix A) are four hours each and cover the whole book.
- What this book did not cover, and where to go for it, is in §10.2.8.

10.4 Exercises

Eight, ending with the capstone. That one is the whole book in one program: predict every stage, solve every stage, and reconcile the two, row by row.

How to check your work. A `CODE` exercise is judged by the suite that ships with it: run `make test` at the root of the companion repository, with `AEL_DIR` pointing at your toolkit. A suite you have not started is reported as `SKIP` rather than as a failure, so the command is worth running from the first day. The hand calculations and designs are checked against the toolkit functions each one names.

The solutions are not in this book. They are published in full in the companion repository, in `lectures/L10/appendix`, including the plausible wrong answer where there is one. Read them once you have your own answers, including the ones you are unsure about, because being unsure and right is a different thing from being unsure and wrong and the solutions say which is which.

Exercise 10.1 What each stage is for

RECALL

1. Name the four stages and say what each is responsible for.
2. Two of them have no voltage gain. Which, and what are they for?
3. Which stage decides the amplifier's input offset, and can a later stage improve it?
4. Which node carries the compensation, and why that one?

Exercise 10.2 The loop

RECALL

1. Write the closed-loop gain in terms of the open-loop gain and the feedback fraction.
2. What is the loop gain of this amplifier at a closed-loop gain of 20?
3. Feedback divides the output resistance by one plus the loop gain. Compute it, then say why the answer is not a real number.
4. The open-loop gain varies by a factor of 14 across a beta spread. By how much does the closed-loop gain vary, and why is that the answer feedback exists to give?

Exercise 10.3 The budget

HAND CALCULATION

The amplifier of §10.1: 2 mA tail, 1 mA buffer, 1 mA voltage stage, 120 mA idle, 8 ohm load, $\beta = 50$.

1. The unloaded gain of stages 1 and 3.
2. The input resistance of stages 2, 3 and 4.
3. Each stage loaded by the next, and the open-loop gain.
4. The difference in decibels between the product of the unloaded gains and your answer to part 3. Where is most of it?

Check yourself: `budget`, `openLoopGain`

Exercise 10.4 The stages that do nothing

HAND CALCULATION

1. What would stage 1's gain be if it drove stage 3 directly? Give the open-loop gain that results.
2. What would stage 3's gain be if the output stage were a single follower rather than a Darlington? Give the open-loop gain that results.
3. State what each of the two no-gain stages is worth, in decibels.
4. Both stages cost signal. State what each costs, and comment on the exchange rate.

Exercise 10.5 The input offset that matters

DESIGN

1. What input offset voltage puts the open-loop output at a rail?
2. Two ordinary transistors match their V_{BE} to about 1 mV. How many times larger is that than your answer to part 1?
3. State what this means for solving the amplifier open-loop.
4. The closed-loop amplifier has an output offset. Compute it for 1 mV of input offset at a closed-loop gain of 20, and say what would reduce it.

Exercise 10.6 The gain you cannot buy

DESIGN

1. Compute stage 1's gain at a tail current of 0.2 mA, 2 mA and 20 mA, with a mirror load in each case. Then do the same for stage 3 at 0.1 mA, 1 mA and 10 mA with a current-source load.
2. Explain the result. Write the gain symbolically and show what cancels.
3. So how *do* you get more gain out of one stage? Compute what your answer gives at 1 mA.
4. The compensation capacitor is 30 pF. Compute the amplifier's slew rate at the 2 mA tail, then the tail current that would give 10 V per microsecond, and say what that change costs in gain.

Check yourself: `mirrorGain`, `intrinsicEmitterResistance`, `cascodeOutputResistance`

Exercise 10.7 PNP support, and the report

CODE

Add `Polarity` to `addBjt` and stamp a PNP as an NPN with every voltage and current negated. Then implement `ael::report` to the specification in §10.2.6.

budget must compose the functions from Chapter 7 to Chapter 9, not restate their formulas. One shipped test checks that its stage-1 figure agrees with `ael::diffpair::mirrorGain` to machine precision, which it can only do by calling it.

Exercise 10.8 The capstone**CROSS-CHECK****Predict the whole amplifier, solve the whole amplifier, and reconcile the two, row by row.**

1. **Predict.** Run budget and record the four stage gains and the open-loop gain.
2. **Solve, stage by stage.** Build each stage as its own netlist, bias it, perturb its input by a small voltage, and measure its output. Load each stage with the next stage's input resistance as a resistor, so that the solver is measuring the same quantity the closed form predicted.
3. **Reconcile.** Print predicted against measured with the difference in per cent, one row per stage.
4. **Then solve the whole amplifier open-loop,** with the four stages connected. Report what happens.
5. **Then close the loop** with a feedback divider for a gain of 20, solve again, and report the closed-loop gain.

What to expect

Every row differs by a few per cent, and every difference has a name. The closed forms take V_T as 26 mV where the device computes $kT/q = 25.87$; they take r_o as V_A/I_C where differentiating the model gives $(V_A + V_{CE})/I_C$; they take β as 50 where the transport model's incremental value differs slightly. **Account for every row.** A reader whose rows agree to six figures has run the same code twice and learned nothing.

Part 4 will not converge on a sensible answer, and that is the correct result. With an open-loop gain of a million and rails of 15 V, an input offset of **15 microvolts** saturates the output. Your solver will report the output at a rail, or fail to converge, and both are true statements about operational amplifiers. Do not tune your solver until it produces a mid-rail answer; there is not one.

Part 5 will converge, and the closed-loop gain should be within a per cent of 20. Notice what you have just demonstrated: the same circuit is unsolvable open-loop and well-behaved closed-loop, and nothing about the transistors changed.

Finally, the exercise that is the book. Repeat part 5 with β at 20 and at 200. The open-loop gain moves by a factor of 14. Report what the closed-loop gain does, and say in one sentence what that is worth.

The Written Papers

This appendix and the four after it hold two complete four-hour papers for *Analog Electronics*, with worked solutions. Appendices B and C are the papers, questions only; Appendices D and E are their model answers, with marks.

What these are for

The book as written has **no exam and nothing is marked**. Assessment is the exercises in every chapter, the shipped test suite each component comes with, and the Cross-check that closes every one of the ten chapters.

These papers do not replace any of that. **They are here for students to check their own knowledge, and nothing else**. They gate nothing, they are not a qualification, and no part of the book requires them. Nothing in the repository depends on them and make test does not know they exist. What they check is the half the test suites cannot: whether you can still produce the derivation with the text out of the room.

Take one only once you have finished the book. Every paper has one question per chapter, so sitting one partway through examines material nobody has taught you yet. The intended point is after Chapter 10, once the capstone's report prints a row for every stage.

The two papers

Both cover the whole book, one question per chapter, and question N carries the same chapter and the same mark in both. They share no question, so the topics below are Paper A's; Paper B's ten are different. Either can be used alone; use both as a main sitting and a resit, or in alternate years.

Question	Topic	Chapter	Marks
1	A divider, its Thevenin resistance, and a load that takes a third	Chapter 1	10
2	One capacitor, two corners, and the resistance nobody drew	Chapter 2	9
3	Two sections, one buffer, and the factor of 1.72	Chapter 3	11
4	Loop gain, the error it divides, and a diode that will not converge	Chapter 4	10
5	Sixty millivolts a decade, and a switch that ignores beta	Chapter 5	10
6	A bias point that loads itself, and drift that is a loop gain	Chapter 6	9
7	The emitter factor, and the node it belongs to	Chapter 7	11
8	Eight ohms, two Darlington's, and a bias that is not two drops	Chapter 8	10
9	The tail, the 520 volts, and the resistor that cancels	Chapter 9	11
10	A gain budget, and the two stages that amplify nothing	Chapter 10	9
			100

The weighting is the book's own. Ten chapters in two parts of four and six, and the paper puts 40 marks on Part 1 and 60 on Part 2, because that is the split the book was designed with and the transistors are the point of it.

The three at 11 are Chapter 3, Chapter 7 and Chapter 9, because each asks for two separate derivations rather than one: a cascade's corner and the buffer that fixes it; the emitter factor and the node it belongs to; the two gains of a pair and the ratio in which the load cancels. The three at 9 are Chapter 2, whose examinable content is one reactance and one corner; Chapter 6, which is a droop and a suppression factor; and Chapter 10, which is a multiplication and one observation about it.

Paper A leans towards the numbers this repository publishes and checks: 1.7085 V from a 33k and a 6.8k, a Thevenin resistance of 5.638 kilohm and a third of the output lost to a 10 kilohm load; a 1001 Hz corner that becomes 151 Hz; a cascade at 375 Hz where a buffer would give 644; 0.01 per cent of gain error from a loop gain of ten thousand; a diode at 0.6965 V that takes 168 Newton iterations without limiting and 7 with it; 287 ohm becoming 270 and a forced beta of 9.4; a base that droops 105 mV and a collector current of 0.934 mA rather than 1.06; an output resistance of 9.89 kilohm where the tempting rule says 100; a Darlington's 21.08 kilohm against a single follower's 411; 51.7 dB of rejection and the 520 V it would take to reach 80; and an open-loop gain of 119.9 dB where the stage gains multiply to 131.4.

Paper B leans towards the results the book derives: why a divider's output resistance is the parallel combination and not either resistor; why a corner moves when nothing in the filter changed; why two identical sections do not give twice the roll-off at the same corner; why every feedback result is a function of the loop gain rather than of the open-loop gain; why saturation is not a case a transport model handles but a thing the equation does; why a thermal argument that moves V_{BE} by 100 mV describes a circuit that does not exist; why the emitter factor multiplies the resistance looking into the collector and not the stage's output resistance; why a model worth 1 per cent in one calculation is worth a factor of sixty in another; why the collector resistor cancels out of a rejection ratio; and why two stages with no voltage gain are worth more decibels than either gain stage.

Neither paper asks a candidate to write C++. The toolkit is how the book teaches; it is not what the book is about, and a written paper that asked for an `ael::ssm` implementation would be examining typing.

Conventions the papers assume

Both papers state these in their own rubric, so a candidate never has to have read this section.

- **The thermal voltage is 26 mV** and every r_e in both papers follows from it. The physical value is 25.85 mV at 300 K, and 25.87 mV at the 300.15 K that Chapter 6 makes the device model's default; the book rounds, says so, and marks to the rounded figure.
- **Beta is 50**, and it is a worst case rather than a typical one. Where a question wants the spread it gives the two values.
- **The constant-drop model is 0.65 V**, except where a question names the exponential, and then $V_{BE} = V_T \ln(I_C/I_S)$ with $I_S = 10^{-14}$ A. Several answers turn on knowing which of the two a calculation needs. **A saturated switch is the one standing exception at 0.7 V**, which is what §5.1.4 and `baseResistor` both use.
- **Component values land on E12** wherever a question asks for a design, and the answer is the E12 value with the resulting error stated. A design quoted to four figures has not been designed.
- **Twenty for amplitudes, ten for powers.** Both produce a number in dB carrying no label. Every gain in this book is a voltage ratio, so it is $20 \log_{10}$ throughout, and a candidate who uses ten has halved every answer in decibels.
- **A number pinned by a rule is pinned.** The 220 mV degeneration rule, the 26 mV output-stage rule, $r_e = V_T/I_C$, the emitter factor as $(r_e + R_E)/r_e$, and $r_o = V_A/I_C$ with $V_A = 100$ V each fix an answer exactly.
- **An assertion without arithmetic scores nothing** where a question asks which of two terms dominates, or by how much, which is several of them.

Marking

Every solution is written to be marked by somebody who has read the appendices and has not written the toolkit, so each carries the reasoning rather than the answer alone. Marks are shown per part.

- **Method carries the marks.** A correct rule with an arithmetic slip in it is worth more than a correct number with no working, and several questions consume their own earlier answers. Follow through an error rather than penalising it twice.
- **The named traps are worth full marks on their own.** Several parts exist entirely to see whether a candidate avoids one specific mistake. The list, in the order the papers meet them:

Trap	Where	What it looks like
A divider's output resistance taken as the lower resistor	Chapter 1	6.8 kilohm where the answer is 5.638
A loaded divider computed without reforming the divider	Chapter 1	1.71 V quoted for a node sitting at 1.09
An RC corner computed from the resistor that is drawn	Chapter 2	1001 Hz where the circuit gives 151
Two cascaded sections given the single-section corner	Chapter 3	1001 Hz where the answer is 375
A feedback result computed from the open-loop gain alone	Chapter 4	An error that does not move when the feedback network does
Newton started at zero on an exponential	Chapter 4	168 iterations, or no convergence at all
A saturated transistor assumed to obey β	Chapter 5	A base current 5 times smaller than the design needs
A switch designed from the datasheet's h_{FE}	Chapter 5	A design that works on one device and not the next
A divider bias computed unloaded	Chapter 6	1.06 mA where the stage sits at 0.934
The thermal argument that moves V_{BE} by 100 mV	Chapter 6	A factor of 47 described as 10 per cent
The emitter factor applied to the stage's output resistance	Chapter 7	100 kilohm where a parallel combination cannot exceed 10
r_e read from the tail current rather than one side	Chapter 7, Chapter 9	Every gain out by a factor of two
A Miller capacitance left unmultiplied	Chapter 7	A bandwidth 385 times too optimistic
A Darlington's r_e not doubled	Chapter 8	An input resistance 2.6 per cent low, and an identity that fails
A class-AB bias taken as two constant drops	Chapter 8	2 mA of idle current where 120 was wanted
A larger collector resistor prescribed to improve CMRR	Chapter 9	Both gains doubled and the rejection unchanged
An ideal current source read as infinite rejection	Chapter 9	101 dB reported from a mechanism nobody designed
A gain budget computed without loading	Chapter 10	131 dB claimed for a 120 dB amplifier
An open-loop DC solve expected to converge mid-rail	Chapter 10	A solver blamed for an amplifier's own arithmetic

- **The discussion parts are not decoration.** “Say what this establishes” and “say what it does not” is where the book’s actual content is, and a paper marked only on the arithmetic would pass a candidate who has understood none of it. Four of the hardest things in this book carry almost no arithmetic at all: that a source resistance and a load resistance form a divider and that this one subtraction decides six separate results; that a model’s

accuracy is a property of the calculation it appears in rather than of the model; that an idealisation can delete the very quantity a design is about; and that the stages with no gain are the ones that make the gain usable.

Written Paper A

Time: 4 hours. **Closed book.** A basic calculator is permitted. **Total: 100 marks.**

This paper exists to let you check your own knowledge. It is not a qualification and it gates nothing. It has one question per chapter, so **it is meant to be taken once you have finished the book**, after Chapter 10 and the capstone. The model answers are in Appendix D.

Rubric

The thermal voltage is 26 mV. Every intrinsic emitter resistance in this paper follows from $r_e = V_T/I_C$ and that figure.

Beta is 50 unless a question says otherwise, and it is a worst case rather than a typical value.

The constant-drop model is $V_{BE} = 0.65$ V, except where a question names the exponential, and then $V_{BE} = V_T \ln(I_C/I_S)$. Knowing which of the two a calculation needs is examinable.

A saturated switch's base-emitter drop is 0.7 V. A forced beta drives the base ten times harder than the active region would, and at 60 mV a decade the junction sits above the 0.65 V of a small-signal bias point.

Every gain in this paper is a voltage ratio, so decibels are $20 \log_{10}$ throughout.

Where a question asks for a design, give an E12 value and state the error it leaves. A design quoted to four figures has not been designed.

Where a question asks by how much, give the arithmetic. An assertion without it scores nothing.

No question asks for C++.

Device parameters

	Value
Thermal voltage V_T	26 mV
Saturation current I_S	10^{-14} A
Forward current gain β	50
Early voltage V_A	100 V
Collector-base capacitance C_{bc}	4 pF
NMOS transconductance at 1 mA	4 mS
MOSFET threshold	2 V

The E12 series

10, 12, 15, 18, 22, 27, 33, 39, 47, 56, 68, 82, and every decade of them.

Marks

Question	Chapter	Marks
1	Chapter 1	10
2	Chapter 2	9
3	Chapter 3	11
4	Chapter 4	10
5	Chapter 5	10
6	Chapter 6	9
7	Chapter 7	11
8	Chapter 8	10
9	Chapter 9	11
10	Chapter 10	9
		100

Question 1 A divider, its resistance, and the load that takes a third 10 MARKS

A 33 kilohm resistor and a 6.8 kilohm resistor form a divider across a 10 V supply, with the output taken at their junction.

(a) Give the open-circuit output voltage, and the divider's Thevenin resistance.

State in one sentence why the Thevenin resistance is neither of the two resistors, and why the supply does not appear in it. (3 marks)

(b) A 10 kilohm load is connected to that output.

Give the output voltage now, and the fraction of the unloaded value it retains. State the two different ways of getting there and show that they agree.

A colleague reports the node as sitting at 1.71 V with the load connected. Say what they have done. (3 marks)

(c) The requirement is that the load takes no more than 1 per cent of the output.

Give the smallest load resistance that meets it, and the largest divider Thevenin resistance that would meet it with the original 10 kilohm load.

Give one way of meeting the requirement without changing either resistor's ratio, and state its cost. (2 marks)

(d) This same subtraction appears in six separate places in this book.

Name three of them, with the quantity being divided down in each, and say what the six have in common. (2 marks)

Question 2 One capacitor, two corners, and the resistance nobody drew

9 MARKS

A 159 nF capacitor is connected from the output of the divider in Question 1 to ground, through a 1 kilohm series resistor. The intent is a low-pass filter with a corner at about 1 kHz.

(a) Give the capacitor's reactance at 1 kHz, and the corner frequency the 1 kilohm resistor and the capacitor give on their own. (2 marks)

(b) Give the corner frequency the circuit actually has, and the ratio between the two.

State what resistance is responsible and why it does not appear anywhere on the schematic. (3 marks)

(c) Give the attenuation, in decibels, that this filter applies at the frequency it was designed to have as its corner. Say what a measurement at that frequency would look like to somebody who did not know about part (b). (2 marks)

(d) A series LC circuit uses 10 mH and 1 microfarad with 10 ohm of series resistance.

Give its resonant frequency, the reactance of the inductor at resonance, and its Q. State what the voltage across the inductor is at resonance, relative to the applied voltage, and why that is not a violation of anything. (2 marks)

Question 3 Two sections, one buffer, and the factor of 1.72

11 MARKS

(a) A single RC low-pass section uses 1 kilohm and 159 nF. Give its corner frequency and its roll-off in decibels per decade well above it. (1 mark)

(b) Two such sections are connected directly in series, output of the first to input of the second.

Give the combined circuit's -3 dB frequency, and the ratio between it and the answer to part (a).

Now give the -3 dB frequency two such sections would have if a perfect buffer were placed between them, and the ratio between *that* and part (a). (4 marks)

(c) State the ratio between your two answers to part (b), and say precisely what the buffer bought. Say why it did not change the roll-off. (2 marks)

(d) An ideal operational amplifier is used as a Schmitt trigger, with positive feedback from the output to the non-inverting input through a 100 kilohm resistor, and a 10 kilohm resistor from that input to ground. The supplies are ± 12 V.

Give the two thresholds, and the hysteresis between them.

The 10 kilohm resistor is changed to 1 kilohm. Give the new thresholds, and say what has happened to the circuit's immunity to a noisy input. (4 marks)

Question 4 Loop gain, the error it divides, and a diode that will not converge 10 MARKS

(a) An amplifier with an open-loop gain of 10^5 is used with a feedback fraction of 0.1.

Give the ideal closed-loop gain, the loop gain, the actual closed-loop gain, and the error as a percentage.

Repeat for open-loop gains of 10^3 and 10^6 , and state the relationship between the loop gain and the error in one sentence. (4 marks)

(b) State what feedback does to a stage's input resistance, its output resistance, its distortion and its bandwidth, giving the factor in each case.

Say which single quantity all four answers are functions of, and why that makes the open-loop gain on its own an incomplete specification. (2 marks)

(c) A diode with $I_S = 10^{-14}$ A is connected in series with a 1 kilohm resistor across a 5 V supply.

Give the diode voltage and the current, working to four significant figures. State the equation you solved and the fact that makes it unsolvable in closed form. (2 marks)

(d) A Newton-Raphson iteration is started from a diode voltage of zero.

Say what happens on the first step and why. Give the approximate number of iterations the unlimited iteration takes to settle, and the number it takes with the step limiting this book specifies.

State what the limiting does, in one sentence, and why it does not change the answer.

(2 marks)

Question 5 Sixty millivolts a decade, and a switch that ignores beta

10 MARKS

(a) Give the base-emitter voltage a transistor with $I_S = 10^{-14}$ A needs for collector currents of 10 microamps, 1 mA and 10 mA.

State the voltage change per decade of current, and derive it. (3 marks)

(b) Name the three regions this book uses, in terms of which junction is forward biased.

State which region an amplifier lives in and which two a switch uses, and say what the transistor is behaving as in each. (2 marks)

(c) A microcontroller output at 5 V must switch a load drawing 150 mA from a 12 V rail.

Design the base drive using a forced beta of 10. Give the base resistor, the nearest E12 value, the base current that value delivers, and the forced beta it actually produces.

The transistor's datasheet gives h_{FE} as 100 minimum and 400 typical. State where either number entered your design, and what that means for building a thousand of them.

(3 marks)

(d) The book's transport model has no branch in it on which region the device is in.

State how saturation arises from it, and say what a model written with three cases instead would show at the boundaries. (2 marks)

Question 6 A bias point that loads itself, and drift that is a loop gain

9 MARKS

A common-emitter stage runs from 10 V. Its base is biased by 33 kilohm and 6.8 kilohm, its emitter resistor is 1 kilohm and its collector resistor is 4.7 kilohm.

(a) Give the collector current the unloaded divider predicts. (1 mark)

(b) Give the base current at that collector current, the voltage it drops across the divider's Thevenin resistance, and the collector current that actually results.

Give the error between the two answers as a percentage, and name the chapter whose arithmetic this is. (3 marks)

(c) Define the divider's stiffness, compute it for this stage, and state the usual rule of thumb.

Give the divider current a stiffness of 200 would need, and say why that is usually the point at which a designer changes the bias arrangement instead. (2 marks)

(d) A tempting argument for the emitter resistor runs: a one degree rise raises the collector current 10 per cent to 1.1 mA, so the emitter voltage rises to 1.1 V, so with the base held at 1.65 V the base-emitter voltage falls to 0.55 V, which reduces the current again.

Compute what a base-emitter voltage of 0.55 V would actually do to the collector current, relative to 0.65 V.

State what is wrong with the argument, give the correct suppression factor for this stage, and name the quantity that factor is equal to. (3 marks)

Question 7 The emitter factor, and the node it belongs to

11 MARKS

A common-emitter stage runs at 1 mA with a 10 kilohm collector resistor and a 234 ohm emitter resistor.

(a) Give r_e , the emitter factor, the stage's voltage gain, and the gain the same stage would have with the emitter resistor bypassed.

Give the resistance looking into the base. (3 marks)

(b) A tempting rule states that the emitter resistor multiplies the stage's output resistance by the emitter factor, giving 100 kilohm here.

State why that cannot be true, **without computing anything**.

Then give the resistance looking into the collector, and the stage's actual output resistance. (4 marks)

(c) The collector resistor is replaced by a current mirror.

Give the stage's output resistance with and without the emitter resistor, and state what the emitter factor is now worth.

Say what this establishes about why mirror loads are used. (2 marks)

(d) The bypassed stage of part (a) is driven from a 1 kilohm source.

Give the capacitance the source sees, and the corner frequency that produces.

State what a cascode does to that number and what it costs. (2 marks)

Question 8 Eight ohms, two Darlington's, and a bias that is not two drops

10 MARKS

(a) An emitter follower biased at 120 mA drives an 8 ohm loudspeaker.

Give its voltage gain, and the resistance looking into its base.

Give the same two numbers for a Darlington follower at the same current and load, and state the relationship between a Darlington's effective emitter resistance and a single transistor's. (4 marks)

(b) The stage of Question 7(a), with an output resistance of 9.89 kilohm and a gain of 38.5, must drive that loudspeaker.

Give the gain it delivers directly into 8 ohms, through the single follower, and through the Darlington. Express each as a percentage of the unloaded gain. (3 marks)

(c) A class-AB output stage is to idle at 120 mA.

Give the emitter resistors the 26 millivolt rule specifies, the nearest E12 value, and the emitter factor that produces.

Give the bias voltage the stage needs between the two bases, computed from the exponential. Then give the idle current that results if the bias is instead set to two constant drops plus the two resistor drops, and the ratio between the two currents.

Say in one sentence why the constant-drop model, worth about 1 per cent in Question 5, is worth that ratio here. (3 marks)

Question 9 The tail, the 520 volts, and the resistor that cancels 11 MARKS

A differential pair runs from a 2 mA tail with 10 kilohm collector resistors, and the output is taken at one collector.

- (a) Give r_e on each side, and say why it is not V_T divided by the tail current.
Give the differential gain to one collector, and the gain to both. Account for both factors of two. (3 marks)
- (b) The tail is a 10 kilohm resistor to a negative rail.
Give the common-mode gain and the common-mode rejection ratio, as a ratio and in decibels.
Show that the collector resistor cancels out of the rejection ratio, and state what that means for a designer who proposes a larger collector resistor to improve it. (4 marks)
- (c) The requirement is 80 dB of rejection at the same tail current.
Give the tail resistance that would need, and the voltage such a resistor would drop.
Give the output resistance a simple current mirror would present at 2 mA and the rejection that gives, and say what has to be done instead. (2 marks)
- (d) The collector resistors are replaced by a current mirror.
Give the differential gain to the single output, and split the improvement over part (a) into its two independent causes with a number for each. (2 marks)

Question 10 A gain budget, and the two stages that amplify nothing 9 MARKS

An operational amplifier is built from four stages: a mirror-loaded differential pair at a 2 mA tail, a Darlington emitter follower at 1 mA, a common-emitter stage at 1 mA with a current-source load, and a class-AB Darlington output idling at 120 mA into 8 ohms. The rails are ± 15 V.

- (a) Give the unloaded gain of stage 1 and of stage 3.
Give the input resistance of stages 2, 3 and 4. (3 marks)
- (b) Give each stage loaded by the next, and the open-loop gain in decibels.
Give the decibel difference between that and the product of the two unloaded gains, and say which single stage accounts for most of it. (3 marks)
- (c) Give the open-loop gain the amplifier would have with the Darlington follower removed and stage 1 driving stage 3 directly.
State what that stage is worth in decibels, and what it costs in signal. (2 marks)

(d) The amplifier is asked to solve for its DC operating point with the loop open, and the output is reported at the negative rail.

State whether this is a fault, and justify your answer with one number. (1 mark)

Written Paper B

Time: 4 hours. **Closed book.** A basic calculator is permitted. **Total: 100 marks.**

This paper exists to let you check your own knowledge. It is not a qualification and it gates nothing. It has one question per chapter, so **it is meant to be taken once you have finished the book**, after Chapter 10 and the capstone. The model answers are in Appendix E.

Rubric

The thermal voltage is 26 mV. Every intrinsic emitter resistance in this paper follows from $r_e = V_T/I_C$ and that figure.

Beta is 50 unless a question says otherwise.

The constant-drop model is $V_{BE} = 0.65\text{ V}$, except where a question names the exponential, and then $V_{BE} = V_T \ln(I_C/I_S)$. **A saturated switch's base-emitter drop is 0.7 V,** because a forced beta drives the base far harder than the active region would.

Every gain in this paper is a voltage ratio, so decibels are $20 \log_{10}$ throughout.

Where a question says "derive", a stated result scores nothing. The working is the answer.

Where a question asks by how much, give the arithmetic.

No question asks for C++.

Device parameters

	Value
Thermal voltage V_T	26 mV
Saturation current I_S	10^{-14} A
Forward current gain β	50
Early voltage V_A	100 V
Collector-base capacitance C_{bc}	4 pF
MOSFET threshold	2 V
MOSFET transconductance parameter k	8 mA/V ²

The E12 series

10, 12, 15, 18, 22, 27, 33, 39, 47, 56, 68, 82, and every decade of them.

Marks

Question	Chapter	Marks
1	Chapter 1	10
2	Chapter 2	9
3	Chapter 3	11
4	Chapter 4	10
5	Chapter 5	10
6	Chapter 6	9
7	Chapter 7	11
8	Chapter 8	10
9	Chapter 9	11
10	Chapter 10	9
		100

Question 1 Where a source resistance comes from

10 MARKS

(a) Derive the Thevenin equivalent of a two-resistor divider across an ideal supply, giving both the open-circuit voltage and the source resistance, and stating what you did to the supply and why that is legitimate. (3 marks)

(b) A resistor network is driven by two independent sources.

State the superposition principle, the condition a network must satisfy for it to hold, and what you replace each source with while the other is acting.

Say what happens to the *power* dissipated in a resistor if you try to superpose it, and why. (3 marks)

(c) A source with an output resistance R_s drives a load R_L .

Derive the value of R_L that extracts the most power from the source, and state the efficiency at that point.

Then state why almost no circuit in this book is designed that way, and name the two situations in which it is. (2 marks)

(d) A colleague argues that a divider's output resistance cannot matter, because "the load is a million ohms and the divider is only a few kilohms."

State the condition under which they are right, express it as a ratio, and give the error their argument makes when the ratio is 2. (2 marks)

Question 2 Why a corner is a corner

9 MARKS

(a) Derive the corner frequency of a series RC low-pass filter from the condition that defines it, and state that condition in terms of reactance.

Give the phase shift at the corner, and the asymptotic phase shift far above it. (3 marks)

(b) State why a capacitor's impedance is written as a complex number rather than as a resistance, and what the imaginary unit is encoding physically.

State the impedance of an inductor and of a capacitor in that notation, and say what the sign difference means for the current. (2 marks)

(c) Derive the resonant frequency of a series LC circuit from the condition that the two reactances cancel.

Then state the Q of a series RLC circuit in terms of its component values, and say what physical quantity Q is the ratio of. (2 marks)

(d) An ideal transformer has N times as many turns on its secondary as on its primary.

State what it does to voltage, to current and to impedance, giving the factor in each case, and show that the third follows from the first two.

Name two things a real transformer does that the ideal model does not. (2 marks)

Question 3 Cascades, virtual shorts, and what an op-amp assumes

11 MARKS

(a) Two identical RC low-pass sections are connected directly in series.

Explain, without algebra, why the combined circuit's -3 dB frequency is **lower** than that of a buffered pair, and identify the mechanism by name.

Then state the general rule for how the -3 dB frequency of n identical buffered sections relates to the single-section corner, and evaluate it for $n = 2$ and $n = 3$. (4 marks)

(b) State the two assumptions the ideal operational amplifier model makes, and derive the virtual short from them.

State precisely which of the two fails first in a real amplifier, and what that failure is called. (3 marks)

(c) Derive the closed-loop gain of the inverting configuration from the virtual short, naming where each assumption is used.

State the inverting amplifier's input resistance and explain why it is not infinite even though the op-amp's input resistance is. (2 marks)

(d) A Sallen-Key low-pass filter is built with equal resistors and equal capacitors and a unity-gain buffer.

State what sets its corner frequency and what sets its Q , and say why those two are separable in this topology when they are not in a passive cascade. (2 marks)

Question 4 Why everything is a function of the loop gain

10 MARKS

(a) Derive the closed-loop gain of a negative-feedback amplifier from the summing junction, stating each step.

Then derive the fractional gain error against the ideal $1/\beta_f$, and show that it depends on A and β_f only through their product. (4 marks)

(b) State what feedback does to input resistance, output resistance, distortion and bandwidth.

Then explain **why** the same factor appears in all four, in terms of what the loop is doing to a disturbance introduced inside it. (2 marks)

(c) An amplifier has a gain-bandwidth product of 1 MHz.

Give its closed-loop bandwidth at closed-loop gains of 10 and of 1000, and its open-loop gain at 1 kHz.

A specification asks for 0.01 per cent gain accuracy at 1 kHz with a closed-loop gain of 10. State whether this amplifier meets it and justify with a number. (2 marks)

(d) Explain why Newton-Raphson applied to a diode from a zero starting point fails, in terms of the shape of the function rather than the arithmetic.

State what step limiting changes and what it deliberately does not, and say why a limiter that changed the converged answer would be a bug rather than a feature. (2 marks)

Question 5 One equation, three regions

10 MARKS

(a) Write the transport form of the bipolar transistor's collector and base currents.

Show that with the collector reverse biased it collapses to $I_C = \beta_F I_B$, and show what happens as the collector-base junction becomes forward biased. (4 marks)

(b) State why this book prefers a model with no branch in it to one with three cases, giving two consequences of the branching version. (2 marks)

(c) Write the MOSFET's square-law drain current in saturation, and derive its transconductance.

Show that g_m goes as the square root of the drain current, and state what that implies about biasing a MOSFET harder to get more gain. (2 marks)

(d) At 1 mA, a bipolar transistor's transconductance is about ten times a MOSFET's.

Derive both, state where the factor of ten comes from, and name two design decisions elsewhere in this book that follow from it. (2 marks)

Question 6 A rule, a load line, and a factor that appears twice 9 MARKS

(a) Draw or describe the load line of a common-emitter stage with a collector and an emitter resistor, giving its equation and its two intercepts.

State where on it an amplifier sits and why, and what happens at each end. (2 marks)

(b) A common rule is to drop about 220 mV across the emitter resistor.

Derive the emitter resistance that gives at an arbitrary collector current, and the emitter factor it produces.

State why the rule is expressed as a voltage rather than as a resistance, and show that the resulting emitter factor is independent of the current. (3 marks)

(c) Show that the factor by which an emitter resistor suppresses thermal drift is the same factor by which it reduces the stage's gain.

State the consequence in one sentence. (2 marks)

(d) A designer proposes to fix the thermal problem by choosing a transistor with a lower temperature coefficient.

State what the coefficient actually is, why it is not a property the designer can choose, and what the emitter resistor is doing that a better device could not. (2 marks)

Question 7 Three results from one model 11 MARKS

(a) State the four steps that turn a biased stage into its small-signal schematic, and justify the first one.

State what the model is blind to, naming three things. (3 marks)

(b) From the model, derive the common-emitter stage's voltage gain and the resistance looking into its base.

State which of the two depends on β and what that means for specifying it. (3 marks)

(c) Derive the emitter factor as the ratio of two gains, and then state the two quantities it is commonly said to decide.

Show which of the two claims is correct as stated, and give the corrected form of the other, naming the node it belongs to. (3 marks)

(d) This book introduces $r_s = 1/g_m$ so that every result transfers to a MOSFET by substitution.

State the two results that transfer and the one that does not, and say why.

Give r_s at 1 mA, the source factor that 220 mV across a source resistor produces, and explain why it is 2 where the emitter factor is 10. (2 marks)

Question 8 A dead band, a rule, and a thing bolted to a heatsink 10 MARKS

(a) Derive the emitter follower's voltage gain from the small-signal model, and state the one quantity its shortfall from unity depends on.

State the follower's input and output resistances, and describe in one sentence what the stage is for. (3 marks)

(b) State the three amplifier classes by what conducts when, and give the maximum theoretical efficiency of two of them.

Explain why crossover distortion is more objectionable than its percentage suggests, in terms of where in the signal it sits. (3 marks)

(c) The output-stage rule is to drop one thermal voltage across each emitter resistor.

Show that this makes the emitter factor exactly 2, whatever the idle current.

State the two costs the rule is balancing, and what each would look like if the rule were moved by a decade in either direction. (2 marks)

(d) The bias generator of a class-AB stage is two diodes bolted to the output devices' heatsink.

Explain what they are doing there. State what happens to a stage whose diodes are on the circuit board instead, and why the two schematics are identical. (2 marks)

Question 9 The tail, the tanh, and the output you choose 11 MARKS

(a) Explain why a resistance R_{tail} in the tail of a differential pair behaves as $2R_{tail}$ to one half of the pair.

Then derive the common-mode gain and the common-mode rejection ratio, and show that the collector resistor cancels. (4 marks)

(b) State what a current source is, in terms of the two quantities that make it useful in a tail, and explain why that is the property no resistor can have.

Give the general principle in one sentence, and name two other places in this book where the same trade appears. (2 marks)

(c) The pair's difference current is $I_{tail} \tanh(v_d/2V_T)$.

Show that the small-signal transconductance follows from it, and derive the input at which the tanh falls 1 per cent below its tangent.

Show that this figure does not depend on the tail current, and state what that means for a designer who wants more linearity. (3 marks)

(d) A pair's rejection is quoted as 52 dB by one engineer and 98 dB by another, from the same circuit.

Explain how both can be right, state which measurement each made, and say which quantity limits each. (2 marks)

Question 10 The gain you cannot buy**9 MARKS**

- (a) A common-emitter stage is loaded by a current source rather than a resistor.
 Derive its voltage gain, and show that the result does not depend on the collector current.
 State what this means for a designer who proposes more supply current to get more gain, and give the only lever that does work. (4 marks)
- (b) An operational amplifier's two gain stages each give 65.7 dB, and the amplifier delivers 119.9.
 Explain where the difference goes and why it cannot be recovered by improving either gain stage. (2 marks)
- (c) Explain what a Miller compensation capacitor is compensating.
 State what it does to the amplifier's open-loop bandwidth, and why deliberately slowing an amplifier down in one place makes it usable. (2 marks)
- (d) The output stage's input resistance depends on β^2 , which varies by a factor of a hundred between devices.
 Explain in one sentence why the closed-loop gain does not, and name the quantity that makes it so. (1 mark)

Model Answers to Paper A

Marked out of 100, with the marks shown per part. **Method carries them:** a correct rule with an arithmetic slip in it is worth more than a correct number with no working, and several questions consume their own earlier answers, so an error should be followed through rather than penalised twice.

Code is marked on method, not on presentation: an answer that states the rule and slips a digit keeps most of the marks.

Question 1 A divider, its resistance, and the load that takes a third

10 MARKS

(a) Two numbers, and why one of them is neither resistor

3 MARKS

$$V_{out} = 10 \times \frac{6.8}{33 + 6.8} = 1.7085 \text{ V}$$

$$R_{th} = 33\text{k} \parallel 6.8\text{k} = \frac{33 \times 6.8}{39.8} \text{ k} = 5.638 \text{ k}$$

(1 mark each)

Why it is neither resistor. Looking back into the output with the supply replaced by its internal resistance, which for an ideal supply is zero, the two resistors both run from the output node to ground. They are in parallel. The supply does not appear because setting a voltage source to zero makes it a short, and a short has no resistance to contribute.

(1 mark. An answer of 6.8 kilohm scores nothing: it is the first named trap)

(b) What a 10 kilohm load does**3 MARKS****Two ways, and they must agree.***Reform the divider.* The lower arm becomes $6.8\text{k} \parallel 10\text{k} = 4.048\text{ kilohm}$, so

$$V = 10 \times \frac{4.048}{33 + 4.048} = \mathbf{1.0925\text{ V}}$$

Or use the Thevenin equivalent. A source of 1.7085 V behind 5.638 kilohm, loaded by 10 kilohm:

$$V = 1.7085 \times \frac{10}{10 + 5.638} = \mathbf{1.0925\text{ V}}$$

*(2 marks for both methods and the agreement. 1 mark for either alone)*It retains $1.0925/1.7085 = \mathbf{63.9}$ per cent, so the load has taken **36 per cent** of the output.*(1 mark)***What the colleague has done.** Computed the open-circuit voltage and quoted it for a loaded node. The divider is not a voltage source; it is a source behind 5.638 kilohm, and the moment anything is connected the answer changes. This is the second named trap.**(c) Meeting a 1 per cent requirement****2 MARKS**For the load to take 1 per cent, $R_{load} / (R_{load} + R_{th}) = 0.99$, so $R_{load} = 99R_{th} = \mathbf{558\text{ k}}$.Turning it round, with a 10 kilohm load the divider needs $R_{th} \leq 10\text{k}/99 = \mathbf{101}$.*(1 mark for both)***Without changing the ratio**, scale both resistors down by $5.638\text{k}/101 = 56$, to about 590 ohm and 120 ohm. **The cost is supply current:** the divider goes from 251 microamps to 14 mA, which is fifty-six times the power for a node that does exactly what it did before. That trade, accuracy against current, is why the answer in practice is usually a buffer instead. *(1 mark)***(d) The same subtraction, six times****2 MARKS**

Any three of:

Lecture	What loaded what	Cost
Chapter 1	10 kilohm on this divider	a third of the output
Chapter 2	the divider's own resistance on an RC filter	the corner moves 6.6 times
Chapter 3	two filter sections on each other	the corner moves 1.72 times
Chapter 6	base current on a bias divider	12 per cent of I_C
Chapter 8	a loudspeaker on a voltage amplifier	99.9 per cent of the gain
Chapter 10	each amplifier stage on the next	11.5 dB of open-loop gain

(1 mark for three with their costs)

What they have in common: in every case a source with a finite output resistance drives a finite load, and the two form a divider. It is one subtraction. What changes between the six is only how large the ratio is, and here it costs everything. (1 mark)

Question 2 One capacitor, two corners, and the resistance nobody drew 9 MARKS

(a) Two easy numbers 2 MARKS

$$X_C = \frac{1}{2\pi \times 1000 \times 159 \times 10^{-9}} = 1001$$

$$f_c = \frac{1}{2\pi \times 1000 \times 159 \times 10^{-9}} = 1001 \text{ Hz}$$

(1 mark each. They are the same arithmetic, and noticing that the corner is where the reactance equals the resistance is worth stating)

(b) The corner the circuit actually has 3 MARKS

The capacitor does not see 1 kilohm. It sees the 1 kilohm **in series with the divider's Thevenin resistance**, because that is what the source looks like:

$$R = 1000 + 5638 = 6638 \Omega, \quad f_c = \frac{1}{2\pi \times 6638 \times 159 \times 10^{-9}} = 151 \text{ Hz}$$

(2 marks)

The ratio is $1001/151 = 6.6$.

(1 mark)

Why it is not on the schematic. Nobody drew a 5.638 kilohm resistor; it is what two resistors that *were* drawn look like from this node. A Thevenin resistance is a property of a network rather than a component in it, which is exactly why this error is easy to make and hard to see. This is the third named trap.

(c) What a measurement would show 2 MARKS

At 1001 Hz the filter is far past its real corner. With $f/f_c = 1001/151 = 6.64$:

$$|H| = \frac{1}{\sqrt{1 + 6.64^2}} = 0.1489, \quad 20 \log_{10}(0.1489) = -16.5 \text{ dB}$$

(1 mark)

What it would look like. Not a filter with a slightly wrong corner: a filter that has removed 85 per cent of the signal at the frequency it was supposed to be passing. Somebody who did not know about part (b) would suspect the capacitor's tolerance, and no capacitor is wrong by a factor of 6.6. **The symptom points at the wrong component**, which is the reason the error is worth a question of its own. (1 mark)

(d) A series LC**2 MARKS**

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{10^{-2} \times 10^{-6}}} = \mathbf{1592 \text{ Hz}}$$

$$X_L = 2\pi f_0 L = \mathbf{100}, \quad Q = \frac{X_L}{R} = \frac{100}{10} = \mathbf{10}$$

(1 mark for all three)

The inductor voltage is **Q** times the applied voltage, so ten times it. It is not a violation of anything because the capacitor's voltage is equal and opposite: the two cancel, the circuit presents 10 ohm, and the energy is being exchanged between L and C rather than supplied. A circuit that appears to produce more voltage than it is given is a circuit storing energy, and the store is what Q measures. *(1 mark)*

Question 3 Two sections, one buffer, and the factor of 1.72**11 MARKS****(a) One section****1 MARK**

1001 Hz, and 20 dB per decade.

*(1 mark for both)***(b) Two sections, loaded and buffered****4 MARKS**

Directly in series: 375 Hz. The second section loads the first, so the two do not behave as two independent single poles. Solving $|H|^2 = 1/2$ for the loaded cascade gives 0.374 times the single-section corner. *(2 marks)*

With a buffer between them: 644 Hz. Two independent identical poles give $|H| = 1/(1 + (f/f_c)^2)$, and setting that to $1/\sqrt{2}$ gives

$$f_{-3} = f_c \sqrt{\sqrt{2} - 1} = 1001 \times 0.6436 = \mathbf{644 \text{ Hz}}$$

(2 marks)

The ratios against one section are **0.374** and **0.644**. **A candidate who gives 1001 Hz for either has treated the sections as independent and given them the single-section corner, which is the fourth named trap.**

(c) What the buffer bought

2 MARKS

$$\frac{644}{375} = 1.72$$

The buffer bought back a factor of 1.72 in corner frequency, and nothing else. (1 mark)

It did not change the roll-off, which is 40 dB per decade either way, because that is set by the number of poles and there are two of them in both circuits. Loading moves *where* the response falls, not how fast. (1 mark)

Note what the buffer is doing, because it is Question 1 again: the first section has an output resistance, the second has an input impedance, and the two form a divider. The buffer's only job is to make that divider's ratio one.

(d) A Schmitt trigger

4 MARKS

The non-inverting input sits at the output divided by the divider:

$$V_{th} = \pm 12 \times \frac{10}{100 + 10} = \pm 1.091 \text{ V}$$

so the hysteresis is 2.18 V. (2 marks)

With 1 kilohm instead:

$$V_{th} = \pm 12 \times \frac{1}{100 + 1} = \pm 0.119 \text{ V}$$

hysteresis 0.238 V. (1 mark)

Its noise immunity has fallen by a factor of nine. The circuit now switches on any disturbance above 119 mV where it previously needed 1.09 V, so an input with a few hundred millivolts of noise on it will chatter. The trade is that the thresholds are closer to the real crossing, so the comparator is more accurate about *when* the input crossed; hysteresis buys immunity and pays in timing. (1 mark)

Question 4 Loop gain, the error it divides, and a diode that will not converge

10 MARKS

(a) Three open-loop gains

4 MARKS

Ideal closed-loop gain is $1/\beta_f = 10$ in all three cases. (1 mark)

A	Loop gain $A\beta_f$	$A_{cl} = A/(1 + A\beta_f)$	Error
10^3	100	9.90099	0.990 %
10^5	10,000	9.99900	0.0100 %
10^6	100,000	9.99990	0.00100 %

(2 marks for the table)

The error is one part in one plus the loop gain, so a loop gain of ten thousand gives an error of one part in ten thousand. Every factor of ten in loop gain is a factor of ten in error.

(1 mark)

(b) What feedback does, and to what

2 MARKS

Quantity	Factor
Gain error	divided by $1 + A\beta_f$
Distortion	divided by $1 + A\beta_f$
Output resistance	divided by $1 + A\beta_f$ (series-shunt)
Input resistance	multiplied by $1 + A\beta_f$
Bandwidth	multiplied by $1 + A\beta_f$

(1 mark)

All five are functions of the loop gain, not of the open-loop gain.

(1 mark)

So an open-loop gain quoted on its own says nothing about any of them: the same amplifier used at a closed-loop gain of 1 and of 1000 has loop gains differing by a factor of a thousand, and every result above differs by that factor. **A specification that gives A without saying what β_f it will be used at is incomplete**, which is the fifth named trap.

(c) A diode and a resistor

2 MARKS

The equation is Kirchhoff's voltage law with the diode's exponential in it:

$$\frac{5 - V_d}{1000} = I_S (e^{V_d/V_T} - 1)$$

(1 mark for stating it)

$$V_d = 0.6965 \text{ V}, \quad I = \frac{5 - 0.6965}{1000} = 4.304 \text{ mA}$$

(1 mark)

It has no closed form because V_d appears both linearly and inside an exponential; that is a transcendental equation and it must be solved numerically. (The Lambert W function expresses it, which is a name for the answer rather than a way of computing it.)

(d) Newton from a cold start

2 MARKS

On the first step the iteration linearises the diode at $V_d = 0$, where its conductance is $I_S/V_T \approx 4 \times 10^{-13} \text{ S}$. That is essentially an open circuit, so the linear solve puts almost the whole 5 V across the diode. Evaluating the exponential at 5 V asks for e^{192} , which overflows, and the next Jacobian is meaningless.

(1 mark)

Unlimited: about 170 iterations. With the log-domain step limiting: 7.

(0.5 marks)

The limiting caps how far the junction voltage may move in one step, replacing a jump into the exponential with a logarithmic advance. It does not change the answer because

it only alters the *path* to the root, not the equation being solved; the converged point still satisfies the same Kirchhoff equation to the same tolerance. (0.5 marks)

Question 5 Sixty millivolts a decade, and a switch that ignores beta

10 MARKS

(a) Three voltages and a slope

3 MARKS

$$V_{BE} = V_T \ln\left(\frac{I_C}{I_S}\right)$$

I_C	V_{BE}
10 microamps	0.5388 V
1 mA	0.6585 V
10 mA	0.7184 V

(2 marks)

59.9 mV per decade, and the derivation is one line:

$$\Delta V_{BE} = V_T \ln(10) = 0.026 \times 2.3026 = 59.9 \text{ mV}$$

(1 mark. "About 60 mV" with the $V_T \ln 10$ shown is full marks; 60 mV asserted is not)

(b) Three regions

2 MARKS

Region	Base-emitter	Base-collector
Cutoff	reverse	reverse
Active	forward	reverse
Saturation	forward	forward

(1 mark)

An amplifier lives in the active region, where the transistor behaves as a **current source** controlled by its base-emitter voltage. A switch uses cutoff and saturation, where it behaves as an **open circuit** and as a **resistor** respectively. (1 mark)

(c) A switch

3 MARKS

$$I_B = \frac{I_{load}}{\beta_{forced}} = \frac{150 \text{ mA}}{10} = 15 \text{ mA}, \quad R_B = \frac{5 - 0.7}{15 \text{ mA}} = 287$$

Nearest E12: 270 ohm.

(1 mark. The 0.7 V is the rubric's, and it is the figure baseResistor uses: a base driven ten times harder than the active region needs sits above the 0.65 V of a bias point. Using 0.65 gives 290 ohm and the same E12 value, and costs nothing)

$$I_B = \frac{5 - 0.7}{270} = 15.9 \text{ mA}, \quad \beta_{forced} = \frac{150}{15.9} = 9.4$$

(1 mark. Note the E12 value moves the drive the safe way, harder into saturation)

Neither datasheet number entered the design. The forced beta of 10 was chosen; h_{FE} appears nowhere. (1 mark)

For a thousand of them, that is the whole point. A design that used the typical 400 would need 0.375 mA of base drive and would fail on every device near the 100 minimum. A design at a forced beta of 10 works on any device whose h_{FE} exceeds 10, which is all of them by a factor of ten, and its saturation voltage does not depend on which device was fitted. **A switch designed from h_{FE} is the sixth named trap.**

(d) Saturation without a branch

2 MARKS

The transport model carries two exponentials, one for each junction:

$$I_C = I_S (e^{V_{BE}/V_T} - e^{V_{BC}/V_T}) - \frac{I_S}{\beta_R} (e^{V_{BC}/V_T} - 1)$$

With the collector reverse biased the second exponential is negligible and this collapses to $I_C = \beta_F I_B$. Drive the base hard enough that the collector falls below it and the second exponential grows, the collector current stops rising, and the device is saturated. **Saturation is not a case the model handles; it is what the equation does.** (1 mark)

A model written with three cases would show discontinuities at the boundaries, because the three expressions do not meet with matching values and slopes. Those show up as a solver that will not converge near the boundary, and as kinks in any curve that crosses it, which is exactly where a switch operates. (1 mark)

Question 6 A bias point that loads itself, and drift that is a loop gain

9 MARKS

(a) The naive answer

1 MARK

$$V_B = 1.7085 \text{ V}, \quad V_E = 1.7085 - 0.65 = 1.0585 \text{ V}, \quad I_C \approx \frac{1.0585}{1000} = \mathbf{1.059 \text{ mA}}$$

(1 mark)

(b) The answer the circuit gives

3 MARKS

The base draws I_C/β , and that current flows out of the divider through its Thevenin resistance of 5.638 kilohm. It is self-referential, so it must be solved rather than substituted: the base voltage depends on the base current, which depends on the collector current, which depends on the base voltage.

Converged:

$$I_B = \frac{0.934 \text{ mA}}{50} = 18.7 \mu\text{A}, \quad \Delta V_B = 18.7 \mu\text{A} \times 5.638 \text{ k}\Omega = \mathbf{105 \text{ mV}}$$

$$V_B = 1.7085 - 0.105 = 1.603 \text{ V}, \quad I_E = \frac{1.603 - 0.65}{1000} = 0.953 \text{ mA}$$

$$I_C = \frac{\beta}{\beta + 1} I_E = \mathbf{0.934 \text{ mA}}$$

(2 marks. 1 mark for a single non-iterated pass that lands near 0.93. Taking the emitter resistor to carry I_C rather than I_E gives 0.951 mA and does not lose a mark, but it is the $1/\beta$ the rest of the question is about)

The error is $(1.059 - 0.934)/1.059 = \mathbf{12 \text{ per cent}}$, and it is **Chapter 1's loading arithmetic**: the divider has a source resistance and the base is a load on it.

(1 mark. Computing the unloaded answer only is the seventh named trap)

(c) Stiffness

2 MARKS

$$\text{stiffness} = \frac{I_{\text{divider}}}{I_B} = \frac{10/(33\text{k} + 6.8\text{k})}{18.7 \mu\text{A}} = \frac{251 \mu\text{A}}{18.7 \mu\text{A}} = \mathbf{13.4}$$

The rule of thumb is ten, and this stage meets it, which is why the error is 12 per cent rather than 1: a stiffness of ten *accepts* about a tenth of the droop. (1 mark)

A stiffness of 200 needs a divider current of $I_C \times 200/\beta = \mathbf{4.1 \text{ mA}}$, **four times what the stage amplifies with. That is the point at which a designer stops**, because the bias network is now the dominant power consumer, and uses a Darlington or a MOSFET input instead so that the base current is negligible in the first place.

(1 mark. Scaling the present base current instead, $200 \times 18.7 \mu\text{A} = 3.7 \text{ mA}$, keeps the mark: it is the right order and the right conclusion. It is also 8 per cent light, and for (b)'s reason: a stiffer divider droops less, so I_C climbs back towards 1.06 mA and I_B with it, and 3.7 mA is a stiffness of 181)

(d) An argument that does not survive its own numbers**3 MARKS**

$$\frac{I_C(0.55)}{I_C(0.65)} = e^{-0.100/0.026} = \frac{1}{47}$$

The collector current would fall by a factor of 47, not by 10 per cent. (1 mark)

What is wrong is not the physics but the size of the step: **the argument applies an open-loop change to a closed-loop circuit.** The circuit never reaches a base-emitter voltage of 0.55 V, because long before the emitter voltage has risen 100 mV the current has been pulled back. A story about voltages moving in sequence has no place to put a loop gain. (1 mark)

The correct treatment is a local feedback loop. The disturbance is V_{BE} drifting 2 mV per degree, the emitter follows, and

$$\Delta I_C = \frac{2 \text{ mV}}{1 \text{ k}\Omega} = 2 \text{ }\mu\text{A on } 0.934 \text{ mA} = 0.21 \text{ per cent per degree}$$

against about 8 per cent without the resistor, so the suppression factor is

$$1 + \frac{R_E}{r_e} = 1 + \frac{1000}{27.8} = 37$$

and that quantity is the emitter factor, the same number that divides the stage's gain. The stability costs gain one for one.

(1 mark. The factor of 47 without naming the emitter factor is 2 of the 3)

Question 7 The emitter factor, and the node it belongs to**11 MARKS****(a) Four numbers and a resistance****3 MARKS**

$$r_e = \frac{26 \text{ mV}}{1 \text{ mA}} = 26, \quad EF = \frac{26 + 234}{26} = 10$$

$$A_v = -\frac{R_C}{r_e + R_E} = -\frac{10000}{260} = -38.5, \quad A_{v(\text{bypassed})} = -\frac{10000}{26} = -385$$

$$Z_{in(\text{base})} = \beta(r_e + R_E) = 50 \times 260 = 13 \text{ k}$$

(3 marks, one per pair)

(b) Why the tempting rule cannot be right

4 MARKS

Without computing anything: a stage's output resistance is the collector resistor **in parallel** with whatever the transistor presents at that node. A parallel combination is always smaller than either of its parts. So the answer cannot exceed 10 kilohm, and 100 kilohm does.

(2 marks. This is the argument, and it is worth two marks precisely because it needs no arithmetic. A candidate who reaches the right answer only by computing has not made the point)

What the emitter factor does multiply is the resistance **looking into the collector**:

$$R_{into\ collector} = r_o [1 + g_m(R_E \parallel r_\pi)] + (R_E \parallel r_\pi)$$

with $r_o = V_A/I_C = 100$ kilohm and $r_\pi = \beta r_e = 1.3$ kilohm:

$$R_E \parallel r_\pi = 234 \parallel 1300 = 198\ \Omega, \quad R_{into\ collector} = 100k \left[1 + \frac{198}{26} \right] + 198 = 863\ k$$

$$R_{out} = 10k \parallel 863k = 9.89\ k$$

(2 marks)

Against **9.09 kilohm** for the same stage with no emitter resistor: **degeneration bought 9 per cent**, not a factor of ten. **This is the eighth named trap and it is the largest disagreement in the book.**

(c) A mirror load

2 MARKS

The mirror presents its own r_o , 100 kilohm, instead of a 10 kilohm resistor:

$$R_{out} = 100k \parallel 863k = 89.6\ k, \quad \text{without } R_E : 100k \parallel 100k = 50\ k$$

The emitter factor is now worth a factor of 1.8 rather than 1.09. *(1 mark)*

What this establishes. With a resistive load the degeneration boost is real and invisible, because R_C swamps it. A mirror load is how you stop throwing it away. So the correction is not a demolition of the emitter factor; it supplies the missing reason for mirror loads, which otherwise appear from nowhere. *(1 mark)*

(d) Miller

2 MARKS

$$C_{in} = C_{bc}(1 + |A_v|) = 4\ pF \times 386 = 1.54\ nF$$

$$f = \frac{1}{2\pi \times 1000 \times 1.54\ nF} = 103\ kHz$$

(1 mark. Quoting 4 pF and 40 MHz is the ninth named trap)

A cascode removes the multiplication entirely, because the lower transistor's collector no longer swings, so the input sees 4 pF and the corner moves to about 40 MHz. **The cost** is one more transistor, one more bias voltage, and about 0.7 V of output swing. *(1 mark)*

Question 8 Eight ohms, two Darlington's, and a bias that is not two drops

10 MARKS

(a) One follower and two

4 MARKS

At 120 mA, $r_e = 26 \text{ mV}/120 \text{ mA} = 0.217 \Omega$.

Single follower:

$$A_v = \frac{8}{0.217 + 8} = 0.974, \quad Z_{in} = 50 \times 8.217 = 411$$

Darlington:

$$A_v = \frac{8}{0.433 + 8} = 0.949, \quad Z_{in} = 50^2 \times (0.433 + 8) = 21.1 \text{ k}$$

(3 marks)

A Darlington's effective emitter resistance is exactly twice a single transistor's, and β does not appear in that statement. The output device runs at the output current and contributes r_e ; the input device runs at the output device's base current, which is β times smaller, so its own r_e is β times larger, and it is seen through the output device's current gain, so it contributes r_e again. The two betas cancel.

(1 mark. Using $\beta^2(r_e + R)$ rather than $\beta^2(2r_e + R)$ is the tenth named trap and costs this mark only)

(b) What the loudspeaker costs

3 MARKS

	What the stage sees	Gain	Kept
8 ohm direct	8 ohm	0.031	0.08 %
Through the follower	411 ohm	1.53	4.0 %
Through the Darlington	21.1 kilohm	26.2	68 %

(3 marks, one per row, each needing the loading divider $R_{load}/(R_{out} + R_{load})$)

One follower is not enough, and that is not a small shortfall to be tuned away: 4 per cent is better than 0.08 by a factor of fifty and still useless.

(c) A class-AB bias

3 MARKS

$$R_E = \frac{26 \text{ mV}}{120 \text{ mA}} = 0.217 \Omega \rightarrow 0.22 \text{ (E12)}, \quad EF = \frac{0.217 + 0.22}{0.217} = 2.0$$

(1 mark)

$$V_{bias} = 2(V_{BE}(120 \text{ mA}) + I_q R_E) = 2(0.783 + 0.026) = 1.619 \text{ V}$$

The constant-drop version is $2(0.65 + 0.026) = 1.353 \text{ V}$, and applying that to the real stage gives an idle current of **1.96 mA**, a ratio of **61**.

(1 mark)

Why the model changes value. Every earlier use computed a **current from a voltage across a resistor**: subtract 0.65 from a base voltage and divide by R_E . A 133 mV error in a subtraction that yields a volt is 12 per cent, and it enters linearly. Here the calculation runs the other way: the bias lands almost entirely across two **junctions**, so inverting it means going backwards through an exponential of scale 26 mV, and 133 mV is $e^{5.1}$. **A model's accuracy is a property of the calculation it appears in, not of the model.**

(1 mark. This is the eleventh named trap and the sentence is the answer)

Question 9 The tail, the 520 volts, and the resistor that cancels

11 MARKS

(a) Half the tail

3 MARKS

$$r_e = \frac{V_T}{I_{tail}/2} = \frac{26 \text{ mV}}{1 \text{ mA}} = 26$$

Each side carries half the tail, so r_e follows from 1 mA and not from 2. Reading it from the tail current gives 13 ohm and halves every gain below.

(1 mark. This is the twelfth named trap)

$$A_{dm(one collector)} = -\frac{R_C}{2r_e} = -\frac{10000}{52} = -192, \quad A_{dm(both)} = -385$$

(1 mark)

The two factors of two. The first is the **input split**: a differential input of v_d puts only $v_d/2$ on each base, and that is what "differential" means. The second is the **output**: taking one collector discards the other half, which moves oppositely. The first is unavoidable; the second is a choice, and a current-mirror load recovers it.

(1 mark)

(b) Rejection, and the term that cancels

4 MARKS

The tail carries both currents, so one half sees it as $2R_{tail}$ of degeneration:

$$A_{cm} = -\frac{R_C}{2R_{tail} + r_e} = -\frac{10000}{20026} = -0.499$$

(1 mark)

$$CMRR = \frac{A_{dm}}{A_{cm}} = \frac{R_C/2r_e}{R_C/(2R_{tail} + r_e)} = \frac{2R_{tail} + r_e}{2r_e} = \frac{20026}{52} = 385 = 51.7 \text{ dB}$$

(2 marks)

R_C **cancels**, exactly, because it is in the numerator of both gains.

(1 mark)

So the designer proposing a larger collector resistor is wrong. It doubles both gains and rejects exactly as badly as before. Rejection is a property of the tail and the operating current, and no choice of load affects it. **This is the thirteenth named trap, and the instinct behind it is close to universal.**

(c) Eighty decibels**2 MARKS**

$$CMRR = \frac{2R_{tail} + r_e}{2r_e} = 10^4 \Rightarrow R_{tail} = 260 \text{ k}$$

$$V = 260 \text{ k}\Omega \times 2 \text{ mA} = 520 \text{ V}$$

(1 mark)

A simple mirror is not enough either. Its output resistance is $r_o = V_A/I_{tail} = 50 \text{ kilohm}$, giving $50000/26 = 1923 = 65.7 \text{ dB}$, which falls 14 dB short. **The answer is a cascoded mirror**, whose output resistance is $\beta r_o = 2.5 \text{ megohm}$, giving about 100 dB for one extra transistor and 0.7 V of headroom.

*(1 mark)***(d) A mirror load****2 MARKS**

$$A_{dm} = -\frac{r_o \parallel r_o}{r_e} = -\frac{50 \text{ k}\Omega}{26} = -1923$$

(1 mark)

A factor of ten over part (a), and it is two independent factors:

- **A factor of two from the mirror as a mirror.** It turns the idle side's current around and adds it to the output node, recovering the half a single-ended output discards.
- **A factor of five from the load.** $r_o \parallel r_o$ is 50 kilohm where the resistor was 10.

(1 mark. Both, with the numbers. Naming only one scores nothing, because the question asks for the split)

Question 10 A gain budget, and the two stages that amplify nothing

9 MARKS**(a) Six numbers****3 MARKS**

Stage 1 is loaded by $r_o \parallel r_o$ at 1 mA per side, which is 50 kilohm, so $A = 50000/26 = 1923$. **Stage 3** is the same arithmetic at 1 mA: **1923**.

(1 mark)

Stage	Input resistance
3, common-emitter at 1 mA	$\beta r_e = 1.3 \text{ k}$
2, Darlington follower into that	$\beta^2(2r_e + 1300) = 3.4 \text{ M}$
4, Darlington output into 8 ohm	$\beta^2(2 \times 0.217 + 8) = 21.1 \text{ k}$

(2 marks)

(b) The budget

3 MARKS

Stage	Loaded gain
1	$1923 \times 3.4\text{M} / (50\text{k} + 3.4\text{M}) = \mathbf{1895}$
2	$1300 / (52 + 1300) = \mathbf{0.962}$
3	$1923 \times 21.1\text{k} / (50\text{k} + 21.1\text{k}) = \mathbf{570}$
4	$8 / (0.433 + 8) = \mathbf{0.949}$

$$A_{open} = 1895 \times 0.962 \times 570 \times 0.949 = 986,000 = \mathbf{119.9 \text{ dB}}$$

(2 marks)

The two unloaded gains multiply to $1923^2 = 3.70$ million, which is 131.4 dB, so **loading costs 11.5 dB. Stage 3 accounts for 10.6 of it**, driving the output stage; everything else together is under a decibel.

(1 mark. A budget quoted at 131 dB is the fourteenth named trap)

(c) The stage that does nothing

2 MARKS

Without the buffer, stage 1 drives 1.3 kilohm from 50 kilohm and keeps 2.5 per cent:

$$1923 \times 0.0253 = 48.7, \quad A_{open} = 48.7 \times 570 \times 0.949 = 26,400 = \mathbf{88.4 \text{ dB}}$$

(1 mark)

The buffer is worth $119.9 - 88.4 = 31.5$ dB, and it costs 0.34 dB of signal. An exchange rate of about a hundred to one, from a stage with a voltage gain below one, which is why the answer to almost every loading problem in analog design is another follower. (1 mark)

(d) A rail is not a fault

1 MARK

Not a fault. With an open-loop gain of 986,000 and rails of ± 15 V, an input offset of

$$\frac{15}{986000} = \mathbf{15 \text{ V}}$$

saturates the output, and no pair of transistors matches to 15 microvolts. **The amplifier has no open-loop operating point**, so there is no mid-rail answer for a solver to find.

(1 mark for the number and the conclusion. Blaming the solver is the fifteenth named trap)

Model Answers to Paper B

Marked out of 100, with the marks shown per part. **Method carries them:** a correct rule with an arithmetic slip in it is worth more than a correct number with no working, and several questions consume their own earlier answers, so an error should be followed through rather than penalised twice.

Where a question says “derive”, the working carries the marks and a stated result scores nothing. Several parts carry no arithmetic at all; they are not worth fewer marks for it, and a paper marked only on the numbers would pass a candidate who has understood none of the book.

Question 1 Where a source resistance comes from 10 MARKS

(a) The Thevenin equivalent, derived 3 MARKS

Open-circuit voltage. With nothing connected, the same current flows through both resistors, so

$$V_{oc} = V_{supply} \frac{R_2}{R_1 + R_2} \quad (1 \text{ mark})$$

Source resistance. Look back into the output terminal with every independent source set to zero. An ideal voltage source set to zero is a short circuit, because a source that holds zero volts whatever current flows through it is a short. With the supply shorted, R_1 runs from the output node to ground and so does R_2 :

$$R_{th} = R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}$$

(2 marks: 1 for the answer, 1 for justifying the shorting rather than asserting it)

Why that is legitimate. The source resistance is defined as $\partial V / \partial I$ at the terminal, and a fixed source contributes nothing to a *derivative*. Setting it to zero is not an approximation; it is what differentiating means for a linear network.

(b) Superposition**3 MARKS**

The principle. In a linear network the response to several independent sources acting together is the sum of the responses to each acting alone. (1 mark)

The condition is linearity: every element's response must be proportional to its excitation. Resistors, capacitors and inductors qualify; a diode, a transistor's exponential and anything with a product of two signals in it do not. (1 mark)

While one source acts, replace the others by their internal resistances: voltage sources by short circuits, current sources by open circuits.

Power does not superpose. Power in a resistor is V^2/R , and

$$(V_1 + V_2)^2 = V_1^2 + V_2^2 + 2V_1V_2$$

The cross term is not zero, so the total power is not the sum of the individual powers. **Power is a quadratic function of the response, and superposition applies only to linear ones.**

(1 mark. This is the whole answer and it fits in one equation)

(c) Maximum power transfer**2 MARKS**

$$P = I^2 R_L = \left(\frac{V}{R_s + R_L} \right)^2 R_L$$

Differentiating with respect to R_L and setting the result to zero:

$$\frac{dP}{dR_L} = V^2 \frac{(R_s + R_L)^2 - 2R_L(R_s + R_L)}{(R_s + R_L)^4} = 0 \Rightarrow R_s + R_L = 2R_L \Rightarrow \boxed{R_L = R_s}$$

At that point the two resistances drop equal voltages, so **the efficiency is 50 per cent:** as much power is burnt in the source as delivered to the load. (1 mark)

Almost nothing in this book is designed that way, because a circuit that throws away half its power to maximise the absolute quantity delivered is optimising the wrong thing. Voltage amplification wants $R_L \gg R_s$, which delivers less power and more signal.

The two situations where matching is right are **radio-frequency work**, where an unmatched line reflects and the reflection is the problem rather than the loss, and **anything where the source's power is fixed and small**, such as an antenna or a transducer, where there is no more to be had and efficiency is meaningless. (1 mark)

(d) When the colleague is right**2 MARKS**

They are right when $R_L \gg R_{th}$, and the useful form is the ratio:

$$\text{fraction retained} = \frac{R_L}{R_L + R_{th}} = \frac{1}{1 + R_{th}/R_L}$$

so the error is about R_{th}/R_L for small ratios. A megohm on a few kilohm is a ratio of about 500, so the error is 0.2 per cent and they are right. (1 mark)

At a ratio of 2, the node retains 2/3 and the error is **33 per cent**. (1 mark)

The argument is not wrong; it is **unquantified**. The same sentence is correct at a ratio of 500 and catastrophic at 2, and it contains nothing that distinguishes them. That is why this book computes the ratio every time rather than judging it.

Question 2 Why a corner is a corner

9 MARKS

(a) The corner, derived

3 MARKS

$$H(j\omega) = \frac{1/j\omega C}{R + 1/j\omega C} = \frac{1}{1 + j\omega RC}$$

$$|H| = \frac{1}{\sqrt{1 + (\omega RC)^2}} = \frac{1}{\sqrt{2}} \Rightarrow \omega RC = 1 \Rightarrow \boxed{f_c = \frac{1}{2\pi RC}}$$

(2 marks)

The condition is that the capacitor's reactance equals the resistance. $X_C = 1/\omega C$, and setting $X_C = R$ gives the same $\omega = 1/RC$. That is what a corner *is*: the frequency at which the two impedances in the divider are equal, so the output is the input divided by $\sqrt{2}$ and shifted by 45 degrees. (1 mark)

Phase: $\arg H = -\arctan(\omega RC)$, which is **-45** degrees at the corner and tends to **-90** degrees far above it.

(b) Why complex

2 MARKS

Because a capacitor's current and voltage are not proportional; they are related by a derivative. For a sinusoid, differentiating multiplies by ω and advances the phase by 90 degrees, and j is exactly "multiply by one, rotate by 90 degrees". So the imaginary unit encodes the **quarter-cycle phase relationship**, and writing the impedance as a complex number lets a differential equation be solved with the algebra of a resistor divider. (1 mark)

$$Z_L = j\omega L, \quad Z_C = \frac{1}{j\omega C} = -\frac{j}{\omega C}$$

The signs are opposite, so in an inductor the current **lags** the voltage and in a capacitor it **leads**. That opposition is why they can cancel, which is resonance. (1 mark)

(c) Resonance and Q**2 MARKS**

$$X_L = X_C \Rightarrow \omega L = \frac{1}{\omega C} \Rightarrow \omega^2 = \frac{1}{LC} \Rightarrow f_0 = \frac{1}{2\pi\sqrt{LC}}$$

(1 mark)

$$Q = \frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

Q is the ratio of energy stored to energy dissipated per radian of oscillation. Equivalently it is the ratio of the resonant frequency to the -3 dB bandwidth, and it is why the voltage across either reactance at resonance is Q times the applied voltage: that voltage is the store, not a gain.

(1 mark)

(d) The transformer**2 MARKS**

	Factor
Voltage	$\times N$
Current	$\times 1/N$
Impedance seen at the primary	$\times 1/N^2$

The third follows from the first two. An impedance Z on the secondary carries V_s/Z . The primary supplies N times that current at $1/N$ times that voltage, so

$$Z_{\text{primary}} = \frac{V_p}{I_p} = \frac{V_s/N}{NI_s} = \frac{1}{N^2} \cdot \frac{V_s}{I_s} = \frac{Z}{N^2}$$

(1 mark for the table, 1 for the derivation)

Two things a real transformer does: any two of leakage inductance (flux that does not couple), winding resistance, core loss, magnetising current (a finite primary inductance, so it will not pass DC and rolls off at low frequency), and saturation.

Question 3 Cascades, virtual shorts, and what an op-amp assumes

11 MARKS

(a) Why the loaded cascade is worse

4 MARKS

Without algebra: the second section is a load on the first. At any frequency where the second section's input impedance is comparable with the first section's output impedance, the two form a divider, and the first section delivers less than it would into an open circuit. That loss grows with frequency, because the second section's shunt capacitor's reactance falls. So the combined response is lower everywhere above DC, and the frequency at which it reaches -3 dB comes sooner. **The mechanism is loading**, and it is the same one as Question 1.

(2 marks)

For n identical buffered sections:

$$|H| = \left(\frac{1}{\sqrt{1 + (f/f_c)^2}} \right)^n = \frac{1}{\sqrt{2}} \Rightarrow f_{-3} = f_c \sqrt{2^{1/n} - 1}$$

(1 mark)

n	Factor	With $f_c = 1001$ Hz
2	0.644	644 Hz
3	0.510	510 Hz

(1 mark)

Note that even buffered, cascading lowers the corner. Two sections at 1001 Hz do not give a corner at 1001 Hz with a steeper slope; they give 644 Hz. Loading then takes it further, to 375 Hz. Both effects are real and they are different effects.

(b) The two assumptions

3 MARKS

1. The open-loop gain is infinite.
2. The input current is zero, so the input resistance is infinite.

(1 mark)

The virtual short, derived. The output is $A(V_+ - V_-)$. For the output to be finite with A infinite, $V_+ - V_-$ must be zero, so $V_+ = V_-$. The inputs are at the same voltage without being connected, which is why it is a virtual short and not a short: no current flows between them.

(1 mark. An answer that asserts $V_+ = V_-$ without the limiting argument gets nothing: the point is that it is a consequence of the gain, and it fails exactly when the gain does)

The gain assumption fails first, and its failure has a name in this book: the **gain error**, one part in one plus the loop gain. It fails at every frequency (the gain rolls off), at every signal level (finite output swing) and at every closed-loop gain (the loop gain falls as the

closed-loop gain rises). The input-current assumption fails much more gently, as an input bias current of tens of nanoamps. (1 mark)

(c) The inverting configuration

2 MARKS

With V_+ grounded, the virtual short puts V_- at ground too. No current enters the input, so all the current through R_{in} continues through R_f :

$$\frac{V_{in} - 0}{R_{in}} = \frac{0 - V_{out}}{R_f} \Rightarrow A = -\frac{R_f}{R_{in}}$$

The virtual short set $V_- = 0$ and the zero input current let the two currents be equated. (1 mark)

The input resistance is R_{in} , not infinity, because the source does not drive the op-amp's input; it drives the left end of R_{in} , whose right end is held at ground by the loop. The op-amp's own input resistance is behind that node and the source never sees it. (1 mark)

(d) Sallen-Key

2 MARKS

The corner is set by the RC product, $f_c = 1/2\pi RC$ for the equal-component case. The Q is set by the amplifier's gain, which for the unity-gain equal-component form is fixed at 0.5, and which is adjusted in practice either by making the gain greater than one or by making the two resistors or two capacitors unequal. (1 mark)

They are separable because the amplifier supplies energy. In a passive cascade every section loads every other, and both the corner and the damping are consequences of the same component values; there is nothing left to adjust independently. A Sallen-Key stage feeds part of the output back through a capacitor, and how much it feeds back is a free parameter that moves the poles' angle without moving their radius. Q is a property of positive feedback here, and passive networks have none available. (1 mark)

Question 4 Why everything is a function of the loop gain

10 MARKS

(a) The closed loop, derived

4 MARKS

The amplifier sees the input minus the fed-back fraction of the output:

$$V_{out} = A(V_{in} - \beta_f V_{out})$$

$$V_{out}(1 + A\beta_f) = AV_{in} \Rightarrow A_{cl} = \frac{A}{1 + A\beta_f}$$

(2 marks)

The error against the ideal. As $A \rightarrow \infty$, $A_{cl} \rightarrow 1/\beta_f$. So

$$\frac{1/\beta_f - A_{cl}}{1/\beta_f} = 1 - \frac{A\beta_f}{1 + A\beta_f} = \boxed{\frac{1}{1 + A\beta_f}}$$

(2 marks)

A and β_f enter only as their product, which is the loop gain. Neither appears alone anywhere in the result. That is the whole content of the derivation and it is why the loop gain is the quantity worth naming.

(b) Why one factor for four results

2 MARKS

Quantity	Factor
Gain error	$\div (1 + A\beta_f)$
Distortion	$\div (1 + A\beta_f)$
Output resistance	$\div (1 + A\beta_f)$
Input resistance	$\times (1 + A\beta_f)$
Bandwidth	$\times (1 + A\beta_f)$

(1 mark)

Why the same factor. Every one of them is a **disturbance introduced inside the loop**: a distortion product generated by the amplifier, a voltage drop caused by load current, a gain that is not what it should be. The loop measures the output, compares it with what the input asked for, and drives the difference back in with a gain of $A\beta_f$. Any disturbance originating inside therefore appears at the output divided by one plus that. **It is one mechanism seen four times, and the four are not independent properties that happen to share a formula.**

(1 mark)

(c) Gain-bandwidth

2 MARKS

Closed-loop gain	Bandwidth
10	100 kHz
1000	1 kHz

Open-loop gain at 1 kHz: $10^6/10^3 = 1000$.

(1 mark)

It does not meet the specification. At 1 kHz with a closed-loop gain of 10, the loop gain is $1000 \times 0.1 = 100$, so the error is $1/101 = 0.99$ per cent against a requirement of 0.01. It is short by a factor of a hundred, and it is short at 1 kHz on an amplifier whose datasheet says 1 MHz. **Closed-loop accuracy has a bandwidth of its own and it is far below the closed-loop bandwidth.**

(1 mark)

(d) Newton on an exponential**2 MARKS**

The shape is the problem. Newton replaces the function with its tangent and jumps to where the tangent crosses zero. At $V_d = 0$ the diode's slope is I_S/V_T , about 4×10^{-13} siemens, so the tangent is almost horizontal and crosses zero a very long way away. The method is built on the function being locally straight, and an exponential is locally straight only over a few V_T . (1 mark)

Step limiting caps how far a junction voltage may move in one iteration, replacing a jump into the exponential with a logarithmic advance. **It changes the path and not the equation.** The converged point is where the residual is zero, and the residual is computed from the same Kirchhoff equation either way.

A limiter that changed the answer would be a bug, because it would mean the solver was reporting the solution to a problem nobody posed. The whole contract of a nonlinear solver is that the answer depends on the circuit and the tolerance, never on how it got there. (1 mark)

Question 5 One equation, three regions**10 MARKS****(a) The transport form****4 MARKS**

$$I_C = I_S (e^{V_{BE}/V_T} - e^{V_{BC}/V_T}) - \frac{I_S}{\beta_R} (e^{V_{BC}/V_T} - 1)$$

$$I_B = \frac{I_S}{\beta_F} (e^{V_{BE}/V_T} - 1) + \frac{I_S}{\beta_R} (e^{V_{BC}/V_T} - 1)$$

(2 marks)

Collector reverse biased. V_{BC} is negative and large, so $e^{V_{BC}/V_T} \rightarrow 0$:

$$I_C \rightarrow I_S e^{V_{BE}/V_T}, \quad I_B \rightarrow \frac{I_S}{\beta_F} e^{V_{BE}/V_T} \Rightarrow \frac{I_C}{I_B} = \beta_F$$

(1 mark)

Collector junction forward biased. e^{V_{BC}/V_T} grows. In I_C it appears with a **minus** sign, so the collector current stops rising and then falls; in I_B it appears with a **plus** sign, so the base current rises. The ratio I_C/I_B collapses. **That is saturation, and it is what the equation does rather than a case it handles.** (1 mark)

(b) Why no branch**2 MARKS**

Two consequences of the three-case version, either of which is worth the marks:

- **Discontinuities at the boundaries.** Three separately written expressions do not meet with matching values and slopes, so a curve that crosses a boundary has a kink in it, and a solver whose Jacobian jumps there may not converge. A switch operates exactly at that boundary.

- **The model stops being one model.** Three cases have three sets of parameters that can drift apart, and a correction applied to one is not applied to the others. The transport form has one saturation current and two betas, and every region is the same expression.

(2 marks, 1 each)

(c) The square law

2 MARKS

$$I_D = \frac{k}{2}(V_{GS} - V_{th})^2$$

$$g_m = \frac{dI_D}{dV_{GS}} = k(V_{GS} - V_{th})$$

and substituting $V_{GS} - V_{th} = \sqrt{2I_D/k}$:

$$g_m = \sqrt{2kI_D}$$

(1 mark)

So g_m goes as $\sqrt{I_D}$. Biasing a MOSFET four times harder doubles its transconductance, where a bipolar transistor's would quadruple. Current is therefore an expensive way to buy gain from a MOSFET, and the cheaper lever is width.

(1 mark)

(d) The factor of ten

2 MARKS

$$g_{m(BJT)} = \frac{I_C}{V_T} = \frac{1 \text{ mA}}{26 \text{ mV}} = 38.5 \text{ mS}$$

$$g_{m(MOS)} = \sqrt{2 \times 8 \text{ mA/V}^2 \times 1 \text{ mA}} = 4.0 \text{ mS}$$

Ratio 9.6.

(1 mark)

Where it comes from: the bipolar transistor's current is *exponential* in its control voltage and the MOSFET's is *quadratic*. A derivative of an exponential is the exponential itself divided by 26 mV; a derivative of a square is linear. The scale V_T is small and fixed by physics, whereas the MOSFET's overdrive is hundreds of millivolts and chosen by the designer.

Two decisions that follow (any two): the 220 mV degeneration rule gives an emitter factor of 10 and a source factor of 2, because r_s is ten times r_e ; a source follower's gain is 0.80 into 1 kilohm where an emitter follower gives 0.97; MOSFETs go at the input of a differential pair, where infinite input resistance matters and gain does not, and bipolars go at the output, where the reverse is true.

(1 mark)

Question 6 A rule, a load line, and a factor that appears twice

9 MARKS

(a) The load line

2 MARKS

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

A straight line with intercepts at $V_{CE} = V_{CC}$ when $I_C = 0$, and at $I_C = V_{CC}/(R_C + R_E)$ when $V_{CE} = 0$. (1 mark)

An amplifier sits near the middle, because it must be able to move both ways and “the middle” means whichever point leaves equal headroom above and below the wanted swing. **Near the supply end the positive half clips; near saturation the negative half clips**, and both waste the stage. (1 mark)

(b) The 220 millivolt rule

3 MARKS

$$R_E = \frac{220 \text{ mV}}{I_C}$$

(1 mark)

$$EF = \frac{r_e + R_E}{r_e} = \frac{V_T/I_C + 0.220/I_C}{V_T/I_C} = 1 + \frac{0.220}{0.026} = 9.46$$

The current cancels, so the emitter factor is 9.46 at any collector current whatever.

(1 mark)

Why a voltage rather than a resistance. Because that is what makes it current-free. A rule stating a resistance would give a different emitter factor at every bias current and would have to be restated for each; a rule stating a *drop* fixes the ratio R_E/r_e directly, since r_e is itself a voltage over the current.

(1 mark. A candidate who notes that 220 mV also lands the answer on the E12 grid across the useful decade earns credit but not the mark; the cancellation is the point)

(c) One factor, two jobs

2 MARKS

Gain. With and without the emitter resistor,

$$\frac{A_v(\text{no } R_E)}{A_v(\text{with } R_E)} = \frac{R_C/r_e}{R_C/(r_e + R_E)} = \frac{r_e + R_E}{r_e}$$

Drift. The disturbance is V_{BE} falling with temperature, which is equivalent to the base rising. With the base held, the emitter follows, so $\Delta I_C = \Delta V/R_E$ instead of $\Delta I_C = \Delta V/r_e$, and the ratio is

$$\frac{r_e + R_E}{r_e}$$

the same expression.

(1 mark for both derivations)

The consequence: the emitter resistor divides the drift by exactly the factor by which it divides the gain. **The stability is not free and it is not cheap; it costs gain one for one.**

(1 mark)

(d) A better transistor will not help

2 MARKS

The coefficient is about -2 mV per degree in V_{BE} at a fixed collector current, and it comes from the saturation current's temperature dependence, which is set by the bandgap. **It is not a parameter a designer chooses:** every silicon bipolar transistor has very nearly the same figure, because they are all made of silicon.

(1 mark)

What the emitter resistor does that no device could. It is not reducing the drift; it is **feeding the drift back.** The resistor converts a change in collector current into a change in emitter voltage that opposes it, which is a local feedback loop with a loop gain of R_E/r_e . A better device would still drift; a loop divides whatever drift there is by its own loop gain, and that is a property of the circuit rather than of the transistor.

(1 mark)

Question 7 Three results from one model

11 MARKS

(a) Four steps

3 MARKS

1. Every DC source becomes a short to ground.
2. Every coupling and bypass capacitor becomes a short.
3. **The bias network disappears** wherever it is now in parallel with something much smaller.
4. **The transistor becomes r_e from base to emitter and a current source from collector to emitter,** carrying the current r_e passes.

(2 marks)

Justifying the first. A small-signal model describes *changes*. A rail that holds a constant voltage has no change on it, and a node with no change on it is indistinguishable from ground as far as signals are concerned. It is not an approximation about the supply being stiff; it is what "small-signal" means.

(1 mark)

Blind to (any three): everything nonlinear, so clipping, distortion and slew rate; the fact that the device turns off; capacitance and therefore every frequency; the Early effect, until it is added; noise.

(b) Two results**3 MARKS**

Gain. The input drives a current $v_{in}/(r_e + R_E)$ through the emitter branch. That same current comes out of the collector and flows through R_C :

$$A_v = -\frac{R_C}{r_e + R_E}$$

(1 mark)

Input resistance. The current drawn at the base is the emitter current divided by β , and the voltage is that current times the emitter branch:

$$Z_{in(base)} = \beta(r_e + R_E)$$

(1 mark)

The input resistance depends on β and the gain does not. So the gain can be specified; the input resistance can only be bounded. A datasheet quoting β as 100 minimum and 400 typical is quoting an input resistance with a factor of four in it, which is why every stage in this book that needs a defined input resistance gets it from a bias network or a MOSFET rather than from a base.

(1 mark)

(c) The emitter factor, and the node**3 MARKS**

Dividing the two gain expressions:

$$\frac{A_v(\text{no } R_E)}{A_v(\text{with } R_E)} = \frac{r_e + R_E}{r_e} \equiv EF$$

(1 mark)

It is often said to decide two things: the factor by which the stage's gain **decreases**, and the factor by which the stage's **output resistance increases**.

The gain claim is correct as stated, and the derivation above is its proof. (1 mark)

The output-resistance claim is not. A stage's output resistance is R_C **in parallel** with whatever the transistor presents, and a parallel combination is smaller than either part, so no amount of degeneration can push it above R_C . The corrected statement is that the emitter factor multiplies the **resistance looking into the collector**:

$$R_{into\ collector} = r_o [1 + g_m(R_E \parallel r_\pi)] + (R_E \parallel r_\pi) \approx r_o \cdot EF$$

and the stage's output resistance is that in parallel with R_C .

(1 mark)

(d) The substitution

2 MARKS

Transfers: the gain, $A_v = -R_D/(r_s + R_S)$, and the **source factor**, $SF = (r_s + R_S)/r_s$. Output resistance transfers too.

Does not transfer: the input resistance. A gate draws no current, so there is no $\beta(r_e + R_E)$ term to carry across; a common-source stage's input resistance is its bias network and nothing else. (1 mark)

$$r_s = \frac{1}{g_m} = \frac{1}{4 \text{ mS}} = 250, \quad SF = 1 + \frac{220}{250} = 1.88$$

Why 2 where the emitter factor is 10. The same 220 mV is divided by a different intrinsic resistance. r_e is 26 ohm and r_s is 250, a factor of 9.6, because a MOSFET's transconductance is about ten times lower at the same current. **The correspondence between the two factors follows from that one ratio.** (1 mark)

Question 8 A dead band, a rule, and a thing bolted to a heatsink

10 MARKS

(a) The follower

3 MARKS

The input drives r_e and R_E in series; the output is taken across R_E alone, which is a divider:

$$A_v = \frac{R_E}{r_e + R_E}$$

(1 mark)

The shortfall from unity is $r_e/(r_e + R_E)$, and $r_e = V_T/I_C$, so the only quantity it depends on is the current. Not the device, not the load, which is given. A follower that is not good enough is a follower that is not biased hard enough. (1 mark)

$$Z_{in} = \beta(r_e + R_E), \quad Z_{out} = r_e + \frac{R_{source}}{\beta}$$

What the stage is for: it is an **impedance transformer**. It shows the driving stage β times what is on its emitter, and shows the load the driving resistance divided by β , while passing the signal at a gain of about one. (1 mark)

(b) Three classes

3 MARKS

Class	What conducts	Max efficiency
A	one device, the whole cycle	25 % (resistive load)
B	two devices, half each	78.5 %, which is $\pi/4$
AB	two devices, both near zero, one taking over	just below B

(2 marks: 1 for the definitions, 1 for two efficiencies)

Why crossover distortion is worse than its size. The dead band is about 1.3 V wide, which in a stage swinging 30 V is 4 per cent and sounds tolerable. But it sits **at the origin**, and a music signal spends most of its time near the origin. So the distortion is **worst on quiet passages and vanishes at full output**, which is the opposite of every other distortion mechanism in an amplifier and is why it is audible far below the level its percentage suggests.

(1 mark. "It is nonlinear" scores nothing; the position is the answer)

(c) The 26 millivolt rule

2 MARKS

$$R_E = \frac{V_T}{I_q} \Rightarrow EF = \frac{r_e + R_E}{r_e} = \frac{V_T/I_q + V_T/I_q}{V_T/I_q} = 2$$

exactly, and at any idle current, because the rule sets R_E equal to r_e by construction.

(1 mark)

The two costs being balanced are thermal sensitivity, which the resistors halve, and **output power**, which they take because they sit in series with the load.

At **260 mV** the emitter factor is 11 and the stage is very stable, and 2.2 ohm in series with an 8 ohm loudspeaker has thrown away a fifth of the output power. At **2.6 mV** the resistors are 0.022 ohm, they cost nothing, and they have stopped suppressing anything. **The 26 mV point is where the two costs cross.**

(1 mark)

(d) Two diodes on a heatsink

2 MARKS

They are not setting a voltage. They are tracking one. The output transistors' V_{BE} falls 2 mV per degree as they warm, so at a fixed bias the idle current rises, which raises the dissipation, which raises the temperature: a runaway. Diodes at the same temperature drop 2 mV per degree less as well, so the bias falls by exactly as much as the outputs stop needing, and the drift cancels to first order.

(1 mark)

A stage with the diodes on the circuit board is correct at 25 degrees and destroys itself at 60. The emitter resistors halve the drift and do not stop it: 3.8 per cent per degree compounds to a factor of three over thirty degrees.

And the two schematics are identical, because thermal contact is not a circuit property. Nothing about the netlist distinguishes a working amplifier from one that fails in ten minutes, which is why this is the most common way a first output stage is destroyed.

(1 mark)

Question 9 The tail, the tanh, and the output you choose

11 MARKS

(a) The tail, doubled

4 MARKS

Why $2R_{tail}$. For a common-mode input both halves change their current by the same amount i . The tail carries the **sum**, so it carries $2i$, and the voltage it develops is $2iR_{tail}$. One half sees a voltage of $2iR_{tail}$ in response to its own current i , which is a resistance of $2R_{tail}$ in its own emitter. (1 mark)

It is then Chapter 7's degenerated stage, unchanged:

$$A_{cm} = -\frac{R_C}{2R_{tail} + r_e}$$

(1 mark)

$$CMRR = \frac{A_{dm}}{A_{cm}} = \frac{R_C/2r_e}{R_C/(2R_{tail} + r_e)} = \frac{2R_{tail} + r_e}{2r_e} \approx \frac{R_{tail}}{r_e}$$

(1 mark)

R_C **cancels exactly**, because it is in the numerator of both gains. **Rejection is a property of the tail and the operating current, and no choice of load affects it.** (1 mark)

(b) What a current source is

2 MARKS

A current source is a device with a large incremental resistance and a small voltage across it. Those are the two quantities, and they are independent: $r_o = V_A/I_C$ has nothing to do with the voltage the device is sitting at.

No resistor can have both, because a resistor's incremental resistance *is* its resistance, so a large one carrying the tail current necessarily drops a large voltage. 260 kilohm at 2 mA is 520 V. (1 mark)

The general principle: wherever a circuit needs a large resistance without a large voltage across it, the answer is a current source. (1 mark for the principle plus two of: the mirror load of Chapter 7, which needs r_o rather than a large R_C that would take all the headroom; the mirror load of Chapter 9's pair; the current-source load of Chapter 10's gain stage; the emitter sink under Chapter 10's buffer.)

(c) The tanh

3 MARKS

The small-signal transconductance is the slope at the origin. Since $\tanh x \rightarrow x$ for small x :

$$\frac{d}{dv_d} \left[I_{tail} \tanh \frac{v_d}{2V_T} \right]_{v_d=0} = \frac{I_{tail}}{2V_T} = \frac{I_{tail}/2}{V_T} = \frac{1}{r_e}$$

which is $1/r_e$ and not $1/2r_e$: the two in the *gain to one collector* comes from taking one collector, and the difference current has no two in it. (1 mark)

The 1 per cent point. Solve $\tanh(x)/x = 0.99$ with $x = v_d/2V_T$, which gives $x = 0.1743$ and

$$v_d = 2 \times 26 \text{ mV} \times 0.1743 = 9.06 \text{ mV}$$

(1 mark)

The tail current does not appear. It multiplies both the tanh and its tangent, so it cancels out of the ratio. **Biasing the pair harder buys transconductance and buys no linearity at all**, which is the opposite of the pattern followers establish. The only lever on linearity is degeneration, which trades gain for range one for one. (1 mark)

(d) Fifty-two decibels and ninety-eight

2 MARKS

Both are right, and they measured different outputs.

52 dB is the single-ended figure, taken at one collector. A common-mode input genuinely moves that collector, by $R_C/2R_{tail}$, and **the tail alone decides how much**. Matching does not enter at first order.

98 dB is the differential figure, taken between the two collectors. The common-mode motion is identical on both, so it subtracts out exactly when the halves match, and what survives is the **load mismatch** acting on it: $CMRR = 2R_{tail}/\delta r_e$, which for 1 per cent resistors is 98 dB.

(1 mark for the distinction, 1 for naming the limit in each)

So “the CMRR of a differential pair” is not a statement about a circuit unless it says which output is taken. The two differ by 46 dB at ordinary component tolerances, from the same transistors.

Question 10 The gain you cannot buy

9 MARKS

(a) The gain that does not depend on current

4 MARKS

The load is the current source’s own output resistance in parallel with the transistor’s:

$$R_{load} = r_o \parallel r_o = \frac{V_A}{2I_C}$$

and the transconductance is $g_m = I_C/V_T$, so

$$|A_v| = g_m R_{load} = \frac{I_C}{V_T} \cdot \frac{V_A}{2I_C} = \boxed{\frac{V_A}{2V_T}} = \frac{100}{0.052} = \mathbf{1923} = \mathbf{65.7 \text{ dB}}$$

(3 marks: 2 for the derivation, 1 for showing the cancellation explicitly)

The collector current cancels exactly. The load falls as $1/I$ and the transconductance rises as I .

So the designer proposing more supply current gets nothing. Gain here is a property of the process, through V_A , and not of the design, which is why every gain stage in every operational amplifier lands within a few decibels of the same figure.

The only lever that works is a cascode, which multiplies r_o by β and the gain with it: $\beta V_A/2V_T = \mathbf{96,000}$, or **99.7 dB from one stage**. (1 mark)

(b) Where 11.5 dB goes**2 MARKS**

Into loading. Each stage's output resistance and the next stage's input resistance form a divider, and the product of the *unloaded* gains counts none of it. The largest single term is the gain stage driving the output stage: 50 kilohm of output resistance into 21 kilohm of input resistance keeps 30 per cent, which is 10.6 dB. (1 mark)

It cannot be recovered by improving either gain stage, because the loss is not in either of them. Doubling a gain stage's own gain doubles what arrives at the divider and the divider still takes 70 per cent of it. **The fix is at the interface, not at the stages**, and it is another follower: the output stage is already a Darlington, and a third follower is what a real power amplifier uses. (1 mark)

(c) Compensation**2 MARKS**

It is compensating the amplifier's own poles. Four stages give at least four poles, each contributing up to 90 degrees of phase lag, and a loop with 180 degrees of lag and a loop gain above one oscillates. (1 mark)

The capacitor makes one pole dominant, deliberately, at a few tens of hertz, so the open-loop gain is already falling at 20 dB per decade long before any other pole contributes phase. By the time the others matter the loop gain is below one, and a loop with a gain below one cannot oscillate.

Slowing it down is what makes it usable because stability is about the *order* in which the gain and the phase arrive, not about speed. An amplifier fast enough to have all four poles inside its loop bandwidth is an oscillator; one slowed at a single point is an amplifier.

(1 mark. *It is worth noting that this is the Miller effect of Chapter 7, which ruined a stage's bandwidth there, used on purpose here*)

(d) Why the closed loop does not care**1 MARK**

Because the closed-loop gain is set by the feedback network, and the forward path only has to be large. With a loop gain of 49,000 the error is one part in 49,000, and a factor of fourteen change in the open-loop gain moves that error from 0.0007 per cent to 0.0095 per cent: both are nothing. **The quantity that makes it so is the loop gain.** (1 mark)